

ERDŐS PROBLEM 1112, FROM SCRATCH: A COMPLETE PROOF FOR YOUNG MATHEMATICIANS

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ABSTRACT. In 1980 Erdős and Graham asked a question about dodging: can an infinite sequence of numbers with prescribed small gaps always be placed so that the sums of k of its members — the “ k -fold sums” — avoid a given rapidly growing sequence? This document proves the complete answer — yes precisely when $d_2 \geq k + 1$ — assuming nothing beyond high school algebra. Every term in this question, including this d_2 and k , is defined from scratch in the text; every proof is given in full; and the two genuinely deep ingredients (a theory of “balanced words” and a finite subset-sum theorem) are developed as self-contained mini-courses. It is long because it is honest: this is the whole proof, slowly.

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This is the *friendly* version of the paper *A dichotomy for Erdős problem 1112* (paper/erdos1112.pdf in the same repository, https://github.com/beetree/math_erdos_1112). Both documents prove the same theorems; that one is written for professional mathematicians, this one for a strong high school student willing to work. Nothing here is new relative to that paper, and where the two disagree, that one is authoritative.

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Part 1. The problem

1. THE PROBLEM, FROM SCRATCH

1.1. What this document is. This document proves a theorem — a real one, first posed by Paul Erdős and Ronald Graham in 1980 and solved only in 2026. The solution was written up as a research paper for professional mathematicians; you are holding the other version. Both prove exactly the same result. The difference is the audience: here, nothing is assumed beyond school algebra and a willingness to think hard.

Some ground rules, so you know what you are signing up for.

- *Everything is defined.* If a word like “sumset” or “lacunary” appears, it was defined earlier in this document. If you hit an undefined word, that is our mistake, not a gap in your education.
- *Every proof is complete.* When a step is genuinely hard, we slow down and build the tools — we do not say “it can be shown”. Two parts of the argument need real machinery (a theory of infinite “words” in Part III, and a finite subset-sum theorem in Part IV); each gets a self-contained mini-course.
- *Checkpoints are part of the deal.* Scattered through the text are *Checkpoints*: small exercises. They are chosen so that if you can do them, you have understood the last few pages. Doing them is not optional in spirit — mathematics is not a spectator sport — but no later proof secretly depends on your solutions.
- *It is long.* A research paper compresses; this document decompresses. The length is the price of honesty, and the parts can be read somewhat independently (a map is given in §1.8).

One notational convention up front: $\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the positive integers (natural numbers), and when we say *number* without qualification we mean a positive integer.

1.2. Sequences and gaps. The two main characters of this story are infinite sequences of positive integers. Let us fix the vocabulary.

Definition 1.1 (Sequence, gaps). A *sequence* is an infinite list of positive integers

$$a_1 < a_2 < a_3 < \dots,$$

strictly increasing, going on forever. We write $A = \{a_1 < a_2 < \dots\}$ and think of A interchangeably as a list or as an infinite set of positive integers. The *gaps* of A are the differences

between consecutive terms:

$$g_n := a_{n+1} - a_n \quad (n = 1, 2, 3, \dots).$$

Example 1.2. The even numbers $A = \{2, 4, 6, 8, \dots\}$ have all gaps equal to 2. The squares $\{1, 4, 9, 16, 25, \dots\}$ have gaps $3, 5, 7, 9, \dots$, growing without bound. The powers of ten $\{10, 100, 1000, \dots\}$ have gaps $90, 900, 9000, \dots$, growing very fast indeed.

Our first character is a sequence whose gaps are *constrained to a window*: they may wobble, but only between two fixed bounds.

Definition 1.3 ((d_1, d_2) -sequence). Let $d_1 \leq d_2$ be positive integers. A (d_1, d_2) -sequence is a sequence A all of whose gaps lie between d_1 and d_2 :

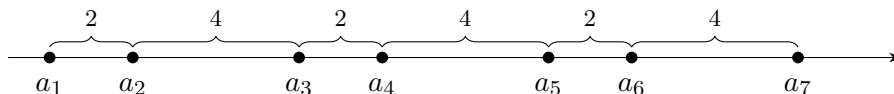
$$d_1 \leq a_{n+1} - a_n \leq d_2 \quad \text{for every } n \geq 1.$$

Example 1.4. Take $(d_1, d_2) = (2, 4)$. Then

$$A = \{1, 3, 7, 9, 13, 15, 19, \dots\}$$

(gaps $2, 4, 2, 4, 2, 4, \dots$) is a $(2, 4)$ -sequence, and so is $\{5, 8, 11, 14, \dots\}$ (all gaps 3), and so is any sequence you build by walking right along the number line taking steps of size 2, 3 or 4, in any order you like, forever.

That last phrasing is the right mental picture: a (d_1, d_2) -sequence is a walk along the number line whose steps are all between d_1 and d_2 . You choose the starting point and you choose each step, subject only to the window.



Checkpoint 1.5. (a) Write down the first six terms of a $(2, 4)$ -sequence that is *not* eventually periodic in its gaps — that is, keep the gap pattern from ever settling into a repeating cycle. Convince yourself you could continue it forever. (b) Show that if A is a (d_1, d_2) -sequence then $a_1 + (n - 1)d_1 \leq a_n \leq a_1 + (n - 1)d_2$ for every n . (Add up the gaps.)

1.3. Sumsets. Our second ingredient is addition: we will be interested not in A itself but in the set of all numbers you can make by adding up k members of A .

Definition 1.6 (k -fold sumset). Let A be a sequence and $k \geq 1$ an integer. The k -fold sumset of A is

$$kA := \{x_1 + x_2 + \dots + x_k : x_1, \dots, x_k \in A\},$$

where the x_i need *not* be distinct — you may use the same member of A several times.

Two warnings about this (standard) notation. First, kA is *not* the set $\{k \cdot a : a \in A\}$ of multiples — though it contains all of those (take $x_1 = \dots = x_k = a$). Second, repetitions are allowed: for $A = \{1, 3, 7, \dots\}$ and $k = 3$, the sums $1 + 1 + 1 = 3$ and $1 + 1 + 7 = 9$ and $3 + 7 + 7 = 17$ all belong to $3A$.

Example 1.7. Let $A = \{5, 8, 11, 14, \dots\}$, the numbers of the form $5 + 3j$ with $j \geq 0$. What is $3A$? Each element is a sum

$$(5 + 3j_1) + (5 + 3j_2) + (5 + 3j_3) = 15 + 3(j_1 + j_2 + j_3),$$

and since $j_1 + j_2 + j_3$ can be any integer ≥ 0 , we get exactly $3A = \{15, 18, 21, 24, \dots\}$: again an arithmetic progression with the same common difference 3, starting at 15. This tidiness

is special to arithmetic progressions — and, as we will see much later, it is precisely what makes them *bad* at dodging.

Checkpoint 1.8. (a) For $A = \{1, 3, 7, 9, 13, \dots\}$ of Example 1.4, list every element of $3A$ up to 25. (b) Show that for any sequence A , the smallest element of kA is ka_1 . (c) Show that if A is a (d_1, d_2) -sequence, then between consecutive “diagonal” elements ka_n and ka_{n+1} of kA the distance is at most kd_2 .

1.4. Lacunary sequences. The sequence to be dodged grows *multiplicatively*: each term is at least a fixed multiple of the one before.

Definition 1.9 (Lacunary sequence). Let $r \geq 1$ be an integer. A sequence $B = \{b_1 < b_2 < \dots\}$ is *lacunary with ratio r* if

$$b_{i+1} \geq r b_i \quad \text{for every } i \geq 1.$$

Example 1.10. The powers $\{r, r^2, r^3, \dots\}$ are lacunary with ratio r — the minimal example, growing as slowly as the definition allows. But a lacunary sequence need not be this regular: $\{1, 7, 100, 1000, 10^{12}, \dots\}$ is lacunary with ratio 7, as long as every step multiplies by at least 7. The word *lacunary* comes from the Latin *lacuna*, a gap or hollow: on the number line, such a sequence is almost all gap.

The picture to hold: a lacunary sequence with a large ratio is *extremely sparse*. Below any number N it has at most about $\log_r N$ terms. Between consecutive terms b_i and $b_{i+1} \geq r b_i$ yawns an empty stretch r times longer than everything that came before. Sparseness is what should make B *easy* to dodge — that is the intuition the problem tests.

Checkpoint 1.11. Let B be lacunary with ratio $r \geq 2$. Show that $b_i \geq b_1 r^{i-1}$ for all i , and conclude that B has at most $1 + \log_2 N$ terms below any $N \geq b_1$.

1.5. The question, as a game. We can now state the problem. Here it is verbatim, as recorded on erdosproblems.com as problem 1112:

Let $1 \leq d_1 < d_2$ and $k \geq 3$. Does there exist an integer r such that if $B = \{b_1 < \dots\}$ is a lacunary sequence of positive integers with $b_{i+1} \geq r b_i$ then there exists a sequence of positive integers $A = \{a_1 < \dots\}$ such that $d_1 \leq a_{i+1} - a_i \leq d_2$ for all $i \geq 1$ and $(kA) \cap B = \emptyset$, where kA is the k -fold sumset?

Every phrase in this is now something we know. But sentences with three quantifiers (“there exists $r \dots$ such that if [every] $B \dots$ then there exists A ”) are hard to hold in the head, so let us restate it as a *game* between you and an Adversary. The numbers $k \geq 3$ and $1 \leq d_1 < d_2$ are fixed in advance and known to both players.

- (1) **You** announce an integer r — once and for all.
- (2) **The Adversary**, knowing r , builds a lacunary sequence B with ratio r : an infinite sequence with $b_{i+1} \geq r b_i$. The Adversary is infinitely clever and builds B specifically to hurt you.
- (3) **You**, knowing B completely, build a (d_1, d_2) -sequence A : a walk with steps in $[d_1, d_2]$.

You win if kA avoids B entirely: no sum of k members of A ever equals a member of B . In symbols, $(kA) \cap B = \emptyset$.

The problem asks: *do you have a winning strategy?* That is, is there a single choice of r such that, whatever B the Adversary produces, you can answer with a winning A ?

The order of moves is the soul of the statement, so let us dwell on it.

- You choose A *after* seeing all of B . You may tailor your walk to the particular sequence you must dodge. (If instead you had to choose A first, the Adversary would simply pick some element of kA — it is an infinite set — and start B there. You would lose always, and the problem would be boring.)
- But you choose r *before* seeing B . One fixed r must defeat every lacunary-ratio- r sequence at once. Making r enormous makes the Adversary's sequence extremely sparse — intuitively, easier to dodge. The question is whether *any* r , however large, is enough.

Why would this be hard? Your walk A is dense — steps at most d_2 , so you have lots of members — and it is only kA , the k -fold sums, that must dodge B . The tension: A 's gaps are *bounded*, so kA is forced to be rather evenly spread, and an evenly spread set has a hard time missing even a very sparse one, because “evenly spread” means it shows up everywhere at every scale. Whether the k -fold sums have enough slack to be steered around a sparse B — that is exactly what the problem is asking, and the answer will turn out to hinge on a precise comparison between k and d_2 .

1.6. A first fight: tiny r loses. To get blood flowing, let us play the game and prove something: announcing $r = 1$ loses. This tiny result will not be needed later, but it makes the rules concrete, and it is our first complete proof.

Proposition 1.12. *Let $k \geq 3$ and $1 \leq d_1 < d_2$. If you announce $r = 1$, the Adversary wins: there is a lacunary sequence B with ratio 1 such that every (d_1, d_2) -sequence A has $(kA) \cap B \neq \emptyset$.*

Proof. With $r = 1$ the lacunarity condition reads $b_{i+1} \geq b_i$, which any increasing sequence satisfies. So the Adversary may play

$$B = \{1, 2, 3, 4, 5, \dots\},$$

all positive integers. Now let A be any (d_1, d_2) -sequence whatsoever. The sumset kA is not empty — it contains, for instance, ka_1 (add a_1 to itself k times), which is a positive integer, hence a member of B . So $(kA) \cap B \neq \emptyset$: the Adversary wins. $\square$

The same trivially covers $r \leq 0$ (for B positive, $b_{i+1} \geq rb_i$ is then no constraint at all). So any winning r must be at least 2, and the real question starts there: the problem page asks about “an integer r ”, and we now see this means, in effect, an integer $r \geq 2$.

Checkpoint 1.13. Suppose you announce $r = 2$ and, feeling lazy, you resolve to always play an arithmetic progression: $A = \{c, c + d, c + 2d, \dots\}$ for some starting point c and fixed gap $d \in [d_1, d_2]$ of your choosing. Show that the Adversary can beat this strategy. (Hint: Example 1.7 shows kA is itself an arithmetic progression — it contains all sufficiently large multiples of d ... at least when gcd-considerations cooperate. First try to see why the Adversary can defeat each *single* progression, then worry about defeating all of them with one B . The second part is genuinely harder; we will meet it seriously in Part III as the “certificate trick”.)

1.7. The answer. Here is the theorem this document proves. It answers the question completely, and the answer is a genuine dichotomy: a sharp line through parameter space, with a winning strategy on one side and a losing verdict on the other.

Theorem A (The dichotomy). Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers.

- (1) (Existence.) If $d_2 \geq k + 1$, you have a winning strategy. In fact announcing any $r \geq 192 d_2$ works: every lacunary B with that ratio can be dodged by some (d_1, d_2) -sequence A with $(kA) \cap B = \emptyset$.
- (2) (Non-existence, in a strong form.) If $d_2 \leq k$, no announcement wins. Worse: even if the fixed ratio r is replaced by any prescribed sequence $r_1, r_2, r_3, \dots$ of lower bounds — the Adversary now being required to satisfy $b_{i+1} \geq r_i b_i$ at every step, with the r_i growing as fast as you like — the Adversary can still build a *single* sequence B that defeats *every* (d_1, d_2) -sequence A at once.

So the answer to Erdős problem 1112 is: *yes precisely when $d_2 \geq k + 1$.*

Pause on how lopsided the two conclusions are. Above the threshold, one fixed ratio $192 d_2$ tames every lacunary B . Below it, the collapse is total: you may demand $b_{i+1} \geq r_i b_i$ with $r_i = 10^{10^i}$ — forcing B to be unimaginably sparse — and still a single B meets the k -fold sums of *every* admissible walk. Sparseness, which looked like your best friend in §1.5, turns out to be completely irrelevant below the threshold: no amount of it saves you.

Notice also what the threshold does *not* depend on: d_1 . Only the largest allowed step d_2 matters, measured against the number of summands k . Why the combination “ d_2 versus $k+1$ ”? That is the subject of the next section, where one picture explains the whole dichotomy — and the rest of the document is the long, happy labor of making that picture rigorous.

1.8. Map of the document. Part I (§1–2) states the problem and gives the one picture — windows of width k striding at pace γ — from which everything grows; §2.4 turns that picture into an honest ledger of four gaps, which the rest of the document closes. **Part II** (§3–6) proves the existence half: floor toolbox, Beatty walks, the safe/unsafe dial, the free-gap lemma, nested intervals, the ratio $192 d_2$. **Part III** (§7–14) reduces the non-existence half to walks (the certificate trick), then proves every admissible walk tail-covering: easy cases first, then the two-letter campaign — band and sweep, the Morse–Hedlund classification, the Sturmian endgame — and finally the slot lemma, which prices the corner at exactly the subset-sum lemma. **Part IV** (§15–23) proves that lemma: machinery, reduction to the hard core, six branches. **Part V** (§24–25) assembles the theorem and explains the computer verification.

Three reading routes. The *spine*: §1, §2, then the Big idea boxes and theorem statements of each part — an afternoon, and you will know what is true and why it is plausible. The *existence route*: Parts I–II are self-contained and fully rigorous — a complete, checkable proof of the positive half in about twenty pages. The *whole climb*: everything in order; the two mini-courses (§10, §12) and Part IV are steepest. Checkpoints throughout are the footholds.

Correspondence with the research paper. Part I $\leftrightarrow$ its §1; Part II $\leftrightarrow$ its §2; Part III $\leftrightarrow$ its §3; Part IV $\leftrightarrow$ its §4; Part V $\leftrightarrow$ its §5–§6 and appendices. Every theorem here bears the same content as a numbered result there (and through its Appendix C, a machine-checked Lean declaration); this document adds pedagogy, examples and checkpoints, and inherits all authority from the parent.

2. THE ONE PICTURE THAT EXPLAINS THE THRESHOLD

Why should the dividing line fall at $d_2 = k + 1$? Before any rigor, this section shows you the picture from which the entire proof grows. Nothing here is officially part of the proof — everything will be re-done carefully later — but every later section is best read with this picture in mind. The section ends with an honest list of exactly what the picture does *not* establish; that list is the table of contents of the rest of the document.

2.1. Steady walks, and rounding error. Among all (d_1, d_2) -sequences, the simplest are the *steady* ones: walks that try to move at a constant average speed γ per step. If γ is an integer d , that is an arithmetic progression, $a_i = c + id$. But γ need not be an integer. To walk at average speed $\gamma = 3.5$, alternate steps of 3 and 4; and in general, the canonical way to walk at any real speed γ is to place your i -th footfall at the integer just below $i\gamma$:

$$a_i = c + \lfloor i\gamma \rfloor \quad (i = 1, 2, 3, \dots),$$

where $c \geq 0$ is an integer offset and $\lfloor x \rfloor$ denotes the *floor* of x , the greatest integer $\leq x$ (so $\lfloor 3.5 \rfloor = 3$, $\lfloor 7 \rfloor = 7$). We will call this the *Beatty walk* with slope γ and offset c .¹

Example 2.1. Take $\gamma = 3.5$ and $c = 0$. Then $\lfloor 1 \cdot 3.5 \rfloor = 3$, $\lfloor 2 \cdot 3.5 \rfloor = 7$, $\lfloor 3 \cdot 3.5 \rfloor = 10$, and so on:

$$A = \{3, 7, 10, 14, 17, 21, 24, 28, 31, 35, \dots\},$$

with gaps $4, 3, 4, 3, 4, 3, \dots$ — a perfectly legal $(d_1, 4)$ -sequence for any $d_1 \leq 3$.

Now the question the whole problem turns on: *where do the k -fold sums of a steady walk live?* Take k members of the walk, say with indices $i_1, \dots, i_k$, and write $s := i_1 + \dots + i_k$ for the index total. If there were no floors, the sum would be exactly $kc + s\gamma$. Each floor rounds down by less than 1 — $\lfloor x \rfloor$ always lies in the half-open interval $(x - 1, x]$ — so k floors together round down by less than k :

$$a_{i_1} + \dots + a_{i_k} = kc + \sum_{t=1}^k \lfloor i_t \gamma \rfloor \in (kc + s\gamma - k, kc + s\gamma].$$

Every k -fold sum therefore lands in a *window* of width k hanging just below one of the values $kc + s\gamma$. As s runs through the integers, these anchor values march along the number line in equal strides of γ :

Big idea. The k -fold sums of a steady walk of speed γ are confined to windows of width k , one hanging below each multiple of γ (shifted by kc). The windows stride along the line at pace γ . Everything — both halves of the dichotomy — follows from comparing the stride γ to the window width k .

Checkpoint 2.2. Prove the inequality just used: for any real numbers $x_1, \dots, x_k$,

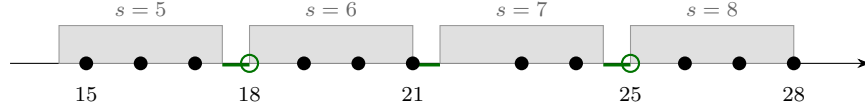
$$x_1 + \dots + x_k - k < \lfloor x_1 \rfloor + \dots + \lfloor x_k \rfloor \leq x_1 + \dots + x_k.$$

(Start from $x - 1 < \lfloor x \rfloor \leq x$ for a single x , and add.)

2.2. Above the threshold: daylight. Suppose the stride exceeds the window width: $\gamma > k$. Then consecutive windows do not touch — between the end of one and the start of the next there is a stretch of length $\gamma - k$ that no k -fold sum of the walk can ever reach. Call these stretches *daylight*. They recur forever, once per stride, making up a fixed positive fraction of the number line at every scale.

Figure 1 shows this for Example 2.1, where $k = 3$ and $\gamma = 3.5$: the windows are $(3.5s - 3, 3.5s]$ and the daylight stretches have width 0.5. The integer 25 sits in daylight — so no three members of that walk sum to 25, a fact you can (and should) also confirm by hand.

¹After Samuel Beatty, whose name is attached to sequences of the form $\lfloor i\gamma \rfloor$. Part II develops their basic properties from scratch; here we only need the rough picture.



windows $(3.5s - 3, 3.5s]$ for $k = 3$, $\gamma = 3.5$, $c = 0$; circled integers sit in daylight

FIGURE 1. The window picture for the walk of Example 2.1. Each 3-fold sum lands in a shaded window; the integers 18 and 25 fall in the daylight between windows, so they are *unreachable*: no three members of A sum to them. (The window $(18, 21]$ is open at its left end, which is why 18 escapes.)

Checkpoint 2.3. (a) Using the first ten terms listed in Example 2.1, verify by direct search that no three members of A (repetitions allowed) sum to 18 or to 25. (b) Explain why the window containment makes the search finite: any sum of three terms involving a term > 25 is itself > 25 . (c) Find the next two unreachable integers after 25.

Now the strategic point. When can an admissible walk have stride $\gamma > k$? The steps available to us are integers between d_1 and d_2 , so the fastest we can go is $\gamma = d_2$, and to get a *fractional* stride strictly between k and an achievable pair of step sizes we want to mix steps $d_2 - 1$ and d_2 , giving any speed γ with $d_2 - 1 < \gamma < d_2$. Such a γ exceeds k exactly when $d_2 - 1 \geq k$, that is, when

$$d_2 \geq k + 1.$$

There is the threshold, appearing for the first time — not yet as a theorem, but as the exact condition under which the daylight picture is available to us at all.

Above the threshold, then, here is the plan for winning the game. The daylight stretches sit at predictable places, determined by the slope γ ; *turning the dial γ slides the whole ladder of windows*, and with it the daylight. Given the Adversary's sequence B , we will tune γ (and the offset c) so that every single element of B falls into daylight. One dial must dodge infinitely many targets simultaneously — that sounds ambitious, but the targets b_i are lacunary, hence ever sparser, and it will turn out (Part II) that each successive target only forbids a tiny range of dial positions, leaving plenty for the next. The tool that makes “tiny” precise is the *free-gap lemma*, and the tool that finishes infinitely many rounds of tuning is the *nested interval theorem*. The mysterious ratio $192d_2$ of Theorem A is nothing deeper than the bookkeeping constant that keeps each round of tuning from exhausting the dial room the previous round left behind.

2.3. Below the threshold: gridlock. Now suppose $d_2 \leq k$, so every step is at most k and every stride is at most k : windows of width k marching at pace $\gamma \leq k$. Consecutive windows now overlap, or at best touch. There is no daylight anywhere. The picture suggests that the k -fold sums of such a walk eventually swallow everything in their path — and for steady integer strides this is not merely a picture but a two-line calculation.

Example 2.4 (Steady walks are hopeless). Let A be the arithmetic progression $a_i = c + id$ with integer gap $d \leq k$. A k -fold sum is

$$(c + i_1 d) + \cdots + (c + i_k d) = kc + sd, \quad s = i_1 + \cdots + i_k,$$

and s ranges over *all* integers $\geq k$. So

$$kA = \{kc + sd : s \geq k\} :$$

from $kc + kd$ onwards, kA contains *every* number that is $\equiv kc \pmod{d}$. We say kA contains a *tail of a congruence class*. A set this fat cannot dodge: however sparse the Adversary is forced to be, the class contains arbitrarily large numbers, and the Adversary simply builds B out of them — choosing each b_{i+1} to be some member of the class beyond $r_i b_i$. Sparseness costs the Adversary nothing, which is exactly the brutal flavor of part (2) of Theorem A.

Checkpoint 2.5. In Example 2.4, why does s range over all integers $\geq k$ and no others? (Show: each $i_t \geq 1$ gives $s \geq k$; conversely, realize any $s \geq k$ by choosing $i_1 = s - k + 1$ and $i_2 = \dots = i_k = 1$.) Note this uses repetitions — Definition 1.6 earning its keep.

So a walk that wants to survive below the threshold must *wobble*: vary its steps, forever, in a way that never settles into an arithmetic progression. The entire difficulty of the non-existence half is the question: *can clever wobbling recreate daylight?* The window picture pushes toward “no” — wobbling changes where the windows sit but not their width, and there is no room between them — but the picture reasons about steady walks, and a wobbling walk has no single stride γ . That gap in the reasoning is not small print. Closing it is most of this document.

2.4. What the picture does not prove. Here is the honest ledger — four gaps between the picture and the theorem, and where each one is closed.

- (1) *Above the threshold, one dial setting must defeat all of B at once.* The daylight argument dodges one target at a time; to dodge every b_i simultaneously we tune γ through infinitely many rounds and must show the dial room never runs out. This is Part II: the free-gap lemma (§5) and the nested-interval construction (§6), yielding the explicit ratio $192d_2$.
- (2) *Below the threshold, one B must defeat all walks at once.* Example 2.4 beats each steady walk with a B built from *its* congruence class — but different walks use different classes, and the theorem demands a single B that beats every admissible walk. The device that accomplishes this is the *certificate trick* (§7), a diagonalization.
- (3) *Below the threshold, wobbling walks must be shown hopeless too.* The plan: show every admissible walk’s sumset contains a tail of a congruence class (“tail-covering”), by classifying how a walk can wobble — the analysis of *gap words* occupying §§8–14. The delicate walks are those that wobble between two step sizes as evenly and aperiodically as possible; these are the *Sturmian* walks, and taming them (§§12–13) is the deepest water in the document.
- (4) *The corner $d_2 = k$ is razor-thin.* When the largest step equals k exactly, the windows touch without overlapping, and the walk has just barely too little room; converting “barely” into a proof costs a genuinely new piece of finite mathematics about subset sums, stated in §15 and proved across Part IV.

One more promissory note. In Example 2.4 the offset “ kc ” depended on the walk, and in gap (2) we waved at defeating all offsets and classes simultaneously; when we do this properly, the notion of *tail-covering* will be defined to absorb all of it. From here on, the document simply executes the ledger: Part II closes gap (1); Part III closes gaps (2) and (3), reducing the corner case to a finite lemma; Part IV proves that lemma, closing gap (4); Part V assembles the theorem and describes the computer verification.

Part 2. Existence: winning when $d_2 \geq k + 1$

3. FLOORS AND BEATTY WALKS

Part II proves the existence half of the dichotomy: if $d_2 \geq k + 1$, announcing $r = 192 d_2$ wins the game. The plan was sketched in §2.2; from here on we execute it rigorously. **Standing assumptions for all of Part II:** k, d_1, d_2 are integers with

$$k \geq 2, \quad 1 \leq d_1 < d_2, \quad d_2 \geq k + 1.$$

(The problem poses $k \geq 3$, but the existence proof works verbatim from $k = 2$, so we prove the slightly more general statement at no extra cost. Note $d_2 \geq 3$ follows.)

This section builds the two tools everything else stands on: the floor function, and the fact that a Beatty walk with slope between $d_2 - 1$ and d_2 is a legal (d_1, d_2) -sequence.

3.1. The floor function.

Definition 3.1 (Floor). For a real number x , the *floor* $\lfloor x \rfloor$ is the unique integer m satisfying

$$m \leq x < m + 1.$$

So $\lfloor 3.5 \rfloor = 3$, $\lfloor 7 \rfloor = 7$, and (watch this one) $\lfloor -0.5 \rfloor = -1$: the floor is the integer just *below* x (or x itself), not “the integer part”. That an integer $m \leq x < m + 1$ exists is visually clear — walk down the number line from above x until you first pass an integer — and we take it as known; but uniqueness deserves its one-line proof: if $m \leq x < m + 1$ and $m' \leq x < m' + 1$ with, say, $m < m'$, then $m + 1 \leq m'$ (they are integers), so $x < m + 1 \leq m' \leq x$, a contradiction.

Everything we ever use about floors is in the next lemma.

Lemma 3.2 (Floor toolbox). *For all real x, y and every integer n :*

- (1) $x - 1 < \lfloor x \rfloor \leq x$;
- (2) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$;
- (3) if $x \leq y$ then $\lfloor x \rfloor \leq \lfloor y \rfloor$;
- (4) for reals $x_1, \dots, x_k$: $x_1 + \dots + x_k - k < \lfloor x_1 \rfloor + \dots + \lfloor x_k \rfloor \leq x_1 + \dots + x_k$.

Proof. Write $m = \lfloor x \rfloor$, so $m \leq x < m + 1$ by definition.

(1) The right inequality is $m \leq x$ directly; the left is $x < m + 1$ rearranged.

(2) The integer $m + n$ satisfies $m + n \leq x + n < (m + n) + 1$, which is exactly the defining property of $\lfloor x + n \rfloor$; by uniqueness, $\lfloor x + n \rfloor = m + n$.

(3) Suppose $x \leq y$. Then $\lfloor x \rfloor \leq x \leq y$, so $\lfloor x \rfloor$ is an integer $\leq y$. But $\lfloor y \rfloor$ is the *greatest* integer $\leq y$: any integer $m' \leq y$ has $m' \leq \lfloor y \rfloor$, since otherwise $m' \geq \lfloor y \rfloor + 1 > y$. Hence $\lfloor x \rfloor \leq \lfloor y \rfloor$.

(4) Add the k instances of (1): summing $x_t - 1 < \lfloor x_t \rfloor \leq x_t$ over $t = 1, \dots, k$ gives the claim. (This is Checkpoint 2.2, now on the official record.) $\square$

Checkpoint 3.3. (a) Compute $\lfloor 5.999 \rfloor$, $\lfloor -3 \rfloor$, $\lfloor -3.01 \rfloor$. (b) Show by example that $\lfloor x + y \rfloor$ is *not* always $\lfloor x \rfloor + \lfloor y \rfloor$ — and prove that it always equals $\lfloor x \rfloor + \lfloor y \rfloor$ or $\lfloor x \rfloor + \lfloor y \rfloor + 1$. (c) Where does the proof of Lemma 3.2(2) break if n is not an integer?

3.2. Beatty walks, officially.

Definition 3.4 (Beatty walk). For a real number $\gamma > 1$ and an integer $c \geq 0$, the *Beatty walk* with slope γ and offset c is the sequence

$$A_{\gamma, c} := \{c + \lfloor i\gamma \rfloor : i = 1, 2, 3, \dots\}, \quad \text{i.e. } a_i = c + \lfloor i\gamma \rfloor.$$

We met $A_{3.5,0} = \{3, 7, 10, 14, 17, 21, \dots\}$ in Example 2.1. One more, with a slope that is not a half-integer: for $\gamma = \frac{10}{3}$ and $c = 0$,

$$\lfloor \frac{10}{3} \rfloor = 3, \quad \lfloor \frac{20}{3} \rfloor = 6, \quad \lfloor \frac{30}{3} \rfloor = 10, \quad \lfloor \frac{40}{3} \rfloor = 13, \dots$$

so $A_{10/3,0} = \{3, 6, 10, 13, 16, 20, 23, 26, 30, \dots\}$, with gap pattern $3, 4, 3, 3, 4, 3, 3, 4, \dots$: the gap 4 arrives once every three steps, keeping the average speed at exactly $\frac{10}{3}$.

The single structural fact we need is that a Beatty walk's gaps take at most two values, and we can say which ones.

Lemma 3.5 (Two-value gaps). *Let $\gamma > 1$ be real. Every gap of the Beatty walk $A_{\gamma,c}$ lies in $\{\lfloor \gamma \rfloor, \lfloor \gamma \rfloor + 1\}$:*

$$\lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor \in \{\lfloor \gamma \rfloor, \lfloor \gamma \rfloor + 1\} \quad \text{for every } i \geq 1.$$

Proof. The offset c cancels from every gap, so consider $g := \lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor$, a difference of two integers, hence an integer. We trap it in an open interval of length 2. By Lemma 3.2(1),

$$\lfloor (i+1)\gamma \rfloor \leq (i+1)\gamma \quad \text{and} \quad \lfloor i\gamma \rfloor > i\gamma - 1,$$

so $g < (i+1)\gamma - (i\gamma - 1) = \gamma + 1$. Similarly $\lfloor (i+1)\gamma \rfloor > (i+1)\gamma - 1$ and $\lfloor i\gamma \rfloor \leq i\gamma$ give $g > \gamma - 1$. So g is an integer with

$$\gamma - 1 < g < \gamma + 1.$$

Which integers lie strictly between $\gamma - 1$ and $\gamma + 1$? Only $\lfloor \gamma \rfloor$ and $\lfloor \gamma \rfloor + 1$ can. To see this, split on whether $g \leq \gamma$ or $g > \gamma$.

- If $g \leq \gamma$: then g is an integer $\leq \gamma$, so $g \leq \lfloor \gamma \rfloor$ ($\lfloor \gamma \rfloor$ being the greatest such). On the other hand $\lfloor \gamma \rfloor \leq \gamma$ gives $\lfloor \gamma \rfloor - 1 \leq \gamma - 1 < g$. So $\lfloor \gamma \rfloor - 1 < g \leq \lfloor \gamma \rfloor$, and the only integer in that range is $g = \lfloor \gamma \rfloor$.
- If $g > \gamma$: then $g - 1$ is an integer with $\gamma - 1 < g - 1 < \gamma$, so the previous case applies to $g - 1$ and yields $g - 1 = \lfloor \gamma \rfloor$, i.e. $g = \lfloor \gamma \rfloor + 1$.

□

(If γ is an integer, only the value $\lfloor \gamma \rfloor = \gamma$ actually occurs, and the walk is an arithmetic progression — Checkpoint below. For non-integer γ , both values occur infinitely often, though we will not need that.)

3.3. Beatty walks are admissible. Now aim the slope at the top of our permitted window.

Proposition 3.6 (Admissibility). *Let γ be any real number with $d_2 - 1 < \gamma < d_2$, and $c \geq 0$ any integer. Then $A_{\gamma,c}$ is a (d_1, d_2) -sequence.*

Proof. Three things to check: the terms are positive, the sequence is strictly increasing, and the gaps lie in $[d_1, d_2]$.

Since $d_2 - 1 < \gamma < d_2$ and $d_2 - 1, d_2$ are integers, the defining property of the floor gives $\lfloor \gamma \rfloor = d_2 - 1$. By Lemma 3.5, every gap of $A_{\gamma,c}$ is $d_2 - 1$ or d_2 . Both lie in $[d_1, d_2]$: indeed $d_1 \leq d_2 - 1$ because $d_1 < d_2$ are integers. As $d_2 \geq k + 1 \geq 3$, every gap is at least $2 > 0$, so the sequence is strictly increasing; and its first term is $a_1 = c + \lfloor \gamma \rfloor \geq 0 + (d_2 - 1) \geq 2 > 0$. All three checks pass. □

Two remarks worth pausing on. First, *the slope may be rational*: nothing anywhere in Part II needs γ irrational, and the final winning slope will simply be some real number produced by the nested-interval theorem — we will neither know nor care whether it is rational. Second, notice how little of the freedom of (d_1, d_2) -sequences we are using: only

steps $d_2 - 1$ and d_2 , the two largest. The lower bound d_1 plays no role beyond $d_1 \leq d_2 - 1$, which is automatic. This is the promised explanation of why d_1 is absent from the threshold: the existence strategy never has any reason to take a small step.

Checkpoint 3.7. (a) Show that if $\gamma = d$ is an integer, $A_{d,c}$ is the arithmetic progression $c + d, c + 2d, c + 3d, \dots$ (b) For $\gamma = \frac{7}{2}$, which i have gap $g_i = 4$ and which have 3? Find the pattern and prove it. (Hint: $\lfloor (i+1)\gamma \rfloor - \lfloor i\gamma \rfloor$ depends on whether i is even or odd.) (c) Show that for any real $\gamma > 1$, among the first n gaps of $A_{\gamma,c}$, the number of gaps equal to $\lfloor \gamma \rfloor + 1$ is $\lfloor (n+1)\gamma \rfloor - \lfloor \gamma \rfloor - n\lfloor \gamma \rfloor$. (Telescope the sum of the gaps.)

4. WHERE THE k -FOLD SUMS LIVE

This section turns the window picture of §2 into theorems, and then performs the change of viewpoint on which the whole existence proof runs: instead of asking *which numbers* the walk's sums can reach, we ask *which dial positions* γ are safe for a given target. By the end of the section, dodging a single element of B will be a clean geometric condition on γ alone.

4.1. The window containment.

Proposition 4.1 (Windows). *Let $\gamma > 1$ be real, $c \geq 0$ an integer, and $k \geq 1$. Then*

$$kA_{\gamma,c} \subseteq \bigcup_{s \geq k} (kc + s\gamma - k, kc + s\gamma],$$

the union over integers $s \geq k$.

Proof. A member of $kA_{\gamma,c}$ is a sum of k terms of the walk, say with indices $i_1, \dots, i_k \geq 1$ (repetitions allowed):

$$\sum_{t=1}^k (c + \lfloor i_t \gamma \rfloor) = kc + \sum_{t=1}^k \lfloor i_t \gamma \rfloor.$$

Set $s := i_1 + \dots + i_k$; since each $i_t \geq 1$, we have $s \geq k$. By the sum-of-floors bound (Lemma 3.2(4)) applied to the numbers $x_t = i_t \gamma$, whose sum is $s\gamma$,

$$s\gamma - k < \sum_{t=1}^k \lfloor i_t \gamma \rfloor \leq s\gamma,$$

so the member lies in $(kc + s\gamma - k, kc + s\gamma]$. $\square$

Alongside the windows we record where the sums *start*: nothing in $kA_{\gamma,c}$ is small.

Lemma 4.2 (Sums start high). *If $d_2 - 1 < \gamma < d_2$, the smallest element of $kA_{\gamma,c}$ is $k(c + d_2 - 1)$. In particular every integer $b \leq kc$ satisfies $b \notin kA_{\gamma,c}$: targets at or below kc are dodged automatically.*

Proof. Every term of the walk is at least its first term $a_1 = c + \lfloor \gamma \rfloor = c + d_2 - 1$, so every k -fold sum is at least $ka_1 = k(c + d_2 - 1)$ — and this value is attained, by summing a_1 with itself k times. Finally $k(c + d_2 - 1) > kc$, since $d_2 \geq 2$. $\square$

4.2. The avoidance criterion. Fix a target $b > kc$ that we wish to keep out of $kA_{\gamma,c}$, and write

$$t := b - kc \geq 1$$

for the *shifted target*. Subtracting kc throughout, Proposition 4.1 says: if $b \in kA_{\gamma,c}$, then $t \in (s\gamma - k, s\gamma]$ for some integer $s \geq k$ — equivalently, $s\gamma \in [t, t + k)$ for some $s \geq k$. Turning this around gives the criterion we will use forever after.

Definition 4.3 (Safe slope). Let $t \geq 1$ be an integer. A real $\gamma > 0$ is *safe for t* if

$$s\gamma \notin [t, t + k] \quad \text{for every integer } s \geq 1.$$

Proposition 4.4 (Avoidance criterion). *Let $d_2 - 1 < \gamma < d_2$, let $c \geq 0$, and let $b > kc$ be an integer. If γ is safe for $t = b - kc$, then $b \notin kA_{\gamma,c}$.*

Proof. Contrapositive. If $b \in kA_{\gamma,c}$, the discussion above produces an integer $s \geq k \geq 1$ with $s\gamma \in [t, t + k) \subseteq [t, t + k]$, so γ is not safe for t . $\square$

Look closely: Definition 4.3 demands *more* than the proof of Proposition 4.4 needs, in two ways, and both are deliberate.

- It forbids $s\gamma \in [t, t + k]$ for *all* $s \geq 1$, though only $s \geq k$ can actually occur in a window. Ruling out extra values of s is a stronger demand on γ — harmless for the conclusion, and it frees every later argument from tracking which s are realizable.
- It uses the *closed* interval $[t, t + k]$, though the windows are half-open. Again this only makes “safe” harder to achieve, never easier — and closed intervals are more comfortable to work with (their endpoints belong to them, so “the least s with...” arguments have no edge cases).

The cost of this generosity is real but affordable: some slopes that would in fact dodge b are not certified safe. You will meet a concrete instance in Checkpoint 4.5 — the walk of Example 2.1 really does miss 25, yet the criterion declines to vouch for it. We simply will not care: the free-gap lemma of §5 finds slopes that pass even the stricter test.

Big idea. For a Beatty walk, “the sum of k members never equals b ” has become a statement about the dial position alone: no multiple $s\gamma$ may land in the closed window $[t, t + k]$, where $t = b - kc$. The game of dodging an infinite sequence B is now a game of placing one real number γ clear of infinitely many forbidden zones.

Checkpoint 4.5. Take the walk of Example 2.1: $\gamma = 3.5$, $c = 0$, $k = 3$. (a) Show that γ is *not* safe for $t = 25$: find the integer s with $3.5s \in [25, 28]$. (b) Yet $25 \notin 3A$ (Checkpoint 2.3). Explain, using the half-open window $(25, 28]$, why the point $3.5 \cdot 8 = 28$ landing exactly on the closed window’s right endpoint does not put 25 into $3A$ — so the criterion’s verdict is merely conservative, not wrong. (c) Stronger: show that *no* integer $t \geq 1$ is safe for $\gamma = 3.5$. (Consider the smallest multiple u of 3.5 with $u \geq t$; multiples of 3.5 are integers or half-integers, so show $u \leq t + 3$.) So at this particular slope the criterion certifies nothing at all — even though the walk misses 18, 25, 32, ... The slopes the free-gap lemma will select are precisely engineered to avoid this fate. (d) By listing the unsafe intervals $[\frac{46}{s}, \frac{49}{s}]$ that meet $(3, 4)$, find a slope that *is* safe for $t = 46$, and verify your slope directly against $s = 12$ and $s = 13$.



FIGURE 2. The unsafe set U_{100} for $k = 3$, on the stretch $2.95 \leq \gamma \leq 4.18$ of the dial: the interval $[\frac{100}{s}, \frac{103}{s}]$ for $s = 25, 26, \dots, 34$. Safe gaps (two are marked) open up between consecutive unsafe intervals, wider toward the right; toward the left the intervals crowd together and, below $\gamma = 3 = k$, merge into a solid unsafe block (Proposition 4.6).

4.3. The dial, and its unsafe zones. Fix the shifted target t and look at the condition of Definition 4.3 from the dial's point of view. For each single $s \geq 1$,

$$s\gamma \in [t, t+k] \iff \gamma \in \left[\frac{t}{s}, \frac{t+k}{s} \right],$$

so the slopes ruled out by index s form a closed interval, and γ is safe for t exactly when it avoids the *unsafe set*

$$U_t := \bigcup_{s \geq 1} \left[\frac{t}{s}, \frac{t+k}{s} \right].$$

As s grows, these unsafe intervals march down the dial toward 0, shrinking as they go (both endpoints are proportional to $1/s$). Figure 2 shows the portion of U_{100} near the dial region we shop in, for $k = 3$.

Two features of this picture are the pivot of the whole story. First, the safe gaps — the white space between consecutive unsafe intervals — *exist* in the region $\gamma > k$, and we will learn to find them (§5). Second, below $\gamma = k$ there is nothing to find:

Proposition 4.6 (No safety below k). *Let $t \geq 1$ be an integer. Every real γ with $0 < \gamma \leq k$ is unsafe for t .*

Proof. We must find an integer $s \geq 1$ with $s\gamma \in [t, t+k]$, i.e. with

$$s \in \left[\frac{t}{\gamma}, \frac{t+k}{\gamma} \right].$$

This is a real interval of length $k/\gamma \geq 1$, so it contains an integer s (between any two reals at distance ≥ 1 there is an integer: $s = \lfloor t/\gamma \rfloor + 1$ works, as it satisfies $t/\gamma < s \leq t/\gamma + 1 \leq (t+k)/\gamma$). That integer is positive, since $s > t/\gamma > 0$. $\square$

This is the threshold, seen from the dial. A walk whose stride is at most k cannot be certified to dodge *anything*: every single target's unsafe set swallows the entire dial region $(0, k]$. Our standing assumption $d_2 \geq k+1$ is exactly what places the shopping district $(d_2 - 1, d_2)$ of Proposition 3.6 inside the living zone $\gamma > k$. (That the picture “no safety below k ” hardens into the actual non-existence theorem below the threshold — not merely a failure of this particular criterion — is the business of Parts III and IV.)

What remains for the existence half is quantitative: given the target t , *how wide* are the safe gaps near the top of the dial, and how little of the dial does each new target destroy? That is the free-gap lemma, to which we now turn.

Checkpoint 4.7. Let $k = 3$ and $t = 20$. (a) List the unsafe intervals $[\frac{20}{s}, \frac{23}{s}]$ that meet the dial stretch $(3, 4)$, and conclude that the safe slopes there form exactly the two gaps $(\frac{23}{7}, \frac{20}{6})$ and $(\frac{23}{6}, 4)$. (b) Pick a slope in one of them and write down the first eight terms of a $(1, 4)$ -walk none of whose 3-fold sums equals $20 + 3c$ for your chosen offset c .

Checkpoint 4.8. Show that the unsafe intervals for indices s and $s + 1$ overlap or touch — that is, $\frac{t+k}{s+1} \geq \frac{t}{s}$ — if and only if $s \geq t/k$. (This is the precise sense in which the unsafe intervals “merge into a solid block” at the left of Figure 2: from $s = \lceil t/k \rceil$ onward, consecutive intervals chain together without gaps.)

5. THE FREE-GAP LEMMA

We now build the engine of the existence proof. The task left over from §4.3: given a target t , find a safe stretch of dial — and find it *quantitatively*, because we will have to do this infinitely often, once per element of B , each time inside whatever room the previous rounds left us. The lemma of this section says: inside any dial interval that is not too narrow (compared to $1/t$), there is a safe subinterval that is not too much narrower. The two “not too”s are what the constant 192 will be made of in §6.

One more tool for the toolbox first.

Definition 5.1 (Ceiling). For a real x , the *ceiling* $\lceil x \rceil$ is the unique integer m with $m - 1 < x \leq m$ — the least integer $\geq x$. (Uniqueness: as for the floor. Note $\lceil x \rceil = \lfloor x \rfloor$ when x is an integer, and $\lfloor x \rfloor + 1$ otherwise.)

5.1. The statement.

Lemma 5.2 (Free gap). *Let $t \geq 32d_2^2$ be an integer, and let $I = [lo, hi]$ be a closed real interval with*

$$I \subseteq [d_2 - \tfrac{1}{2}, d_2] \quad \text{and} \quad |I| = hi - lo \geq \frac{4d_2^2}{t}.$$

Then I contains a closed subinterval I' with

$$|I'| \geq \frac{d_2}{48t}$$

such that every $\gamma \in I'$ is safe for t (Definition 4.3): $s\gamma \notin [t, t+k]$ for every integer $s \geq 1$.

Read the shape of this before the proof. The input interval must have width at least $4d_2^2/t$; the output is guaranteed width at least $d_2/(48t)$. Both scale like $1/t$: a bigger target is dodged with proportionally finer dial adjustments. The ratio between them,

$$\frac{4d_2^2/t}{d_2/(48t)} = 192d_2,$$

is the number of times the output of one round fits into the input the next round demands — hold that thought until §6, where it becomes the theorem’s constant.

5.2. The idea: name one gap, count nothing. The unsafe set $U_t = \bigcup_{s \geq 1} [t/s, (t+k)/s]$ is an infinite union, and a first instinct is to estimate how many of its intervals meet I and how much room they eat. That accounting can be made to work, but there is something better: we can point at *one specific gap* between two *specific* consecutive unsafe intervals, and prove that this single gap already avoids every unsafe interval at once, has the width we need, and sits inside I . No counting, no union bounds — three clean checks on one explicitly named object.

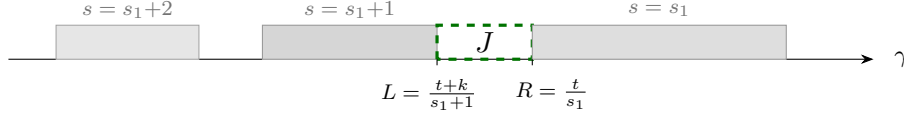
Which gap? Look at Figure 2 again: near the top of the dial the unsafe intervals are the sparsest and the gaps the widest, so we aim as high in I as possible. Let

$$s_1 := \left\lceil \frac{t}{hi} \right\rceil \quad (\text{the least integer } s \text{ with } t/s \leq hi),$$

so that the unsafe interval of index s_1 is the highest-placed one whose left endpoint has not yet escaped past the top of I . Our gap is the space just below that interval and just above the next one:

$$L := \frac{t+k}{s_1+1}, \quad R := \frac{t}{s_1}, \quad J := (L, R)$$

— the open interval between the right end of unsafe interval s_1+1 and the left end of unsafe interval s_1 .



5.3. Three checks. Throughout, remember the standing facts: $2 \leq k \leq d_2 - 1$ (from $d_2 \geq k + 1$), $d_2 \geq 3$, $t \geq 32d_2^2$, and $d_2 - \frac{1}{2} \leq lo \leq hi \leq d_2$.

We first pin down the size of s_1 . Its defining property gives the two-sided estimate

$$\frac{t}{hi} \leq s_1 < \frac{t}{hi} + 1. \quad (1)$$

From the left half and $hi \leq d_2$:

$$s_1 \geq \frac{t}{d_2} \geq \frac{32d_2^2}{d_2} = 32d_2. \quad (2)$$

From the right half and $hi \geq d_2 - \frac{1}{2} \geq \frac{d_2}{2}$ (using $d_2 \geq 1$):

$$s_1 + 1 < \frac{t}{hi} + 2 \leq \frac{2t}{d_2} + 2 \leq \frac{2t}{d_2} + \frac{t}{d_2} = \frac{3t}{d_2} \leq \frac{5t}{d_2}, \quad (3)$$

where $2 \leq t/d_2$ holds because $t/d_2 \geq 32d_2 \geq 96$.

Check (i): every point of J is safe for t . Let $\gamma \in J$, i.e. $L < \gamma < R$, and let $s \geq 1$ be any integer; we show $s\gamma \notin [t, t+k]$, splitting on the size of s .

- If $s \leq s_1$: then $t/s \geq t/s_1 = R > \gamma$, so $s\gamma < t$. The product undershoots the window.
- If $s \geq s_1 + 1$: then $(t+k)/s \leq (t+k)/(s_1+1) = L < \gamma$, so $s\gamma > t+k$. The product overshoots it.

This is the whole trick, so pause to admire it: the unsafe intervals $[t/s, (t+k)/s]$ move monotonically leftward as s grows, so the single gap between intervals s_1 and s_1+1 is automatically clear of *all* of them — those with small s lie entirely to its right, those with large s entirely to its left. One comparison per side replaces an infinite enumeration.

Check (ii): J is wide enough: $|J| = R - L \geq \frac{d_2}{16t}$. Put the difference over a common denominator:

$$R - L = \frac{t}{s_1} - \frac{t+k}{s_1+1} = \frac{t(s_1+1) - (t+k)s_1}{s_1(s_1+1)} = \frac{t - ks_1}{s_1(s_1+1)}.$$

So the width question is: how much does t exceed ks_1 ? Intuitively $s_1 \approx t/hi \approx t/d_2$ while $k \leq d_2 - 1$, so $ks_1 \approx t(d_2 - 1)/d_2$ leaves a healthy margin of about t/d_2 ; the following makes that precise with explicit constants. Using $s_1 < t/hi + 1 \leq t/(d_2 - \frac{1}{2}) + 1$ and $k \leq d_2 - 1$:

$$\left(k + \frac{15}{32}\right) s_1 \leq \left(d_2 - \frac{17}{32}\right) \left(\frac{t}{d_2 - \frac{1}{2}} + 1\right) = t \frac{d_2 - \frac{17}{32}}{d_2 - \frac{1}{2}} + \left(d_2 - \frac{17}{32}\right).$$

The fraction equals $1 - \frac{1/32}{d_2 - 1/2}$, so the right side is

$$t - \frac{t}{32(d_2 - \frac{1}{2})} + \left(d_2 - \frac{17}{32}\right) \leq t,$$

provided the middle term dominates the last, i.e. provided $\frac{t}{32(d_2 - 1/2)} \geq d_2 - \frac{17}{32}$. And it does: from $t \geq 32d_2^2$,

$$\frac{t}{32(d_2 - \frac{1}{2})} \geq \frac{d_2^2}{d_2 - \frac{1}{2}} \geq d_2 + \frac{1}{2} > d_2 - \frac{17}{32},$$

where the middle inequality is $d_2^2 \geq (d_2 + \frac{1}{2})(d_2 - \frac{1}{2}) = d_2^2 - \frac{1}{4}$. Altogether $(k + \frac{15}{32})s_1 \leq t$, i.e.

$$t - k s_1 \geq \frac{15}{32} s_1.$$

Feeding this into the width formula, then using (3):

$$R - L \geq \frac{\frac{15}{32} s_1}{s_1(s_1 + 1)} = \frac{15}{32} \cdot \frac{1}{s_1 + 1} \geq \frac{15}{32} \cdot \frac{d_2}{5t} = \frac{3}{32} \cdot \frac{d_2}{t} \geq \frac{d_2}{16t}.$$

Check (iii): J sits inside I . The top is easy: $R = t/s_1 \leq hi$ by the defining property of s_1 . For the bottom we need $L \geq lo$, and we get it by showing J hangs only a little below hi . First, R is close below hi : from $s_1 < t/hi + 1$,

$$R = \frac{t}{s_1} > \frac{t}{t/hi + 1} = \frac{hit}{t + hi} = hi \left(1 - \frac{hi}{t + hi}\right) > hi - \frac{hi^2}{t} \geq hi - \frac{d_2^2}{t}.$$

Second, J itself is short: using (1),

$$R - L = \frac{t - ks_1}{s_1(s_1 + 1)} \leq \frac{t}{s_1^2} \leq \frac{t}{(t/hi)^2} = \frac{hi^2}{t} \leq \frac{d_2^2}{t}.$$

Adding: $L = R - (R - L) > hi - 2d_2^2/t$. Finally the hypothesis $hi - lo \geq 4d_2^2/t$ says exactly that $2d_2^2/t \leq \frac{1}{2}(hi - lo)$, so

$$L > hi - \frac{1}{2}(hi - lo) \geq lo.$$

Conclusion. Let I' be the closed middle third of J . Then $I' \subseteq J \subseteq I$; every $\gamma \in I'$ is safe for t by check (i); and $|I'| = |J|/3 \geq d_2/(48t)$ by check (ii). (This proves Lemma 5.2.) $\square$

Why the middle third, and not J itself? Because the lemma promises a *closed* subinterval — the next round of the construction will want to apply the lemma again *inside* I' , and closed intervals are what the nested-interval theorem of §6 consumes. The open gap J safely contains its closed middle third, and a third is the cheapest way to buy closedness with an explicit constant.

Checkpoint 5.3. Run the whole proof numerically for $d_2 = 4$, $k = 3$, $t = 10000$, $I = [3.5, 4]$. (a) Check the hypotheses ($t \geq 32d_2^2$, $|I| \geq 4d_2^2/t$). (b) Compute s_1 , R , L , and the width $R - L$; compare with the promised $d_2/(16t)$. (c) Verify safety at the midpoint of J directly against $s = s_1$ and $s = s_1 + 1$. (d) How many unsafe intervals meet I ? (Estimate — then appreciate that the proof never needed to know.)

Checkpoint 5.4. Where does the proof use $t \geq 32d_2^2$? Find all three places, and test the lemma's limits: for $d_2 = 4$, $k = 3$, find a small t for which the interval J defined above pokes out of $[d_2 - \frac{1}{2}, d_2]$ or has the wrong width.

6. NESTED INTERVALS, AND THE RATIO $192d_2$

We can now assemble the existence half. Here is the theorem, stated for the record.

Theorem 6.1 (Existence). *Let $k \geq 2$ and $1 \leq d_1 < d_2$ with $d_2 \geq k + 1$. Then announcing $r = 192d_2$ wins the game: for every lacunary B with ratio $192d_2$ there is a (d_1, d_2) -sequence A with $(kA) \cap B = \emptyset$.*

The plan: fix the Adversary's B , choose a good offset c , convert each element of B into a shifted target, and then apply the free-gap lemma over and over — each application shrinking our interval of candidate slopes, each shrunken interval still wide enough for the next application *because B is lacunary*. A limit point of the shrinking intervals is a slope safe for every target at once.

6.1. One more axiom: the nested interval theorem. The construction produces an infinite chain of intervals, each inside the last, and needs a point lying in all of them. That such a point exists is a fact about the real number line itself.

Nested interval theorem. If $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$ are nonempty closed bounded intervals of real numbers, then there is at least one real number γ^* belonging to every I_j .

Why believe it? Write $I_j = [lo_j, hi_j]$. The left endpoints $lo_0 \leq lo_1 \leq lo_2 \leq \cdots$ form a nondecreasing sequence, bounded above (by hi_0); such a sequence crawls upward toward a limiting value γ^* — think of the decimal expansions stabilizing digit by digit — and this γ^* is $\geq$ every lo_j and $\leq$ every hi_j (if $\gamma^* > hi_{j_0}$ for some j_0 , then some lo_j would eventually exceed $hi_{j_0} \geq hi_j$, impossible). The honest fine print: the existence of that limiting value is the *completeness* of the real numbers — the defining property that separates $\mathbb{R}$ from the rationals, where the theorem is false (nested intervals can squeeze down onto $\sqrt{2}$, which a rational line does not contain). We take completeness as an axiom of the real line, as every treatment must at some level; a rigorous construction of $\mathbb{R}$ is a beautiful topic that lives one course away from this document. Note the theorem needs *closed* intervals: the open intervals $(0, 1/n)$ are nested with empty intersection — this is why the free-gap lemma delivered a closed I' .

6.2. Clearing the small elements of B . Fix $B = \{b_1 < b_2 < \cdots\}$, lacunary with ratio $r = 192d_2$. The dial criterion of §4 handles targets above kc ; elements of B at or below kc are dodged automatically (Lemma 4.2). So we choose the offset to split B cleanly. Since B is infinite and increasing, some index j_0 has

$$b_{j_0} \geq 4d_2^2;$$

fix such a j_0 and set

$$c := b_{j_0} + 1.$$

Every element b_i with $i \leq j_0$ satisfies $b_i \leq b_{j_0} < kc$, hence is dodged automatically, whatever slope we pick. For the remaining elements define the shifted targets

$$t_i := b_i - kc \quad (i > j_0).$$

Three facts about these targets, each a short computation with the lacunarity.

(1) *The targets are positive — indeed the first one is already huge.* Using $b_{j_0+1} \geq 192d_2b_{j_0}$ and $kc = k(b_{j_0} + 1) \leq d_2(b_{j_0} + 1) \leq 2d_2b_{j_0}$ (as $b_{j_0} \geq 1$):

$$t_{j_0+1} = b_{j_0+1} - kc \geq 192d_2b_{j_0} - 2d_2b_{j_0} = 190d_2b_{j_0} \geq 190d_2 \cdot 4d_2^2 \geq 32d_2^2,$$

comfortably meeting the free-gap lemma's threshold $t \geq 32d_2^2$. Later targets are larger still, since B increases.

(2) *Shifting preserves the ratio.* For $x \geq y > \tau \geq 0$ one has

$$\frac{x - \tau}{y - \tau} \geq \frac{x}{y} \quad (\text{cross-multiply: } xy - \tau x \geq xy - \tau y \iff \tau y \leq \tau x).$$

Apply this with $x = b_{i+1}$, $y = b_i$, $\tau = kc$ (valid: for $i > j_0$, $b_i \geq b_{j_0+1} \geq 192d_2b_{j_0} > 2d_2b_{j_0} \geq kc$, so $y > \tau$):

$$\frac{t_{i+1}}{t_i} \geq \frac{b_{i+1}}{b_i} \geq 192d_2 \quad (i > j_0). \quad (4)$$

Subtracting the same amount from both terms of a ratio only helps the ratio — the targets inherit the lacunarity of B , undiluted.

(3) *If every target is dodged, we win.* Suppose $\gamma^* \in (d_2 - 1, d_2)$ is safe for every t_i , $i > j_0$. Then by the avoidance criterion (Proposition 4.4), no b_i with $i > j_0$ lies in $kA_{\gamma^*,c}$; and no b_i with $i \leq j_0$ does either, by the automatic clearance above. So $(kA_{\gamma^*,c}) \cap B = \emptyset$, and $A_{\gamma^*,c}$ is a legal (d_1, d_2) -sequence by Proposition 3.6.

6.3. The nest. Everything is now in place. Start from the closed interval

$$I_0 := [d_2 - \tfrac{1}{2}, d_2 - \tfrac{1}{4}],$$

of width exactly $\frac{1}{4}$. (We stop at $d_2 - \frac{1}{4}$ rather than d_2 out of tidiness only; any closed subinterval of $[d_2 - \frac{1}{2}, d_2]$ of width $\frac{1}{4}$ would do. Note $I_0 \subseteq (d_2 - 1, d_2)$ since $d_2 \geq 3$.)

Round 1. Apply the free-gap lemma to the first target t_{j_0+1} and the interval I_0 . Its hypotheses hold: $t_{j_0+1} \geq 32d_2^2$ by fact (1), and

$$|I_0| = \tfrac{1}{4} \geq \frac{4d_2^2}{t_{j_0+1}} \iff t_{j_0+1} \geq 16d_2^2,$$

true again by fact (1). Out comes a closed $I_1 \subseteq I_0$, safe for t_{j_0+1} , with $|I_1| \geq d_2/(48t_{j_0+1})$.

Round $j+1$, given round j . Suppose inductively that the closed interval I_j is safe for the targets $t_{j_0+1}, \dots, t_{j_0+j}$ and satisfies

$$|I_j| \geq \frac{d_2}{48t_{j_0+j}}.$$

We want to apply the free-gap lemma to the next target $t := t_{j_0+j+1}$ inside I_j . The only hypothesis in doubt is the width demand $|I_j| \geq 4d_2^2/t$, which in view of the inductive width bound follows from

$$\frac{d_2}{48t_{j_0+j}} \geq \frac{4d_2^2}{t_{j_0+j+1}} \iff t_{j_0+j+1} \geq 192d_2 \cdot t_{j_0+j},$$

which is precisely (4). *This is the entire ledger of the constant:* $192 = 4 \times 48$, the free-gap lemma's input demand $(4d_2^2/t)$ divided by its output guarantee $(d_2/(48t))$. The lacunarity ratio of B is exactly what refills the width the previous round spent. The lemma returns a closed $I_{j+1} \subseteq I_j$, safe also for t_{j_0+j+1} , of width $\geq d_2/(48t_{j_0+j+1})$ — the inductive hypothesis for the next round.

The limit. The rounds produce nonempty closed bounded nested intervals $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$; the nested interval theorem yields a real γ^* in all of them. For every $i > j_0$, the slope γ^* lies in I_{i-j_0} , which is safe for t_i ; so γ^* is safe for every target simultaneously. By fact (3) of §6.2, the Beatty walk $A_{\gamma^*,c}$ is a legal (d_1, d_2) -sequence with $(kA_{\gamma^*,c}) \cap B = \emptyset$. (This proves Theorem 6.1.) $\square$

Big idea. The existence proof in one breath: aim a Beatty walk’s slope dial at the top of the permitted window; each element of B forbids a sparse family of dial positions, but the free-gap lemma always finds a safe sub-stretch losing only a factor $192 d_2$ of width; lacunarity of B regains exactly that factor before the next element arrives; completeness of the reals hands us a single slope surviving every round.

Two closing remarks. First, nothing was random and nothing was irrational-by-fiat: the winning slope γ^* is whatever real number the nest closes on, and we never ask anything else about it. Second, the constant $192 d_2$ is an artifact of our bookkeeping (the middle third, the choice $\frac{15}{32}$, the crude bound (3)); the parent paper makes no claim that it is anywhere near optimal, and neither do we.

Checkpoint 6.2. Trace the construction for $k = 3$, $d_2 = 4$, against the concrete lacunary sequence $b_i = 800^i$ (ratio $800 > 192 \cdot 4 \dots$ check!). Find j_0 and c , compute t_{j_0+1} , and carry out Round 1 numerically with the recipe of §5: exhibit an explicit closed interval of slopes, every one of which yields a $(1, 4)$ -walk whose 3-fold sums miss b_{j_0+1} .

Checkpoint 6.3. The proof used lacunarity *only* through (4), one ratio per consecutive pair. Convince yourself the argument survives if the Adversary’s B merely satisfies $b_{i+1} \geq 192 d_2 b_i$ *eventually* (for all $i \geq$ some i_0) — and locate the single place in §6.2 that needs adjusting.

Part 3. Non-existence: the reduction

7. THE CERTIFICATE TRICK: ONE B AGAINST EVERY A

Part III begins the non-existence half. **Standing assumptions for Parts III and IV:** k, d_1, d_2 are integers with

$$k \geq 3, \quad 1 \leq d_1 < d_2, \quad d_2 \leq k.$$

The goal is part (2) of Theorem A in its full strength: for *any* prescribed sequence $R(1), R(2), \dots$ of ratio lower bounds, a single B with $b_{i+1} \geq R(i) b_i$ whose intersection with kA is nonempty for *every* (d_1, d_2) -sequence A .

This section performs the first great simplification: it converts “defeat every A ” — a statement about all walks at once — into a property to be checked *one walk at a time*.

7.1. Tail-covering.

Definition 7.1 (Tail-covering). A (d_1, d_2) -sequence A is *tail-covering* (for our fixed k) if there are integers $m \geq 1$, ρ , and X_0 such that

$$kA \supseteq \{x \geq X_0 : x \equiv \rho \pmod{m}\} :$$

the sumset contains *every* sufficiently large number in some fixed congruence class.

We met the phenomenon in Example 2.4: for an arithmetic progression with gap d , the sumset kA is a full tail of a class modulo d . The definition allows any modulus m , and this generosity is necessary — the modulus that works can exceed d_2 , and kA need not contain all large integers.

Example 7.2. Let the gaps of A alternate $3, 5, 3, 5, \dots$ and take $k = 5$ (so $d_2 = 5 \leq k$). The prefix sums (partial sums of the gaps) are $0, 3, 8, 11, 16, 19, 24, \dots$: the numbers $8m$ and $8m + 3$. A 5-fold sum of members of A is $5a_1$ plus a sum of five prefix sums, and modulo 8 such a sum is $3j$ where $j \in \{0, 1, \dots, 5\}$ counts how many of the five had the form $8m+3$.

Since $3j \bmod 8$ takes the values $\{0, 3, 6, 1, 4, 7\}$, the sumset $5A$ lives in six of the eight classes modulo 8 and misses the classes $5a_1 + 2$ and $5a_1 + 5$ entirely. It *is* tail-covering — it eventually contains all of $5a_1 + 8\mathbb{Z}$, for instance — but with modulus $8 > d_2$, and it is far from containing all large integers.

Checkpoint 7.3. Verify the claims of Example 7.2: (a) the prefix sums are exactly the numbers $8m$ and $8m + 3$ ($m \geq 0$); (b) $5A$ contains every sufficiently large multiple of 8, shifted by $5a_1$ (how large is “sufficiently”); (c) $5A$ misses the class $5a_1 + 2 \pmod{8}$ completely.

7.2. One sequence to meet them all. Suppose — this is what the rest of Part III will prove — that *every* admissible walk is tail-covering. Each walk hands us a triple (m, ρ, X_0) : its sumset owns the class ρ modulo m from X_0 onward. The Adversary does not know which walk you will play, hence does not know which class; but there are only countably many candidate pairs (m, ρ) , and the Adversary has infinitely many moves. So the Adversary hedges: build B to visit *every* class of *every* modulus, infinitely often, while growing as fast as anyone demands. Whatever walk you play, its class eventually contains an element of B beyond its X_0 — and that element is a collision.

To organize the hedging we need a list of all pairs.

Lemma 7.4 (The recurring list). *There is a sequence of pairs $(m_1, \rho_1), (m_2, \rho_2), \dots$ in which every pair (m, ρ) with $m \geq 1$ and $0 \leq \rho < m$ appears infinitely often.*

Proof. List in blocks. Block N (for $N = 1, 2, 3, \dots$) lists all pairs with $m \leq N$, say in order of m then ρ : block 2 is $(1, 0), (2, 0), (2, 1)$; block 3 is $(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2)$; and so on. Concatenate the blocks. Each block is finite, so the concatenation is a legitimate infinite list, and a given pair (m, ρ) appears in every block from the m -th onward — infinitely often. $\square$

Lemma 7.5 (The certificate). *Fix the recurring list of Lemma 7.4 and let $R : \mathbb{N} \rightarrow \mathbb{N}$ be an arbitrary prescribed ratio sequence — growing as fast as you like, with no regularity assumed. Define B by $b_1 := 1$ and, for $i \geq 1$,*

$$b_{i+1} := \text{the least integer} > \max(R(i)b_i, b_i, i) \text{ that is } \equiv \rho_{i+1} \pmod{m_{i+1}}.$$

Then:

- (1) B is a strictly increasing sequence of positive integers with $b_{i+1} \geq R(i)b_i$ for every i ;
- (2) if every (d_1, d_2) -sequence is tail-covering, then $(kA) \cap B \neq \emptyset$ for every (d_1, d_2) -sequence A .

Proof. (1) Each b_{i+1} exists: the numbers $\equiv \rho_{i+1} \pmod{m_{i+1}}$ form an arithmetic progression, which contains numbers beyond any bound. By construction $b_{i+1} > b_i$ (strictly increasing, hence positive throughout) and $b_{i+1} > R(i)b_i$. Note also $b_{i+1} > i$, so $b_i \geq i$ for every $i \geq 2$: the sequence grows at least linearly, a small fact we need in a moment.

(2) Let A be any (d_1, d_2) -sequence, and let (m, ρ, X_0) witness its tail-covering. We must find an element of B that is $\equiv \rho \pmod{m}$ and $\geq X_0$ — such an element lies in kA by Definition 7.1. The pair (m, ρ) appears infinitely often in the recurring list, say at positions $i^{(1)} < i^{(2)} < \dots$; at each such position $i \geq 2$ the construction made $b_i \equiv \rho \pmod{m}$. Since $b_i \geq i \rightarrow \infty$, some such position has $b_i \geq X_0$ as well. That b_i belongs to both B and kA . $\square$

Big idea. Sparseness cannot save the walker, because congruence classes are everywhere: however fast B is forced to grow, it can always take its next step *inside* any prescribed class. The certificate B cycles through all classes of all moduli forever, so it eventually lands, arbitrarily late, in whichever class a given walk's sumset owns. One sequence, built blind, defeats every tail-covering walk.

Notice what the certificate does *not* depend on: k , d_1 and d_2 never entered the construction — only the ratio sequence R and the recurring list. So the very same B works simultaneously for all parameter triples below the threshold. This is stronger than the theorem demands; it comes for free.

The non-existence half is now reduced to a statement about individual walks:

Goal of Part III. *Every (d_1, d_2) -sequence with $d_2 \leq k$ (and $k \geq 3$) is tail-covering.*

The rest of Part III proves this, walk by walk, according to how the walk wobbles; one case will lean on the subset-sum lemma of Part IV.

Checkpoint 7.6. Take $R(i) = 10^i$. Compute b_2, b_3, b_4 of the certificate (with the block list of Lemma 7.4: $(m_2, \rho_2) = (1, 0)$, $(m_3, \rho_3) = (2, 0)$, $(m_4, \rho_4) = (2, 1)$). Check the ratios.

8. GAP WORDS, TAIL ALPHABETS, AND THE PLAN

To analyze a single walk we need coordinates adapted to its steps. This section sets them up, dispatches the two easiest kinds of walk, and lays out the battle plan for the rest.

8.1. Prefix sums, and the sumset in step coordinates. Let A be a (d_1, d_2) -sequence with gaps $g_n = a_{n+1} - a_n \in [d_1, d_2]$. Define the *prefix sums*

$$p_0 := 0, \quad p_n := g_1 + g_2 + \cdots + g_n \quad (n \geq 1),$$

so that $a_{n+1} = a_1 + p_n$ for every $n \geq 0$: each term of the walk is the start plus the steps taken so far.

Lemma 8.1 (The sumset in step coordinates).

$$kA = k a_1 + \{ p_{i_1} + p_{i_2} + \cdots + p_{i_k} : i_1, \dots, i_k \geq 0 \}.$$

Proof. A member of kA is $a_{j_1} + \cdots + a_{j_k}$ with each $j_t \geq 1$; substituting $a_{j_t} = a_1 + p_{j_t-1}$ and setting $i_t := j_t - 1 \geq 0$ gives one direction, and the substitution reverses. $\square$

From now on we study the set $kA - k a_1$ of *sums of k prefix sums*. Note that tail-covering is insensitive to the shift: kA contains a class tail if and only if $kA - k a_1$ does (with the residue shifted by $k a_1$).

8.2. The gap word and the tail alphabet. The *gap word* of A is the infinite sequence $g_1, g_2, g_3, \dots$ read as a word — a string of “letters” drawn from the finite alphabet $\{d_1, \dots, d_2\}$. Since the alphabet is finite, some letters occur infinitely often in the word and some (perhaps) only finitely often. Define the *tail alphabet*

$$G_\infty := \{ \delta : g_n = \delta \text{ for infinitely many } n \},$$

which is nonempty, and fix an index T beyond which *only* letters of G_∞ occur (possible: each of the finitely many non-recurring letters has a last occurrence; take T past all of them).

Lemma 8.2 (Tail reduction). *Let $A' := \{a_{T+1} < a_{T+2} < \dots\}$ be the tail of A beyond T — itself a (d_1, d_2) -sequence, whose gap word uses only letters of G_∞ . If A' is tail-covering, so is A .*

Proof. Every sum of k members of A' is a sum of k members of A , so $kA \supseteq kA'$; a class tail inside kA' is a class tail inside kA . $\square$

We will invoke this silently: “pass to the tail” means replace A by A' , so that every gap lies in G_∞ and every letter of G_∞ still occurs infinitely often.

The analysis now splits on the *size* of the tail alphabet, and the first case is immediate:

Lemma 8.3 (One letter). *If $G_\infty = \{\delta\}$, then A is tail-covering.*

Proof. Pass to the tail: all gaps equal δ , so the tail is an arithmetic progression with gap δ , and Example 2.4 (which is now official: the computation there is exact) shows its k -fold sumset is a full class tail modulo δ . $\square$

One more normalization. If all the recurring letters share a common factor, we can divide it out:

Lemma 8.4 (Rescaling). *Suppose (after passing to the tail) every gap of A lies in G_∞ , and let $g := \gcd(G_\infty) > 1$. Define the rescaled walk A' by*

$$a'_n := \frac{a_n - a_1}{g} + 1 \quad (n \geq 1),$$

a sequence of positive integers with gaps $g'_n = g_n/g$, alphabet $G_\infty/g := \{\delta/g : \delta \in G_\infty\}$, $\gcd(G_\infty/g) = 1$, and maximum letter $\leq d_2/g$. If A' is tail-covering modulo m' , then A is tail-covering modulo gm' .

Proof. Every gap of A is a multiple of g , so each $a_n - a_1$ is one too and the a'_n are integers, strictly increasing from $a'_1 = 1$; their gaps are g_n/g by construction. Unwinding the definition, $a_n = (a_1 - g) + g a'_n$, so a sum of k members of A is $k(a_1 - g) + g \cdot$ (a sum of k members of A'):

$$kA = k(a_1 - g) + g \cdot kA'.$$

Now suppose $kA' \supseteq \{x \geq X'_0 : x \equiv \rho' \pmod{m'}\}$. Consider any $y \equiv k(a_1 - g) + g\rho' \pmod{gm'}$ with y large. Then $y - k(a_1 - g)$ is divisible by g (its residue modulo gm' is $g\rho'$), and $x := (y - k(a_1 - g))/g \equiv \rho' \pmod{m'}$ is large along with y ; so $x \in kA'$ and $y = k(a_1 - g) + gx \in kA$. Hence kA contains a full class tail modulo gm' . $\square$

So from now on we may and do assume $\gcd(G_\infty) = 1$ (rescaling first if necessary, and remembering that the final modulus picks up a factor g — harmless, since tail-covering permits any modulus).

8.3. The battle plan. Here is the case tree for the Goal, matching the parent paper’s map; each node names the section that closes it.

Tail alphabet (after rescaling)	Closed by
$ G_\infty = 1$	Lemma 8.3 (arithmetic progression)
$ G_\infty = 2$, gap word eventually periodic	§9
$ G_\infty = 2$, genuinely aperiodic	§§11–13:
— irregular enough (wide band)	the sweep (§11)
— maximally regular (balanced)	Morse–Hedlund (§12), then the Sturmian endgame (§13)
$ G_\infty \geq 3$	§14 (slot lemma) + Part IV

The two-letter case is the deep one: a two-letter walk has exactly one degree of freedom per step, and the walks that use that freedom *most evenly* — neither settling into a period nor drifting irregularly — are precisely the Sturmian walks, where the hardest analysis lives. Three or more letters, counterintuitively, is *easier*: more letters means more kinds of step to combine, and a bounded combinatorial gadget (the slot lemma) plus one finite theorem (Part IV) covers all such walks at once.

9. EVENTUALLY PERIODIC GAP WORDS

A gap word is *eventually periodic* if beyond some point it repeats with a fixed period: there are $N_0 \geq 0$ and $\ell \geq 1$ with

$$g_{n+\ell} = g_n \quad \text{for all } n > N_0.$$

Lemma 9.1 (Eventually periodic). *If the gap word of A is eventually periodic, then A is tail-covering.*

Proof. Pass to the tail beyond N_0 (Lemma 8.2); relabel so that the gap word satisfies $g_{n+\ell} = g_n$ for all $n \geq 1$. Let

$$P := g_1 + g_2 + \cdots + g_\ell$$

be the *period sum* — the distance the walk advances in each full period. Then the prefix sums repeat with drift P : for $0 \leq j < \ell$ and $m \geq 0$,

$$p_{m\ell+j} = mP + p_j,$$

by grouping the first $m\ell$ gaps into m full periods (each summing to P) plus the leftover prefix.

Now build sums of k prefix sums using only the multiples of the period: choosing $i_t = m_t\ell$ (i.e. $j = 0$) for all t gives

$$p_{m_1\ell} + \cdots + p_{m_k\ell} = (m_1 + \cdots + m_k)P,$$

and $m_1 + \cdots + m_k$ ranges over every integer ≥ 0 (Checkpoint in §2.3 showed this ranging argument for $\geq k$; here the i_t may be 0, so even every integer ≥ 0). By Lemma 8.1, $kA \supseteq ka_1 + P\{0, 1, 2, \dots\}$: a full tail of the class of ka_1 modulo P . $\square$

Note how little of the period structure we used — only the sub-walk at the period marks, which is an arithmetic progression with gap P . The other residues $j \neq 0$ would give further classes (as in Example 7.2, which is the case $\ell = 2$, $P = 8$ of this proof), but one class is all tail-covering asks.

Checkpoint 9.2. The walk with repeating gap block $(2, 3, 2)$ (period $\ell = 3$, $P = 7$) and $k = 3$: which classes modulo 7 does $3A - 3a_1$ contain in full, beyond the class of 0? Compute the possible values of $p_{i_1} + p_{i_2} + p_{i_3} \bmod 7$.

10. INTERLUDE: A MINI-COURSE ON WORDS, AND A LITTLE ANALYSIS

The two-letter case will occupy us for four sections, and it runs on a small amount of theory — about *words* (infinite strings of symbols) and about *limits* — that deserves its own unhurried introduction. Nothing here mentions the problem; it is pure toolbox.

10.1. Words. An *alphabet* is a finite set of symbols; a *word* is a finite or infinite string of them. We care about *binary* words, alphabet $\{0, 1\}$, written $h = h_1h_2h_3\cdots$ with $h_n \in \{0, 1\}$. A *window* of h is a run of consecutive letters $h_{i+1}h_{i+2}\cdots h_j$ (which we address by its endpoints $i < j$; its *length* is $j - i$).

The *counting function* of h is

$$q_0 := 0, \quad q_n := h_1 + h_2 + \cdots + h_n \quad (\text{the number of 1s among the first } n \text{ letters}).$$

The count of 1s in the window from i to j is then $q_j - q_i$. Two properties tie h and q together: q is nondecreasing with steps $q_n - q_{n-1} = h_n \in \{0, 1\}$, and conversely any such staircase function comes from exactly one word.

Definition 10.1 (Balanced). A binary word h is *balanced* if any two windows of equal length contain numbers of 1s differing by at most one:

$$j - i = j' - i' \implies |(q_j - q_i) - (q_{j'} - q_{i'})| \leq 1.$$

Balance is a strong uniformity requirement: wherever you cut out two equal stretches of the word, they must contain almost exactly the same number of 1s.

Example 10.2. (a) The constant words $000\cdots$ and $111\cdots$ are balanced, as is the alternating word $010101\cdots$ (a window of length $2m$ always has exactly m ones; length $2m+1$ has m or $m+1$). (b) The periodic word $001001001\cdots$ is balanced (check windows of each length modulo 3). (c) The periodic word $00110011\cdots$ is *not* balanced: the windows 00 and 11 have 1-counts 0 and 2. (d) The *Fibonacci word* $0100101001001010010\cdots$ — built by iterating the substitution $0 \mapsto 01, 1 \mapsto 0$ — is balanced and *not* eventually periodic. It will reappear in §13 as the paradigm of the hardest case. (Proving these two claims about it is not needed for our proof; the classification below works from balancedness alone, whatever word exhibits it.)

Checkpoint 10.3. Verify (b) and (c) of Example 10.2. Then show that in a balanced word, if some window of length Δ contains u ones, every window of length Δ contains $u - 1, u$, or $u + 1$ ones.

10.2. Suprema, and limits. Two notions from analysis, in the exact strength we need.

Supremum. A set S of real numbers that is nonempty and bounded above has a *least upper bound* $\sup S$: an upper bound of S that is $\leq$ every other upper bound. (Its existence is, again, the completeness of the real line — the same axiom behind the nested interval theorem of §6; indeed each implies the other.) Likewise a nonempty set bounded below has a greatest lower bound $\inf S$. Two reflexes to internalize: every element of S is $\leq \sup S$; and for any $\varepsilon > 0$, some element of S exceeds $\sup S - \varepsilon$ (else $\sup S - \varepsilon$ would be a smaller upper bound).

Limits. A sequence $x_1, x_2, \dots$ of reals *converges* to x (written $x_n \rightarrow x$) if for every $\varepsilon > 0$, all but finitely many n have $|x_n - x| < \varepsilon$. We need only three facts, each immediate from the definition: (i) $1/n \rightarrow 0$; (ii) if $x_n \rightarrow x$ and $x_n \leq C$ for all n , then $x \leq C$ (a limit cannot escape a closed bound); (iii) if $x_n \rightarrow x$ and $x_n \geq c$ for all n , then $x \geq c$.

10.3. The discrete intermediate value theorem.

Lemma 10.4 (Discrete IVT). Let $v_0, v_1, \dots, v_N$ be integers with $|v_{i+1} - v_i| \leq 1$ for every i . Then every integer between v_0 and v_N (inclusive) equals some v_i .

Proof. Say $v_0 \leq v_N$ (else reverse the list) and let u be an integer with $v_0 \leq u \leq v_N$. Let i^* be the largest index with $v_{i^*} \leq u$ (it exists: $i = 0$ qualifies; and $i^* < N$ would give $v_{i^*+1} > u$, hence $v_{i^*+1} \geq u + 1$, so $v_{i^*} \geq v_{i^*+1} - 1 \geq u$, forcing $v_{i^*} = u$). If $i^* = N$ then $v_N \leq u \leq v_N$, again $u = v_N$. $\square$

A sequence of integers that never jumps by more than 1 cannot step over an integer. We will use this constantly: it is the discrete shadow of the “no-skipping” arguments that ran through Part II.

11. TWO LETTERS: THE REACHABLE BAND, AND THE SWEEP

We now take up the main case. **Standing setup for §§11–13:** after tail-passing and rescaling (§8), the walk A has tail alphabet $G_\infty = \{\delta < \delta'\}$ with

$$\gcd(\delta, \delta') = 1, \quad e := \delta' - \delta \geq 1, \quad \delta' = \delta + e \leq d_2 \leq k,$$

and both letters occur infinitely often. (The bound $\delta' \leq d_2$ holds because every gap is $\leq d_2$; after rescaling it could only improve. Note $\gcd(\delta, e) = \gcd(\delta, \delta') = 1$.) Encode the gap word in binary: write

$$g_n = \delta + e h_n, \quad h_n \in \{0, 1\},$$

so small steps are 0s and large steps are 1s, and let q_n be the counting function of the binary word h .

11.1. The reachable band. The prefix sums split along the same lines: $p_n = \sum_{j \leq n} (\delta + e h_j) = \delta n + e q_n$. So a sum of k prefix sums, with indices $i_1, \dots, i_k \geq 0$ summing to s , is

$$p_{i_1} + \dots + p_{i_k} = \delta s + e (q_{i_1} + \dots + q_{i_k}),$$

and by Lemma 8.1,

$$kA - k a_1 = \{ \delta s + e w : s \geq 0, w \in W_s \}, \quad W_s := \left\{ \sum_{t=1}^k q_{i_t} : i_t \geq 0, \sum_{t=1}^k i_t = s \right\}. \quad (5)$$

Everything now hangs on the sets W_s : for each “index budget” s , which totals w of 1-counts are achievable? The first surprise is that there are never holes.

Lemma 11.1 (The band is an interval). *For every $s \geq 0$, the set W_s is a full interval of integers: $W_s = [w^-(s), w^+(s)] \cap \mathbb{Z}$ for some integers $w^-(s) \leq w^+(s)$. Moreover both endpoints climb gently:*

$$w^\pm(s+1) - w^\pm(s) \in \{0, 1\}.$$

Proof. Call a multiset $\{i_1, \dots, i_k\}$ of nonnegative integers with sum s a *configuration*, and its *value* the total $\sum_t q_{i_t}$. A *move* takes one unit from one index and gives it to another: it replaces some pair (i_a, i_b) by $(i_a - 1, i_b + 1)$ (allowed when $i_a \geq 1$). A move keeps the sum s and changes the value by

$$(q_{i_b+1} - q_{i_b}) - (q_{i_a} - q_{i_a-1}) = h_{i_b+1} - h_{i_a} \in \{-1, 0, 1\}.$$

Any configuration can be walked to any other by such moves: repeatedly take a unit from an index that is too large relative to the target configuration and give it to one that is too small (formally: induct on the total discrepancy $\sum |i_t - i'_t|$ over matched sorted entries, which each well-chosen move reduces). In particular there is a chain of moves from a minimum-value configuration to a maximum-value one; the values along the chain form an integer sequence with steps in $\{-1, 0, 1\}$ starting at $w^-(s)$ and ending at $w^+(s)$, so by the discrete IVT (Lemma 10.4) every integer in $[w^-(s), w^+(s)]$ is a value. That is the interval claim.

For the endpoint increments, compare budgets s and $s+1$ in both directions. Given any configuration for s , adding one unit to any index i_a produces a configuration for $s+1$ whose value has grown by $h_{i_a+1} \in \{0, 1\}$; applied to a maximizer this gives $w^+(s+1) \geq w^+(s)$, applied to a minimizer, $w^-(s+1) \leq w^-(s) + 1$. Given any configuration for $s+1$, removing

one unit from a positive index drops the value by 0 or 1; applied to a maximizer, $w^+(s) \geq w^+(s+1) - 1$, applied to a minimizer, $w^-(s) \leq w^-(s+1)$. Combining the four inequalities pins both increments into $\{0, 1\}$. $\square$

So for each s the reachable pairs (s, w) form a vertical integer segment, and as s grows these segments drift upward, each endpoint by 0 or 1 per step: a *band* across the (s, w) -plane. Write

$$\text{width}(s) := w^+(s) - w^-(s)$$

for the band's height. One more elementary bound, used twice below: since $q_i \leq i$ always, every configuration's value is at most $\sum_t i_t = s$, so

$$w^+(s) \leq s; \tag{6}$$

and taking the configuration $(s, 0, \dots, 0)$ shows $w^+(s) \geq q_s$, which tends to infinity because the letter 1 occurs infinitely often.

11.2. The sweep. Now the payoff. Recall from (5) that realizing a target integer x means writing $x = \delta s + ew$ with $w \in W_s$. If the band is *wide* — as wide as the alphabet spread requires — then every large x is realized:

Lemma 11.2 (Sweep). *Suppose there is an S_0 with $\text{width}(s) \geq d_2 - 1$ for all $s \geq S_0$. Then $kA - ka_1$ contains every sufficiently large integer; in particular A is tail-covering with modulus $m = 1$.*

Proof. Fix a large integer x (thresholds at the end). Which budgets s could realize x ? From $x = \delta s + ew$ we need $\delta s \equiv x \pmod{e}$, and since $\gcd(\delta, e) = 1$ this pins s to a single residue class modulo e ; call such s *admissible*, and for admissible s set

$$w(s) := \frac{x - \delta s}{e} \in \mathbb{Z},$$

the height at which the band would have to be hit. As s steps up by e (to the next admissible budget), $w(s)$ steps *down* by exactly δ . Meanwhile, by Lemma 11.1, each band endpoint rises by between 0 and e over those e unit steps. A descending target against a rising band: the picture from Figure 3 is that they must cross, and that a wide band cannot be jumped over in a single stride. We make both halves precise.

The target starts above the band and ends below it. At the small end: let $s_{\min}$ be the smallest admissible $s \geq S_0$; then $w(s_{\min}) \geq (x - \delta(S_0 + e))/e$, which for large x exceeds $s_{\min}$, hence exceeds $w^+(s_{\min})$ by (6) — explicitly, $x > (\delta + e)(S_0 + e)$ suffices. At the large end: enlarge S_0 once and for all so that $w^+(s) \geq \delta$ for every $s \geq S_0$ (possible since $w^+(s) \geq q_s \rightarrow \infty$; this depends only on the word, not on x). For the largest admissible s with $w(s) \geq 0$, the next admissible step would take w negative, so $0 \leq w(s) < \delta \leq w^+(s)$: at that budget the target sits at or below the band's top.

No skipping. Suppose, for contradiction, that no admissible $s \geq S_0$ has $w(s) \in [w^-(s), w^+(s)]$. The target descends (by δ per admissible step) from above the band to below its top; since it never lands inside, there must be two *consecutive* admissible budgets s' and $s' + e$ with

$$w(s') > w^+(s') \quad \text{and} \quad w(s' + e) < w^-(s' + e) :$$

the leap over the band happens between them. First, s' is large: from $w(s' + e) \leq w^-(s' + e) \leq w^+(s' + e) \leq s' + e$ (again (6)),

$$\frac{x - \delta(s' + e)}{e} \leq s' + e \implies s' + e \geq \frac{x}{\delta + e},$$

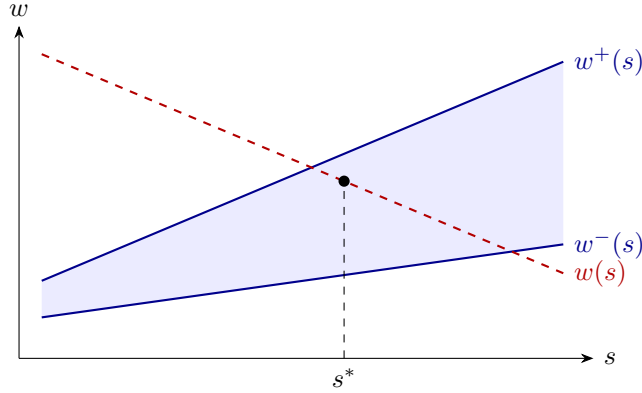


FIGURE 3. The sweep. The reachable band (shaded, endpoints $w^\pm$) rises gently as the budget s grows; the required height $w(s)$ for a fixed target x descends along the admissible budgets. If the band is wide, the descending line cannot skip it: some admissible s^* lands inside, and there x is realized.

so $s' \geq S_0$ holds automatically once $x \geq (\delta + e)(S_0 + e)$ — the same threshold as before, so no new constraint. Now stack the inequalities. The target drops by exactly δ across the leap, and the band's floor rises by at most e :

$$w^+(s') < w(s') = w(s' + e) + \delta \leq w^-(s' + e) - 1 + \delta \leq w^-(s') + e - 1 + \delta.$$

Hence

$$\text{width}(s') = w^+(s') - w^-(s') \leq \delta + e - 2 = \delta' - 2 \leq d_2 - 2,$$

contradicting the hypothesis $\text{width}(s') \geq d_2 - 1$. So some admissible $s^* \geq S_0$ has $w(s^*) \in [w^-(s^*), w^+(s^*)]$; by Lemma 11.1 the integer $w(s^*)$ is actually attained, and $x = \delta s^* + e w(s^*) \in kA - ka_1$. $\square$

Look where the alphabet bound was spent: the contradiction needed $\text{width} \geq \delta' - 1$, and the hypothesis supplies $d_2 - 1 \geq \delta' - 1$. A band exactly one narrower could be leapt over — the sweep operates with no room to spare, a first taste of how tight the corner $d_2 = k$ will be.

11.3. Wide or balanced: the width dichotomy. The sweep shifts the burden: we must now show the band is wide. Width comes from *irregularity* of the word h — if some stretches are 1-rich and others 1-poor, configurations can exploit both. The cleanest measure uses just two indices. For $\sigma \geq 0$ define the *two-index width*

$$W_2(\sigma) := \max\{q_i + q_j : i + j = \sigma, i, j \geq 0\} - \min\{q_i + q_j : i + j = \sigma, i, j \geq 0\}.$$

Lemma 11.3 (Width dichotomy).

- (a) Suppose $W_2(\sigma_0) \geq 2$ for some σ_0 , and suppose k is odd or $d_2 \leq k - 1$. Then $\text{width}(s) \geq d_2 - 1$ for all sufficiently large s — and the sweep applies.
- (b) If instead $W_2(\sigma) \leq 1$ for every σ , then the word h is balanced.

Proof. (a) The engine is additivity: if we split the k indices of a configuration into groups with prescribed subtotal sums, the extreme values over each group add. Fix a pair (i^-, j^-) attaining the minimum in $W_2(\sigma_0)$ and a pair (i^+, j^+) attaining the maximum, so the max exceeds the min by $W_2(\sigma_0) \geq 2$.

k odd. Write $k = 2u + 1$. For any $s \geq u\sigma_0$, build configurations from u pairs, each summing to σ_0 , plus one leftover index $j := s - u\sigma_0 \geq 0$: choosing all u pairs to be (i^+, j^+) gives value $u \cdot \max + q_j$, choosing (i^-, j^-) gives $u \cdot \min + q_j$. Hence

$$\text{width}(s) \geq u W_2(\sigma_0) \geq 2u = k - 1 \geq d_2 - 1,$$

using $d_2 \leq k$.

k even, $d_2 \leq k - 1$. Write $k = 2u$. For $s \geq (u - 1)\sigma_0$, use $u - 1$ prescribed pairs at σ_0 plus one *free* pair with sum $\tau := s - (u - 1)\sigma_0 \geq 0$ (any split $\tau = i + j$): the prescribed pairs alone already force

$$\text{width}(s) \geq (u - 1) W_2(\sigma_0) \geq 2u - 2 = k - 2 \geq d_2 - 1.$$

(b) Take two equal-length windows, from i to i' and from j' to j (so $i' - i = j - j'$, i.e. $i + j = i' + j' =: \sigma$). Then

$$|(q_{i'} - q_i) - (q_j - q_{j'})| = |(q_{i'} + q_{j'}) - (q_i + q_j)| \leq W_2(\sigma) \leq 1,$$

which is precisely balancedness. $\square$

Part (a)'s parity split is not laziness. When k is even and $d_2 = k$, the budget of k indices is exactly wrong: u full pairs use all k indices and give $\text{width} \geq k$ — but only when s has the right form — while $u - 1$ pairs guarantee only $k - 2 = d_2 - 2$, one short of what the sweep needs. This single missing unit of width is a genuine boundary phenomenon, and §13 will close it with a separate argument (Lemma 13.4) rather than a sharper pairing.

Where we stand. If some $W_2(\sigma_0) \geq 2$ and (k odd or $d_2 \leq k - 1$): done, by (a) and the sweep. If every $W_2(\sigma) \leq 1$: the word is balanced, and the next section classifies it — eventually periodic (done, by §9) or Sturmian (handled in §13). The one remaining configuration, k even with $d_2 = k$ and the word unbalanced, is also settled in §13. The ledger stays honest.

Checkpoint 11.4. For the periodic word $h = 00110011 \dots$ (which Example 10.2 showed is unbalanced), find a σ_0 with $W_2(\sigma_0) \geq 2$, and exhibit, for $k = 3$, $s = 12$, two configurations whose values differ by at least 2.

Checkpoint 11.5. W_2 uses exactly two indices while width uses k ; compare them. Prove that $\text{width}(s) \geq W_2(\sigma)$ whenever $s \geq \sigma$ and $k \geq 2$ (pad the two-index configurations with zeros — $q_0 = 0$ costs nothing), and find a word and an s with $\text{width}(s) > W_2(s)$.

12. BALANCED WORDS: THE MORSE–HEDLUND CLASSIFICATION

The width dichotomy left us with the balanced words, and this section proves the structure theorem that tames them: *a balanced binary word is either eventually periodic, or — after discarding at most one initial stretch — is “mechanical”: its counting function is $q_n = \lfloor n\alpha + \beta \rfloor$ for a fixed irrational slope α .* This is (one direction of) a celebrated 1940 theorem of Morse and Hedlund. The proof below is elementary and self-contained: its only imported ingredient is the completeness of the reals (through sup), and the section closes with the one approximation tool (Dirichlet, by pigeonhole) that the endgame will need.

Throughout, h is a balanced binary word with counting function q ($q_0 = 0$).

12.1. Step 1: a balanced word is nearly additive.

Lemma 12.1 (Near-additivity). *For all $n, m \geq 0$: $|q_{n+m} - q_n - q_m| \leq 1$.*

Proof. $q_{n+m} - q_n$ counts the 1s in the window of length m starting after position n ; q_m counts them in the window of length m at the very start. Balance bounds the difference by 1. $\square$

Iterating pins down long-range behavior:

Lemma 12.2. *For all $n \geq 0$, $t \geq 1$: $tq_n - (t-1) \leq q_{tn} \leq tq_n + (t-1)$.*

Proof. Induct on t . Trivial at $t = 1$. If it holds at t , then by Lemma 12.1 (with the pair tn, n), $q_{(t+1)n} \leq q_{tn} + q_n + 1 \leq (t+1)q_n + t$, and similarly from below. $\square$

12.2. Step 2: the slope.

Lemma 12.3 (The slope exists). *There is a real number $\alpha \in [0, 1]$ such that*

$$|D_n| \leq 1 \quad \text{for every } n \geq 0, \quad \text{where } D_n := q_n - n\alpha.$$

Moreover $q_n/n \rightarrow \alpha$.

Proof. The key inequality: for all $n, m \geq 1$,

$$mq_n - nq_m \leq n + m - 2. \tag{7}$$

To see it, bound q_{nm} twice by Lemma 12.2: viewing nm as m copies of n , $q_{nm} \geq mq_n - (m-1)$; viewing it as n copies of m , $q_{nm} \leq nq_m + (n-1)$. Chaining the two gives (7).

Dividing (7) by nm :

$$\frac{q_n - 1}{n} - \frac{q_m + 1}{m} = \frac{mq_n - nq_m - (n + m)}{nm} \leq \frac{-2}{nm} < 0.$$

So every number of the form $(q_n - 1)/n$ is below every number of the form $(q_m + 1)/m$. The set $\{(q_n - 1)/n : n \geq 1\}$ is therefore bounded above; let

$$\alpha := \sup_{n \geq 1} \frac{q_n - 1}{n}.$$

By construction $(q_n - 1)/n \leq \alpha$ for every n ; and $\alpha \leq (q_m + 1)/m$ for every m , because $(q_m + 1)/m$ is an upper bound of the set. The two together read $q_n - 1 \leq n\alpha \leq q_n + 1$, i.e. $|D_n| \leq 1$ (also at $n = 0$, where $D_0 = 0$). Since $0 \leq q_n \leq n$, we get $(q_n - 1)/n < 1$ for all n , so $\alpha \leq 1$; and $\alpha \geq (q_n - 1)/n \geq -1/n$ for every n , so $\alpha \geq 0$. Finally $|q_n/n - \alpha| = |D_n|/n \leq 1/n \rightarrow 0$. $\square$

The number α is the *slope* of h : the limiting density of 1s. The inequality $|D_n| \leq 1$ says the word tracks the line $y = \alpha x$ with error never exceeding 1 — already a strong rigidity statement, extracted from balance alone.

12.3. Step 3: the discrepancy barely oscillates.

Lemma 12.4 (Oscillation bound). *For all $n, m \geq 0$: $|D_n - D_m| \leq 1$.*

Proof. Say $m = n + \Delta$ with $\Delta \geq 1$. For $j \geq 0$ let $W(j) := q_{j+\Delta} - q_j$ count the 1s in the length- Δ window starting after j ; balance gives $|W(j) - W(n)| \leq 1$ for every j . Sum W over the t consecutive disjoint windows starting at $0, \Delta, 2\Delta, \dots, (t-1)\Delta$; the sum telescopes to $q_{t\Delta}$, and each summand lies within 1 of $W(n)$:

$$t(W(n) - 1) \leq q_{t\Delta} \leq t(W(n) + 1).$$

Divide by $t\Delta$ and let $t \rightarrow \infty$: the middle tends to α (Lemma 12.3), so by the limit facts of §10,

$$\frac{W(n) - 1}{\Delta} \leq \alpha \leq \frac{W(n) + 1}{\Delta}, \quad \text{i.e.} \quad |W(n) - \Delta\alpha| \leq 1.$$

And $D_{n+\Delta} - D_n = (q_{n+\Delta} - q_n) - \Delta\alpha = W(n) - \Delta\alpha$. $\square$

So all the discrepancies $D_0, D_1, D_2, \dots$ live in a band of height 1 — they may drift within it, but no two ever differ by more than 1. Everything below flows from this single fact.

12.4. Step 4: rational slope forces eventual periodicity.

Proposition 12.5. *If α is rational, h is eventually periodic.*

Proof. Write $\alpha = p/q$ in lowest terms ($q \geq 1$) and set

$$c_n := q q_n - p n = q D_n,$$

an integer with $|c_n| \leq q$ (Lemma 12.3). Let $w_n := q_{n+q} - q_n$ (the 1-count of the length- q window after n), so that

$$c_{n+q} - c_n = q(w_n - p).$$

Balance confines the w_n to two consecutive values $\{W, W+1\}$ (all length- q windows agree to within one). We claim $p \in \{W, W+1\}$: if every $w_n \geq p+1$, then $c_{n+q} - c_n \geq q$ for every n , so following the indices $n, n+q, n+2q, \dots$ the integer c climbs by at least q per step, exiting $[-q, q]$ — impossible; symmetrically, $w_n \leq p-1$ for all n is impossible. So either $w_n \in \{p, p+1\}$ for all n , or $w_n \in \{p-1, p\}$ for all n .

In the first case, $c_{n+q} - c_n \in \{0, q\}$: along each of the q index classes modulo q , the sequence c is nondecreasing and bounded, hence eventually constant; from some point on, no step of $+q$ ever occurs, i.e. $w_n = p$ for all large n . The second case is the mirror image (nonincreasing), with the same conclusion. So $q_{n+q} = q_n + p$ for all large n , and then

$$h_{n+q+1} = q_{n+q+1} - q_{n+q} = (q_{n+1} + p) - (q_n + p) = h_{n+1}$$

for all large n : the word repeats with period q from some point on. $\square$

12.5. Step 5: irrational slope forces a mechanical tail.

Definition 12.6 (Mechanical word). A binary word h is *mechanical* with slope α and intercept β if

$$q_n = \lfloor n\alpha + \beta \rfloor \quad \text{for every } n \geq 1.$$

Mechanical words with irrational slope are called *Sturmian*.

(A mechanical word is the “digitized straight line” of slope α : its n -th letter records whether the line $y = \alpha x + \beta$ crosses an integer height in the n -th unit interval. Beatty walks and mechanical words are two costumes of the same actor — compare Definition 3.4.)

Proposition 12.7. *If α is irrational, then there is an index $n_0 \geq 0$ such that the shifted word $h'_j := h_{n_0+j}$ is mechanical with slope α (and some intercept β').*

Proof. Set $\beta := \sup_n D_n$, finite since $|D_n| \leq 1$. By the oscillation bound, every $D_n \geq \beta - 1$: for any m , $D_m - D_n \leq 1$, and taking the supremum over m gives $\beta - D_n \leq 1$.

When does the mechanical formula hold at n ? Unwinding the floor’s definition,

$$q_n = \lfloor n\alpha + \beta \rfloor \iff n\alpha + \beta - 1 < q_n \leq n\alpha + \beta \iff \beta - 1 < D_n \leq \beta.$$

The upper half is the definition of β ; the lower half holds at least weakly ($D_n \geq \beta - 1$), so the formula can fail only where $D_n = \beta - 1$ exactly — equivalently where $n\alpha + \beta = q_n + 1$ is an integer. If two distinct indices $n_1 \neq n_2$ both had $n_i\alpha + \beta \in \mathbb{Z}$, subtracting would give $(n_1 - n_2)\alpha \in \mathbb{Z}$, impossible for irrational α . So there is at most one exceptional index; call it n_0 (and set $n_0 := 0$ if there is none).

Discard everything through n_0 : the word $h'_j := h_{n_0+j}$ has counting function $q'_j = q_{n_0+j} - q_{n_0}$, and for every $j \geq 1$ the index $n_0 + j$ is unexceptional, so

$$q'_j = \lfloor (n_0 + j)\alpha + \beta \rfloor - q_{n_0} = \lfloor j\alpha + (n_0\alpha + \beta - q_{n_0}) \rfloor = \lfloor j\alpha + \beta' \rfloor, \quad \beta' := \beta - D_{n_0},$$

where the middle equality moves the integer q_{n_0} through the floor (Lemma 3.2(2)). $\square$

Assembling Steps 4 and 5:

Theorem 12.8 (Balanced classification). *A balanced binary word is either eventually periodic, or agrees beyond some index with a Sturmian word: a mechanical word of irrational slope $\alpha \in (0, 1)$.*

Proof. Steps 2–5; when α is irrational it lies in $(0, 1)$, being in $[0, 1]$ and not equal to 0 or 1. $\square$

For our purposes the theorem is a perfect fork: the periodic branch was closed in §9, so *the only balanced words still standing are the Sturmian tails*, with an explicit formula $q_n = \lfloor n\alpha + \beta \rfloor$ to attack. The attack needs one lever: the ability to find indices n where the fractional part $\{n\alpha + \beta\}$ lands in any prescribed arc. That is a statement about *rotations of a circle*, and it closes this section.

12.6. Rotations, and how to hit an arc on schedule. For a real x write $\{x\} := x - \lfloor x \rfloor \in [0, 1)$ for its *fractional part*. Picture the interval $[0, 1)$ bent into a circle of circumference 1; the map $x \mapsto \{x + \alpha\}$ is rotation by α . The numbers $\{n\alpha + \beta\}$ are the successive positions of a point rotating by α per tick. Write $\|x\|$ for the distance from x to the nearest integer — the circle distance from $\{x\}$ to 0.

Lemma 12.9 (Dirichlet’s approximation theorem). *For every real α and integer $Q \geq 1$ there are integers n_0, m with $1 \leq n_0 \leq Q$ and*

$$|n_0\alpha - m| < \frac{1}{Q}.$$

Proof. Pigeonhole. The $Q + 1$ numbers $\{0 \cdot \alpha\}, \{1 \cdot \alpha\}, \dots, \{Q\alpha\}$ all lie in $[0, 1)$, which we cut into Q boxes $[i/Q, (i + 1)/Q)$. Two of them share a box: $\{j\alpha\}$ and $\{j'\alpha\}$ with $j < j'$, differing by less than $1/Q$. Set $n_0 := j' - j \in [1, Q]$; then $n_0\alpha = (j'\alpha - \lfloor j'\alpha \rfloor) - (j\alpha - \lfloor j\alpha \rfloor) + (\lfloor j'\alpha \rfloor - \lfloor j\alpha \rfloor) = \{j'\alpha\} - \{j\alpha\} + m$ for the integer $m := \lfloor j'\alpha \rfloor - \lfloor j\alpha \rfloor$, and $|n_0\alpha - m| = |\{j'\alpha\} - \{j\alpha\}| < 1/Q$. $\square$

Lemma 12.10 (Uniform hitting). *Let $\alpha \in (0, 1)$ be irrational and $\beta \in \mathbb{R}$. For every $\varepsilon > 0$ there is an integer $L = L(\alpha, \varepsilon)$ — depending only on α and ε , not on β and not on the arc’s position — such that for every arc $J \subseteq (0, 1)$ of length $\geq \varepsilon$ and every $N \geq 0$, some index $n \in \{N + 1, \dots, N + L\}$ has $\{n\alpha + \beta\} \in J$.*

Proof. An explicit *circle walk*. Choose an integer $Q > 2/\varepsilon$ and apply Dirichlet: there are $1 \leq n_0 \leq Q$ and m with $\theta := |n_0\alpha - m| < 1/Q < \varepsilon/2$; and $\theta \neq 0$ since α is irrational. Advancing the index by n_0 rotates the point $\{n\alpha + \beta\}$ by exactly θ around the circle — always in the same direction (the sign of $n_0\alpha - m$ is fixed).

Now fix N , and follow the $\lceil 1/\theta \rceil + 1$ indices

$$n_j := N + (j + 1)n_0 \quad (j = 0, 1, \dots, \lceil 1/\theta \rceil),$$

all lying in $\{N + 1, \dots, N + L\}$ for $L := n_0(\lceil 1/\theta \rceil + 1)$. The successive positions $x_j := \{n_j\alpha + \beta\}$ each step a fixed distance θ around the circle, in a fixed direction, and over $\lceil 1/\theta \rceil$ steps the total distance walked is at least 1: the walk laps the whole circle. If none of the x_j lay in the arc J , then some single step would have to jump clean over J — but each step has length $\theta < \varepsilon/2 < |J|$, too short to clear it. So some $x_j \in J$.

Finally, audit the dependence: Q came from ε ; n_0, m, θ from α and Q ; and $L = n_0(\lceil 1/\theta \rceil + 1) \leq Q(\lceil 1/\theta \rceil + 1)$ from these. Neither β , nor N , nor where J sits was used — only $|J| \geq \varepsilon$. $\square$

The uniformity is the point: in the endgame the arc J will *move* as the target changes, and we need a single window length L serving every position. A subtler equidistribution theorem would say more; this crude walking bound, powered by one pigeonhole, says exactly enough.

Checkpoint 12.11. (a) For the periodic balanced word $001001\cdots$, compute α and check $|D_n| \leq 1$ and the conclusion of Step 4 directly. (b) For $\alpha = \sqrt{2} - 1$ and $Q = 5$, find the $n_0 \leq 5$ and θ that Dirichlet's pigeonhole produces, and the resulting L for $\varepsilon = \frac{1}{5}$.

Checkpoint 12.12. Where exactly did Step 5 use irrationality? (Two places. Find both, and check the argument really does collapse for rational α — as it must, since periodic words are not mechanical-with-irrational-slope.)

13. THE STURMIAN ENDGAME

This section finishes the two-letter case: first the Sturmian walks (the hardest single proof in this document — take it slowly), then the boundary configuration that the width dichotomy could not reach, then the assembled two-letter theorem.

13.1. The Sturmian lemma. We keep the two-letter coordinates of §11 — letters $\delta < \delta' = \delta + e$ with $\gcd(\delta, e) = 1$ and $\delta' \leq d_2 \leq k$, binary word h , counting function q — and now assume, after Theorem 12.8 and a final tail-pass, that the word is Sturmian:

$$q_n = \lfloor n\alpha + \beta \rfloor \quad (n \geq 1)$$

for an irrational $\alpha \in (0, 1)$ and some real β .

Lemma 13.1 (Sturmian). *In this situation $kA - ka_1$ contains every sufficiently large integer; in particular A is tail-covering with modulus 1.*

This is stronger than the sweep gave us — *all* large integers, no residue class needed — and it has to be, because Sturmian words are exactly the words for which the band of §11 is too narrow for the sweep. The proof replaces “wide band” with something better: *exact control*. For a mechanical word the value of a configuration is not merely trapped in a window; it can be steered onto a precise target, because choosing indices means choosing points of an irrational rotation, and the hitting lemma lets us place those points wherever we like.

Proof. Recall (Lemma 8.1 and (5)) that it suffices to realize each large integer x as $x = \delta s + e w$ where w is a sum of k counting-function values $q_{i_1} + \cdots + q_{i_k}$ with $i_t \geq 1$ and $i_1 + \cdots + i_k = s$. The proof has four movements.

Step 0: where the values live. Write, for $n \geq 1$,

$$q_n = n\alpha + \beta - \{n\alpha + \beta\}$$

(floor = number minus fractional part). Summing over k indices with total s :

$$\sum_{t=1}^k q_{i_t} = X_s - \sum_{t=1}^k \{i_t\alpha + \beta\}, \quad X_s := s\alpha + k\beta. \quad (8)$$

Each fractional part lies in $[0, 1)$, so the sum of k of them lies in $[0, k)$: every achievable value sits in the real window $(X_s - k, X_s]$ — the two-letter avatar of Part II's window picture. But now we run the logic *forward*: to achieve a prescribed value w at depth $\theta := X_s - w$ below the window's top, it suffices to make the fractional parts sum to *exactly* θ .

Step 1: the tool. Lemma 12.10: for any arc length $\varepsilon > 0$ there is a window length $L(\alpha, \varepsilon)$ such that every L consecutive indices contain one whose rotation point $\{n\alpha + \beta\}$ lies in any prescribed arc of length ε — uniformly in the arc's position.

Step 2: hitting any depth in the middle of the window. Fix a margin $\eta \in (0, \frac{1}{2})$, to be chosen in Step 3. **Claim:** there is a threshold S_0 (depending on η, α, β, k only) such that for every $s \geq S_0$ and every integer w whose depth $\theta = X_s - w$ lies in $[\eta, k - \eta]$, there are *distinct* indices $i_1 < \dots < i_k$ (all ≥ 1 , summing to s) with

$$\sum_{t=1}^k \{i_t \alpha + \beta\} = \theta \quad \text{exactly,} \quad \text{hence} \quad \sum_{t=1}^k q_{i_t} = w.$$

The mechanism: place the first $k - 1$ points of the rotation in a narrow arc positioned so their sum falls just below θ , then let the forced last index close the books — with an integrality argument snapping the total onto θ exactly.

Set $\kappa := \eta/4$, and consider the open interval of candidate “per-point targets”

$$\mathcal{I} := (\max(0, \theta - 1) + \kappa, \min(k - 1, \theta) - \kappa).$$

Its length is at least $\eta/2$ whatever the regime of θ : for $\eta \leq \theta \leq 1$ it is $\theta - 2\kappa \geq \eta/2$; for $1 < \theta \leq k - 1$ it is $1 - 2\kappa > \eta/2$; for $k - 1 < \theta \leq k - \eta$ it is $(k - \theta) - 2\kappa \geq \eta/2$. Pick any $T^* \in \mathcal{I}$ and set the arc

$$J := \left(\frac{T^*}{k-1} - \frac{\kappa}{k-1}, \frac{T^*}{k-1} + \frac{\kappa}{k-1} \right) \subseteq (0, 1),$$

of length $\varepsilon_0 := 2\kappa/(k-1)$, and let $L := L(\alpha, \varepsilon_0)$ be its hitting constant. The arc’s *position* depends on θ (through T^*), but its *length* does not — and the hitting lemma is uniform in position, so this single L serves every target and every s . (This is exactly why we insisted on uniformity in §12.)

Choosing the free indices. For $j = 1, \dots, k - 1$, the hitting lemma finds an index i_j in the block $\{(j-1)L + 1, \dots, jL\}$ with $\{i_j \alpha + \beta\} \in J$. Different blocks, so $i_1 < i_2 < \dots < i_{k-1}$, with

$$i_j \leq jL, \quad \sum_{j=1}^{k-1} i_j \leq L \frac{k(k-1)}{2}, \quad i_{k-1} \leq (k-1)L. \quad (9)$$

The forced last index. Set

$$S_0 := 1 + L(k-1) \left(\frac{k}{2} + 1 \right)$$

(an integer: $(k-1)(k+2)/2$ has one even factor) and, for $s \geq S_0$, let $i_k := s - (i_1 + \dots + i_{k-1})$. By (9),

$$i_k \geq S_0 - L \frac{k(k-1)}{2} = 1 + L(k-1) \left(\frac{k}{2} + 1 - \frac{k}{2} \right) = 1 + (k-1)L > i_{k-1},$$

so all k indices are distinct and ≥ 1 , with total exactly s .

The snap. Let $F := \sum_{j \leq k-1} \{i_j \alpha + \beta\}$; since each term lies in J ,

$$F \in (T^* - \kappa, T^* + \kappa) \subseteq (\max(0, \theta - 1), \min(k - 1, \theta)),$$

the inclusion because $T^* \in \mathcal{I}$ keeps distance $> \kappa$ from both endpoints. Add the last point: $\Theta := F + \{i_k \alpha + \beta\}$, with the fractional part in $[0, 1)$, satisfies

$$\Theta > \max(0, \theta - 1) \geq \theta - 1 \quad \text{and} \quad \Theta < \min(k - 1, \theta) + 1 \leq \theta + 1.$$

On the other hand, by (8) applied to our indices,

$$\Theta = X_s - \sum_{t \leq k} q_{i_t},$$

and both $X_s - \theta = w$ and $\sum q_{i_t}$ are integers, so Θ differs from θ by an integer. An integer-shift of θ lying strictly between $\theta - 1$ and $\theta + 1$ must be θ itself: $\Theta = \theta$, and $\sum_t q_{i_t} = X_s - \theta = w$. The Claim is proved.

Step 3: the ladder. Now fix the margin

$$\eta := \frac{1}{4} \min(\alpha, 1 - \alpha) \in (0, \frac{1}{8}],$$

and let S_0 be the resulting threshold. Let x be a large integer. As in the sweep, the budgets s able to represent $x = \delta s + ew$ with integer w form one residue class modulo e (the *admissible* s), and for admissible s we set $w(s) := (x - \delta s)/e$ and measure the depth

$$\theta(s) := X_s - w(s) = s\alpha + k\beta - \frac{x - \delta s}{e}.$$

Per admissible step $s \mapsto s + e$, the value X_s grows by $e\alpha$ and $w(s)$ falls by δ , so the depth climbs by exactly

$$\psi := e\alpha + \delta > 0$$

— an arithmetic ladder. Step 2 accepts any depth in the strip $[\eta, k - \eta]$, of height $k - 2\eta$. The ladder cannot step over the strip, because its rung spacing is smaller:

$$\psi = \delta + e\alpha = \delta' - e(1 - \alpha) \leq k - (1 - \alpha) < k - 2\eta,$$

using $\delta' \leq k$, $e \geq 1$, and $2\eta \leq \frac{1}{2}(1 - \alpha) < 1 - \alpha$. Since $\theta(s)$ is very negative for small admissible s (there $w(s) \approx x/e$ dominates) and climbs unboundedly by steps $\psi < k - 2\eta$, some admissible s^* lands in the strip: $\theta(s^*) \in [\eta, k - \eta]$.

The thresholds, explicitly. Multiplying the definition of θ by e and rearranging: $s\psi = x + e\theta(s) - ek\beta$. So any s^* with $\theta(s^*) \in [0, k]$ satisfies

$$\frac{x - ek|\beta|}{\psi} \leq s^* \leq \frac{x + ek(1 + |\beta|)}{\psi}.$$

Two requirements: (a) $s^* \geq S_0$, so Step 2 applies — since $\psi \leq \delta + e = \delta' \leq k$, the left bound gives this once $x \geq kS_0 + ek|\beta|$. (b) $w(s^*) \geq 0$, i.e. $\delta s^* \leq x$ — by the right bound this holds once $\delta(x + ek(1 + |\beta|)) \leq (\delta + e\alpha)x$, i.e. once $x \geq \delta k(1 + |\beta|)/\alpha$ (here $\alpha > 0$ earns its keep). So for

$$x \geq X_0 := 1 + \max\left(kS_0 + ek|\beta|, \frac{\delta k(1 + |\beta|)}{\alpha}\right),$$

Step 2 realizes $w(s^*) \in W_{s^*}$ with legal indices, and

$$x = \delta s^* + ew(s^*) \in kA - ka_1. \quad \square$$

Big idea. For a Sturmian walk the window picture stops being an estimate and becomes an equation: the depth of a value below the window's top *equals* the sum of k rotation points on the circle. The hitting lemma lets us place $k - 1$ of those points at will; integrality snaps the total onto the exact target; and the irrationality of the slope makes the ladder's rung $\psi = \delta + e\alpha$ incommensurable-with-nothing — just strictly shorter than the acceptance strip, so no target escapes.

Example 13.2 (The Fibonacci walk, traced). Take $k = 3$, letters $\{1, 2\}$ (so $\delta = e = 1$, every s admissible), and the Fibonacci gap word: $q_n = \lfloor n\alpha + \beta \rfloor$ with $\alpha = (3 - \sqrt{5})/2 \approx 0.382$ and $\beta = \alpha$. Then $\eta = \alpha/4 \approx 0.095$ and the ladder climbs by $\psi = 1 + \alpha \approx 1.382$ per step — comfortably below the strip height $3 - 2\eta \approx 2.81$. To represent $x = 100$: $\theta(s) = (s + 3)\alpha - 100 + s$ climbs by ψ and first enters $[\eta, 3 - \eta]$ at $s^* = 72$, where $\theta(72) = 75\alpha - 28 \approx 0.647$. So

$w = w(72) = 100 - 72 = 28$, and Step 2 must find three distinct indices with $i_1 + i_2 + i_3 = 72$ and $q_{i_1} + q_{i_2} + q_{i_3} = 28$ — three rotation points summing to depth 0.647. (One valid triple: $i = (10, 26, 36)$, whose counting values are $4 + 10 + 14 = 28$ — check them against $q_n = \lfloor (n+1)\alpha \rfloor$.) Then $100 = 1 \cdot 72 + 1 \cdot 28 \in 3A - 3a_1$.

13.2. The boundary: k even, $d_2 = k$. One configuration escaped the width dichotomy: k even at the exact boundary $d_2 = k$, where the pairing argument came up one unit of width short. The rescue is a lemma that squeezes width out of *unbalancedness itself*, with a beautiful detour through palindromes.

First, a two-line classic:

Lemma 13.3 (Border–period duality). *Let u be a finite word of length τ with a border of length ℓ : its prefix of length ℓ equals its suffix of length ℓ . Then u has period $\tau - \ell$: $u_i = u_{i+\tau-\ell}$ for all $1 \leq i \leq \ell$.*

Proof. The border condition says $u_i = u_{(\tau-\ell)+i}$ for $1 \leq i \leq \ell$ — which is the period condition, re-read. $\square$

Lemma 13.4 (Boundary width). *Let $k \geq 4$ be even. Suppose the binary word h uses both letters infinitely often, is not eventually periodic, and has the property that every tail of h is unbalanced. Then $\text{width}(s) \geq k - 1$ for all sufficiently large s .*

Proof. First, unbalancedness of every tail yields two-index width ≥ 2 infinitely often. Suppose instead $W_2(\sigma) \leq 1$ for all $\sigma \geq \sigma^*$. Take any two equal-length windows of the tail of h beyond position σ^* , say from i to i' and from j' to j ; then $\sigma := i + j = i' + j' \geq \sigma^*$, and exactly as in Lemma 11.3(b), $W_2(\sigma) \leq 1$ bounds the difference of their 1-counts by 1. So that tail would be balanced — contradiction. Hence $W_2(\sigma) \geq 2$ for infinitely many σ ; fix two such, $\sigma_0 < \sigma_1$, and set $\Delta := \sigma_1 - \sigma_0 \geq 1$.

Second, suppose the width bound fails at some large s : $\text{width}(s) \leq k - 2$. Play the two-family game of Lemma 11.3(a), but with a twist. Family one: $k/2 - 1$ extremal pairs at σ_0 plus a free pair at $\tau := s - (k/2 - 1)\sigma_0$; it forces $\text{width}(s) \geq (k - 2) + W_2(\tau)$, so $W_2(\tau) = 0$. Family two ($k \geq 4$ makes room): $k/2 - 2$ pairs at σ_0 , one pair at σ_1 , and a free pair at $\tau - \Delta$; it forces $W_2(\tau - \Delta) = 0$ likewise.

Third, zero two-index width means palindrome. $W_2(\tau) = 0$ says $q_i + q_{\tau-i}$ is constant over $0 \leq i \leq \tau$; differencing consecutive i gives $h_{i+1} = h_{\tau-i}$ for $0 \leq i \leq \tau - 1$ — the prefix $h[1..\tau]$ reads the same forwards and backwards. So both $h[1..\tau]$ and $h[1..\tau - \Delta]$ are palindromes. A palindromic prefix of a palindrome is also its suffix (reverse the whole word: it is unchanged, and the prefix's mirror image — itself — appears at the end). So $h[1..\tau - \Delta]$ is a border of $h[1..\tau]$, and by Lemma 13.3, $h[1..\tau]$ has period Δ .

Finally, infinitely many failures would force global periodicity. If $\text{width}(s) \leq k - 2$ for infinitely many s , then — Δ being fixed while $\tau = \tau(s) \rightarrow \infty$ — we obtain arbitrarily long prefixes of h with the same period Δ . Any fixed position n is eventually covered with room to spare, so $h_n = h_{n+\Delta}$ for every n : the whole word is periodic, contradicting the hypothesis. Hence $\text{width}(s) \geq k - 1$ for all large s . $\square$

Proposition 13.5 (Boundary routing). *Let k be even, $d_2 = k$, and let A have two-letter tail alphabet (the standing setup of §11). Then A is tail-covering.*

Proof. Route by the first case that applies; each closes the matter. If the gap word is eventually periodic: Lemma 9.1. Otherwise, if some tail of the gap word is balanced: pass to that tail (Lemma 8.2) and classify it (Theorem 12.8) — eventually periodic (Lemma 9.1 again) or Sturmian on a further tail (Lemma 13.1); note neither route used any width argument.

Otherwise *every* tail is unbalanced and the word is not eventually periodic (and $k \geq 4$, being even and ≥ 3): Lemma 13.4 gives $\text{width}(s) \geq k - 1 = d_2 - 1$ for large s , and the sweep (Lemma 11.2) finishes. $\square$

13.3. The two-letter theorem.

Theorem 13.6 (Two letters). *Let $k \geq 3$ and $d_2 \leq k$. Every (d_1, d_2) -sequence whose tail alphabet has at most two letters is tail-covering.*

Proof. One letter: Lemma 8.3. Two letters: rescale to $\gcd = 1$ (Lemma 8.4). If k is even and $d_2 = k$: Proposition 13.5. Otherwise k is odd or $d_2 \leq k - 1$, and we consult the two-index widths: if $W_2(\sigma_0) \geq 2$ for some σ_0 , then Lemma 11.3(a) plus the sweep (Lemma 11.2) conclude; if $W_2(\sigma) \leq 1$ for all σ , the word is balanced (Lemma 11.3(b)), hence eventually periodic (Lemma 9.1) or Sturmian on a tail (Theorem 12.8, then Lemma 13.1). Every branch ends in tail-covering. $\square$

Checkpoint 13.7. In Example 13.2, verify the claimed triple: generate the Fibonacci word ($0 \mapsto 01, 1 \mapsto 0$, starting from 0), compute q_{10}, q_{25}, q_{37} , and check the two equations. Then find a different valid triple for the same $x = 100$.

Checkpoint 13.8. Why can the Sturmian lemma not simply reuse the sweep? Locate the precise failure: show that for a Sturmian word, $\text{width}(s) \leq$ some bound that falls short of $d_2 - 1$ in the critical case $\delta' = d_2$. (Hint: Step 0 traps W_s in a real window of length k ; how many integers can it contain, and what does balancedness do to the endpoints?)

14. THREE OR MORE LETTERS: THE SLOT LEMMA

One case of the battle plan remains: tail alphabets with three or more letters. The band-and-sweep machinery does not survive the trip — with three letters, a single move between configurations changes the value by a *difference of letters*, which can exceed 1, so the reachable set W_s can have holes and Lemma 11.1 fails. Instead of repairing the band we change weapons entirely: a small combinatorial gadget — the *slot lemma* — that spends the k summands like a budget, and whose exact price is the finite theorem of Part IV.

14.1. Subset sums of multisets.

Definition 14.1. A *multiset* S drawn from a set G is a collection of elements of G with repetitions allowed ($|S|$ counts repetitions). Its *subset sums* are

$$\text{sums}(S) := \left\{ \sum_{x \in T} x : T \text{ a sub-multiset of } S \right\}$$

(including the empty sum 0). For a finite set G of positive integers with $M := \max G$, define $m(G) := \min \{ |S| : S \text{ a multiset drawn from } G \text{ whose subset sums contain } M \text{ consecutive integers} \}$.

Example 14.2. $G = \{3, 4, 5\}, S = \{3, 3, 4, 5\}$: the subset sums are $\{0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\}$, which contain the run $3, 4, \dots, 12$ — ten consecutive integers, amply covering $M = 5$. So $m(\{3, 4, 5\}) \leq 4$. (Part IV will show 4 is optimal here.)

Part IV is devoted to proving:

The bounded subset-sum covering lemma (SHARP). *Every finite set G of at least three positive integers with $\gcd(G) = 1$ and $\max(G) = M$ has $m(G) \leq M - 1$.*

For now we take it on credit and show exactly how the non-existence proof spends it.

14.2. The slot lemma.

Lemma 14.3 (Slot lemma). *Let A have tail alphabet G_∞ with $|G_\infty| \geq 3$ and $\gcd(G_\infty) = 1$ (after rescaling), and put $M := \max G_\infty \leq d_2$. If $k - 1 \geq m(G_\infty)$, then A is tail-covering.*

Proof. Pass to the tail (Lemma 8.2): all gaps lie in G_∞ , all letters recur infinitely often, and we work with prefix sums via Lemma 8.1. Let S be a multiset realizing $m(G_\infty) =: m \leq k - 1$: $|S| = m$, drawn from G_∞ , with

$$\text{sums}(S) \supseteq \{c, c + 1, \dots, c + M - 1\}$$

for some c . Enumerate S as $\delta_1, \dots, \delta_m$ (values in G_∞ , possibly repeating).

The fine slots. Because each letter of G_∞ occurs infinitely often in the gap word, we can choose m distinct positions $n_1 < n_2 < \dots < n_m$ (all beyond the tail start) with

$$g_{n_t} = \delta_t \quad (t = 1, \dots, m).$$

Summand t of our k -fold sum will be confined to two adjacent choices: index $n_t - 1$ (“off”), contributing the prefix sum p_{n_t-1} , or index n_t (“on”), contributing $p_{n_t} = p_{n_t-1} + \delta_t$. Toggling slot t from off to on adds exactly δ_t . Over all 2^m on/off patterns, the m slots jointly contribute

$$\left(\sum_{t=1}^m p_{n_t-1} \right) + \text{sums}(S) \supseteq \text{base}_0 + [c, c + M - 1], \quad \text{base}_0 := \sum_t p_{n_t-1} :$$

a full block of M consecutive achievable offsets — this is where the multiset’s covering run is cashed in.

The parked summands. We have used m of our k summands. Park the next $k - 1 - m$ at any fixed tail index (repetition of indices is legal in the sumset), folding their fixed contribution into the base: $\text{base} := \text{base}_0 + (\text{parked total})$. Now $k - 1$ summands realize every value of $\text{base} + [c, c + M - 1]$.

The coarse dial. The final summand is an index n of our choosing, contributing p_n . In the tail, consecutive prefix sums differ by a gap $g_{n+1} \in G_\infty \subseteq [1, M]$, so $n \mapsto p_n$ is strictly increasing with steps of size at most M , tending to infinity: a coarse dial that ascends forever but can jump by as much as M at a time.

The first-crossing argument. Fix a large target x and let

$$\phi(n) := x - \text{base} - p_n,$$

which decreases strictly in n , in steps of at most M , from a starting value $\phi(T) > c + M - 1$ (true once x is large, the base and T being fixed) down toward $-\infty$. Let n^* be the first index past T with $\phi(n^*) \leq c + M - 1$. By firstness, $\phi(n^* - 1) > c + M - 1$; and one step drops ϕ by at most M , so

$$\phi(n^*) > (c + M - 1) - M = c - 1, \quad \text{i.e.} \quad \phi(n^*) \geq c.$$

So $\phi(n^*)$ lands *inside* the achievable block $[c, c + M - 1]$: some on/off pattern realizes the offset $\phi(n^*)$, and then

$$x = \text{base} + \phi(n^*) + p_{n^*}$$

exhibits x as a sum of k prefix sums — m slots, $k - 1 - m$ parked, one dial. (The dial index may coincide with a slot or parked index; harmless, repetitions are allowed.) Hence $kA - ka_1$ contains every large integer, and A is tail-covering (with the factor g restored by Lemma 8.4 if rescaling was applied). $\square$

Watch the accounting at the binding corner. The dial can step by M , so the achievable block must be at least M long — a run of $M - 1$ consecutive offsets would let the dial leap clean over it. The block costs $m(G_\infty)$ summands, the dial costs one, and only k are available:

$$m(G_\infty) \leq k - 1.$$

When the Adversary's walk uses the largest legal gap infinitely often and the parameters sit at the corner $k = d_2 = M$, this requirement becomes exactly

$$m(G_\infty) \leq M - 1$$

— SHARP, with nothing to spare. The whole weight of the corner case rests on that finite statement, which is why Part IV exists. Away from the corner there is slack: $m(G_\infty) \leq M - 1 \leq d_2 - 1 \leq k - 1$ holds a fortiori.

For perspective — not needed below — a much cruder bound is easy to prove and shows the mechanism in miniature:

Lemma 14.4 (Base grid). *Let G have $|G| \geq 3$, $\gcd(G) = 1$, $a := \min G$, $M := \max G$, and suppose some $c \in G$ has $\gcd(a, c) = 1$. Then $m(G) \leq a + M - 1 \leq 2M - 3$.*

Proof. Take $a - 1$ copies of c and $p := \lceil ((a - 1)c + M - 1)/a \rceil$ copies of a . For any $n \in [(a - 1)c, pa]$: the residues $0, c, 2c, \dots, (a - 1)c$ modulo a are pairwise distinct (as $\gcd(a, c) = 1$), so a unique $i \in [0, a - 1]$ has $ic \equiv n \pmod{a}$; then $ic \leq (a - 1)c \leq n$, and $(n - ic)/a$ is an integer in $[0, p]$, so $n = ic + a \cdot \frac{n - ic}{a}$ is a subset sum. The interval has length $pa - (a - 1)c + 1 \geq M$ by the choice of p . The size: $c \leq M$ gives $p \leq \lceil ((a - 1)M + M - 1)/a \rceil = \lceil (aM - 1)/a \rceil = M$, so $(a - 1) + p \leq a + M - 1$; and $|G| \geq 3$ squeezes $a \leq M - 2$. $\square$

The gap between $2M - 3$ and SHARP's $M - 1$ is the gap between an evening's exercise and Part IV; note also the extra hypothesis, which genuinely can fail ($G = \{6, 10, 15\}$ has $\gcd 1$ but no element coprime to 6).

14.3. The non-existence half, assembled (modulo SHARP).

Proposition 14.5 (Goal of Part III). *Assume SHARP. Then every (d_1, d_2) -sequence with $d_2 \leq k$ and $k \geq 3$ is tail-covering.*

Proof. Pass to the tail; rescale so $\gcd(G_\infty) = 1$ (Lemmas 8.2, 8.4). If $|G_\infty| \leq 2$: Theorem 13.6. If $|G_\infty| \geq 3$: SHARP applies to G_∞ (three or more elements, $\gcd 1$) and gives $m(G_\infty) \leq \max(G_\infty) - 1 \leq d_2 - 1 \leq k - 1$, so the slot lemma (Lemma 14.3) concludes. $\square$

Combined with the certificate (Lemma 7.5), this proves the strong non-existence half of Theorem A — modulo SHARP. The debt is now fully explicit, and Part IV pays it.

Checkpoint 14.6. (a) Show $m(\{1, 2, 3\}) \leq 2$ and $m(\{2, 3, 4\}) \leq 3$, with explicit multisets. (b) Trace the slot lemma concretely: $k = 4$, $d_2 = 4$, $G_\infty = \{2, 3, 4\}$ (so $M = 4$), gap word cycling $2, 3, 4, 2, 3, 4, \dots$ — pick the multiset from (a), choose slot positions inside the first two periods, and find the dial index and slot pattern realizing $x = 57$ (with a_1 of your choosing).

Checkpoint 14.7. The parked summands were placed at one fixed index. Would the proof survive parking them at index 0 (contributing $p_0 = 0$)? And why does the argument insist the block have length M rather than $M - 1$ — exhibit a concrete failure: a dial whose steps are all M leaping over a block of length $M - 1$.

Part 4. The subset-sum lemma

15. THE SUBSET-SUM LEMMA: STATEMENT, SHARPNESS, AND THE MAP

Part IV is a self-contained piece of finite mathematics. It never mentions walks, sumsets of sequences, or lacunarity; it proves exactly the statement that Part III left as a debt. Recall $m(G)$ from Definition 14.1: the smallest size of a multiset drawn from G whose subset sums contain $\max(G)$ consecutive integers.

Theorem 15.1 (Bounded subset-sum covering — “SHARP”). *Every finite set G of positive integers with $|G| \geq 3$, $\gcd(G) = 1$ and $\max(G) = M$ satisfies*

$$m(G) \leq M - 1.$$

Before proving it (§§16–23), let us be precise about what the bound claims, and check that neither hypothesis nor constant can be weakened.

15.1. First moves, and sharpness. If $1 \in G$, take $M - 1$ copies of 1: the subset sums are $0, 1, \dots, M - 1$ — exactly M consecutive integers at budget $M - 1$. So assume from now on $a := \min G \geq 2$.

The constant cannot be improved to $M - 2$. Take $G = \{3, 4, 5\}$, so $M = 5$. Example 14.2 showed $m(G) \leq 4$ via $\{3, 3, 4, 5\}$; we claim no multiset of size ≤ 3 works, so $m(\{3, 4, 5\}) = 4 = M - 1$ exactly. This is a finite check — there are 3 multisets of size one, 6 of size two, and 10 of size three — and the size-three table settles it (smaller multisets have fewer sums and do no better):

Multiset	Subset sums	Longest run
$\{3, 3, 3\}$	0, 3, 6, 9	1
$\{3, 3, 4\}$	0, 3, 4, 6, 7, 10	2
$\{3, 3, 5\}$	0, 3, 5, 6, 8, 11	2
$\{3, 4, 4\}$	0, 3, 4, 7, 8, 11	2
$\{3, 4, 5\}$	0, 3, 4, 5, 7, 8, 9, 12	3
$\{3, 5, 5\}$	0, 3, 5, 8, 10, 13	1
$\{4, 4, 4\}$	0, 4, 8, 12	1
$\{4, 4, 5\}$	0, 4, 5, 8, 9, 13	2
$\{4, 5, 5\}$	0, 4, 5, 9, 10, 14	2
$\{5, 5, 5\}$	0, 5, 10, 15	1

The best run of consecutive integers achievable with three elements is 3, short of the required $M = 5$.

The hypothesis $|G| \geq 3$ cannot be dropped. Take $G = \{2, 3\}$: $\gcd = 1$, $M = 3$, and the target would be $m(G) \leq 2$. The five nonempty multisets of size ≤ 2 have subset sums

$$\{0, 2\}, \quad \{0, 3\}, \quad \{0, 2, 4\}, \quad \{0, 2, 3, 5\}, \quad \{0, 3, 6\},$$

and none contains 3 consecutive integers. So two-element sets genuinely need more room; three elements is where the uniform budget $M - 1$ becomes available.

Finally, a caution about what SHARP does *not* say: it is not a formula for $m(G)$, which is usually far smaller than $M - 1$ (for $G = \{1, 2, M\}$ it is tiny — see the checkpoint). It is a *uniform budget*, tuned to be exactly what the slot lemma’s corner $k = d_2 = M$ can afford — recall §14: one dial summand plus at most $M - 1$ slot summands is all that $k = M$ pays for.

Checkpoint 15.2. (a) Compute $m(\{1, 2, M\})$ for general $M \geq 3$. (b) A multiset of size s has at most 2^s subset sums, and M consecutive integers are M distinct sums; deduce the

elementary lower bound $m(G) \geq \log_2 M$. Then find a three-element family of sets G (one for each large M) whose $m(G)$ is genuinely logarithmic-ish in M ... or convince yourself that with only three generators this is impossible, and explain which structural feature of G controls how far below $M - 1$ the truth sits.

15.2. The shape of the proof. The whole of Part IV runs on a single covering mechanism, previewed here and built in §16. To cover M consecutive integers with copies of elements of G :

- (1) *Cover the residues.* Using a bounded number of copies of the two largest generators, hit every residue class modulo a (the smallest generator) at least once — a *box* of representatives.
- (2) *Slide and stack.* Add $0, 1, 2, \dots, x$ copies of a : each representative slides up an arithmetic progression of step a , and the progressions interleave into a solid run once every class is represented and the run is long enough.

The cost accounting — how many copies of a suffice, and whether the total stays under $M - 1$ — is the *budget*, and the entire difficulty of Part IV is that near the corner the budget has no slack at all.

The proof organizes as follows.

- §16: the covering machinery — a two-generator interval lemma, the frame lemma (the slide-and-stack step made exact), the staircase (a refined multi-level version), and the λ -lift (a way to certify infinitely many triples with one finite check).
- §17: reduction of the general statement to *hard-core triples* $\{a, b, M\}$ with $\gcd(a, b) = 1$ and b close to M (precisely $1 \leq M - b \leq a - 2$).
- §§18–22: the hard core is carved by simple arithmetic invariants into six branches — D, P, L, E, T, B — four closed by explicit formulas valid across their whole range, two (T and B) closed by formulas *plus* a finite list of exceptional triples, each carrying a small computer-verified certificate. §25 explains exactly what was computed and how it is checked.
- §23: assembly, exhaustiveness of the six branches, and the sharpness proposition.

The six-way split is by two divisibility questions and two size questions, routed in order (evaluate top to bottom, take the first that applies):

Branch	Condition	Mechanism	Closure
D	$a \mid M$	b spans the residues alone	formula
P	$a \mid b + M$	(b, M) pairs as residue-neutral movers	formula
L	$e = h$ (equal spacings)	staircase, extended form	formula
E	$a \geq 12$ and $\mu \geq 12$	the η -box	formula
T	$\mu \leq 11$	staircase lines; 172 exceptions	formula + table
B	$a \leq 11, \mu \geq 12$	178 base certificates, lifted	table + lift

(The symbols e, h, μ are set up in §16; think “ μ small” = the generators are crowded near M , “ a small” = the smallest generator is tiny.)

16. MACHINERY: GRIDS, FRAMES, STAIRCASES, AND THE LIFT

Everything in Part IV covers runs of consecutive integers by the same two-phase mechanism sketched in §15.2. This section builds it once, with exact constants, so the case sections can spend their attention on budgets.

16.1. The two-generator grid. How much of the number line do two coprime generators cover, using boundedly many copies of each? The classic “postage stamp” fact — with stamps of two coprime denominations α, β and unlimited copies, every amount from $(\alpha - 1)(\beta - 1)$ on is payable — has the following bounded-copies version.

Lemma 16.1 (Two-generator interval). *Let $\alpha < \beta$ be coprime positive integers, and let $x \geq \beta - 1$ and $y \geq \alpha - 1$ be integers. Then, with $C := (\alpha - 1)(\beta - 1)$,*

$$\{i\alpha + j\beta : 0 \leq i \leq x, 0 \leq j \leq y\} \supseteq [C, \alpha x + \beta y - C].$$

Note the symmetry of the conclusion: the grid covers everything except a margin of C at each end of its full span $[0, \alpha x + \beta y]$.

Proof. Case $\alpha = 1$. Here $C = 0$. For $n \in [0, x + \beta y]$ put $j := \min(y, \lfloor n/\beta \rfloor)$. If $j = \lfloor n/\beta \rfloor$ then $n - j\beta$ is the remainder, in $[0, \beta - 1] \subseteq [0, x]$; if instead $j = y$ then $n - y\beta \leq x$ directly. Either way $i := n - j\beta \in [0, x]$.

Case $\alpha \geq 2$, base $y = \alpha - 1$. Let $n \in [C, \alpha x + \beta(\alpha - 1) - C]$. Since $\gcd(\alpha, \beta) = 1$, the multiples $0, \beta, 2\beta, \dots, (\alpha - 1)\beta$ hit all residues modulo α ; choose the unique $j \in [0, \alpha - 1]$ with $j\beta \equiv n \pmod{\alpha}$. Then $n - j\beta$ is divisible by α and

$$n - j\beta \geq C - (\alpha - 1)\beta = -(\alpha - 1);$$

a multiple of α that is $\geq -(\alpha - 1)$ is ≥ 0 . And with $i := (n - j\beta)/\alpha$:

$$n - j\beta \leq (\alpha x + \beta(\alpha - 1) - C) - j\beta \leq \alpha x + (\alpha - 1)\beta - C = \alpha x + (\alpha - 1),$$

so $i \leq x + \frac{\alpha-1}{\alpha} < x + 1$, i.e. $i \leq x$.

Induction $y \rightarrow y + 1$ (for $y \geq \alpha - 1$). Only the new stretch $n \in (\alpha x + \beta y - C, \alpha x + \beta(y + 1) - C]$ needs attention. Subtract one β : certainly $n - \beta \leq \alpha x + \beta y - C$; and $n - \beta > \alpha x + \beta y - C - \beta \geq C$, because

$$\alpha x + \beta y \geq \alpha(\beta - 1) + \beta(\alpha - 1) = 2\alpha\beta - \alpha - \beta \geq 2C + \beta$$

— the last inequality simplifies to $\alpha \geq 2$. So $n - \beta$ lies in the already-covered interval: $n - \beta = i\alpha + j\beta$ with $i \leq x, j \leq y$, and $n = i\alpha + (j + 1)\beta$ with $j + 1 \leq y + 1$. $\square$

Checkpoint 16.2. For $(\alpha, \beta) = (3, 5)$, $x = 4$, $y = 2$: list the grid and verify the covered interval $[8, 14]$ predicted by the lemma — and note which numbers outside it are missed. Where is coprimality used in the proof, and what fails for $(\alpha, \beta) = (2, 4)$?

16.2. The frame lemma, and budgets. The second phase — slide and stack — in its general form:

Lemma 16.3 (Frame lemma). *Let ν, g_1, g_2 be positive integers and $L \geq 1$ a target run length. Suppose a box of size (Y, Z) covers all residues modulo ν : for every residue ρ there are $j_\rho \leq Y$, $k_\rho \leq Z$ with $j_\rho g_1 + k_\rho g_2 \equiv \rho \pmod{\nu}$. Let*

$$S := \max_{\rho} (j_\rho g_1 + k_\rho g_2), \quad x := \left\lceil \frac{L - 1 + S}{\nu} \right\rceil.$$

Then the multiset of Y copies of g_1 , Z copies of g_2 and x copies of ν — size $Y + Z + x$ — has subset sums covering the entire integer interval $[S, \nu x]$, which contains $[S, S + L - 1]$: a run of length L .

Proof. Let $n \in [S, \nu x]$ and $\rho := n \bmod \nu$. The representative $r := j_\rho g_1 + k_\rho g_2$ satisfies $r \leq S \leq n$ and $r \equiv n \pmod{\nu}$, so $t := (n - r)/\nu$ is a nonnegative integer, and $t \leq n/\nu \leq x$. Take j_ρ copies of g_1 , k_ρ of g_2 , and t of ν . Finally $\nu x \geq L - 1 + S$ by the choice of x , so the covered interval contains $[S, S + L - 1]$. $\square$

In every use below, $\nu = a = \min G$, $(g_1, g_2) = (b, M)$, and $L = M$. The *budget* of a box (Y, Z) for the triple (a, b, M) is the resulting multiset size,

$$\mathcal{B} = \left\lceil \frac{M-1+S}{a} \right\rceil + Y + Z,$$

and a box *certifies* the triple when it covers all residues, $Y + Z \leq a - 1$, and $\mathcal{B} \leq M - 1$. The case analysis is, at bottom, a hunt for certifying boxes; the exceptional tables of cases T and B are nothing but lists of such boxes, checkable by the two-line computation in this proof.

16.3. Standing notation for triples. From §17 onward we work with triples $G = \{a, b, M\}$, $a < b < M$, $\gcd(a, b) = 1$. Fix, once and for all,

$$\begin{aligned} e &:= b - a, & h &:= M - b, & \mu &:= M - a = e + h, & g &:= \gcd(e, \mu), \\ e' &:= e/g, & \mu' &:= \mu/g, & C' &:= (e' - 1)(\mu' - 1). \end{aligned}$$

(Note $\gcd(e, \mu) = \gcd(e, h)$, and e', μ' are coprime with $\mu' > e'$, so $\mu' \geq 2$ whenever $h \geq 1$.) One fact used repeatedly:

$$\gcd(a, g) = 1 :$$

a common prime of a and g would divide e and a , hence $b = a + e$, contradicting $\gcd(a, b) = 1$.

16.4. Levels and staircases. Consider a multiset of x copies of a , y of b , z of M , with the standing hypotheses

$$y \geq \mu' - 1, \quad z \geq e' - 1;$$

put $c := y + z$ and $V' := ye' + z\mu' - C'$. Group its subset sums by the *number of elements used*: choosing i copies of a , j of b , k of M with $i + j + k = t$ gives

$$ia + jb + kM = ta + je + k\mu,$$

so *level* t ($0 \leq t \leq x + y + z$) consists of $ta + \mathcal{O}_t$ with offset set

$$\mathcal{O}_t = \{je + k\mu : j \leq y, k \leq z, \max(0, t - x) \leq j + k \leq t\}.$$

Each level is a shifted copy of (part of) the same two-generator grid — a *frame* — and consecutive levels are offset by exactly a . Frames merging into solid runs is the entire game:

Lemma 16.4 (Staircase). *In the setting above:*

- (a) For every t with $c \leq t \leq x$: $\mathcal{O}_t \supseteq g \cdot [C', V']$ (meaning $\{gu : C' \leq u \leq V'\}$).
- (b) (Two-frame merge; $g = 1$.) If $x = c + 1$ and $V' - C' \geq a - 1$, the subset sums cover the interval $[ca + C', (c + 1)a + V']$, of length $a + (V' - C') + 1$.
- (c) (Short merge; $g = 1$.) If $x = c$, then $\mathcal{O}_{c+1} \supseteq [\max(C', 1), V']$, and if additionally $V' \geq a + \max(C', 1) - 1$ the subset sums cover $[ca + C', (c + 1)a + V']$.
- (d) (Phase union; $g \geq 2$.) If $x = c + g$, the subset sums cover $[(c + g - 1)a + gC', ca + gV']$ with no further hypothesis (the base form). If moreover $x \geq c + g$ and the frame-merge condition $V' - C' \geq a - 1$ holds, they cover $[(c + g - 1)a + gC', (x - g + 1)a + gV']$ (the extended form).

Proof. (a) For $t \in [c, x]$, every box pair (j, k) ($j \leq y, k \leq z$) is admitted at level t : $j + k \leq c \leq t$ and $i = t - j - k$ lies in $[0, x]$. So $\mathcal{O}_t \supseteq \{je + k\mu\} = g \{je' + k\mu'\}$, and Lemma 16.1 applied to the coprime pair (e', μ') — its hypotheses are exactly $y \geq \mu' - 1, z \geq e' - 1$ — covers $[C', V']$.

(b) Levels c and $c + 1$ both lie in $[c, x]$, so each covers $ta + [C', V']$; two intervals offset by a merge into one exactly when $ca + V' \geq (c + 1)a + C' - 1$, i.e. $V' - C' \geq a - 1$.

(c) At level $c + 1 = x + 1$, the admission constraint is $j + k \geq t - x = 1$: only the pair $(0, 0)$ — offset 0 — is lost. Every offset in $[\max(C', 1), V']$ is nonzero, so all its grid representations survive. The merge then needs $ca + V' \geq (c + 1)a + \max(C', 1) - 1$.

(d) *Base form.* All levels $t \in [c, c + g]$ satisfy (a). Level t 's values $ta + g[C', V']$ lie in the residue class of ta modulo g , and since $\gcd(a, g) = 1$ the map $t \mapsto ta \bmod g$ is a bijection on any g consecutive levels: each residue class modulo g is served by exactly one $t_0 \in [c, c + g - 1]$. Given n in the stated interval, let t_0 be its server and $u := (n - t_0a)/g$; then $u \geq C'$ (since $n \geq (c + g - 1)a + gC' \geq t_0a + gC'$) and $u \leq V'$ (since $n \leq ca + gV' \leq t_0a + gV'$), so level t_0 realizes n .

Extended form. Within the class served by t_0 , the frames at levels $t_0, t_0 + g, t_0 + 2g, \dots$ (all $\leq x$) are arithmetic progressions of step g offset by ga from one another; consecutive frames leave no class-gap exactly when $(t_0 + g)a + gC' \leq t_0a + gV' + g$, i.e. when $V' - C' \geq a - 1$ — the frame-merge condition. Under it, the class is covered solidly from $t_0a + gC'$ up to $(t_0 + Jg)a + gV'$, where $J := \lfloor (x - t_0)/g \rfloor$. Over $t_0 \in [c, c + g - 1]$, all left endpoints are $\leq (c + g - 1)a + gC'$, and maximality of J ($t_0 + Jg > x - g$) makes all right endpoints $\geq (x - g + 1)a + gV'$; so every integer of the stated interval is covered within its own class. $\square$

Remark 16.5 (The merge condition is real). The frame-merge hypothesis in the extended form cannot be dropped. For $(a, b, M) = (5, 7, 9)$: $e = 2$, $\mu = 4$, $g = 2$, $(e', \mu') = (1, 2)$, $C' = 0$. Take $y = 1$, $z = 0$, so $c = 1$, $V' = 1$, and $x = 3 = c + g$. The merge condition fails ($V' - C' = 1 < 4 = a - 1$) — and so does the would-be conclusion: the multiset $\{5, 5, 5, 7\}$ has subset sums $\{0, 5, 7, 10, 12, 15, 17, 22\}$, missing 11 inside the interval $[10, 12]$ that the un-repaired statement would promise. (An early draft of the parent paper's staircase omitted this hypothesis; the formalization caught it, and the counterexample is now permanently pinned in the Lean development as a guard. A cautionary tale told honestly: machine-checking is not decoration.)

16.5. The λ -lift. The final tool lets one finite check certify an infinite family. Triples (a, b, M) and $(a, b + a, M + a)$ have the same a , the same $h = M - b$, and the same residues $b, M \bmod a$; climbing this ladder is the λ -lift.

Lemma 16.6 (λ -lift). *Suppose the box (Y, Z) certifies the triple (a, b, M) : it covers all residues modulo a , has $Y + Z \leq a - 1$, and budget $\mathcal{B} \leq M - 1$. Then the same box certifies $(a, b + a, M + a)$.*

Proof. Modulo a , the generators are unchanged, so the same pairs (j_ρ, k_ρ) still cover all residues. Each representative's height grows by at most $(j_\rho + k_\rho)a \leq (Y + Z)a$, so the new maximum height satisfies $S' \leq S + (Y + Z)a$, and the new budget is

$$\left\lceil \frac{(M + a) - 1 + S'}{a} \right\rceil + Y + Z \leq \left\lceil \frac{M - 1 + S}{a} \right\rceil + 1 + (Y + Z) + Y + Z = \mathcal{B} + (Y + Z) + 1.$$

The budget grew by at most $Y + Z + 1 \leq a$; the target $M - 1$ grew by exactly a . A certificate whose margin was ≥ 0 keeps a margin ≥ 0 — forever, up the whole ladder. $\square$

Checkpoint 16.7. Verify Remark 16.5's subset-sum list by hand. Then find the smallest x for which $\{5^{(x)}, 7\}$ (i.e. x copies of 5 and one 7) *does* contain a run of 5 consecutive integers among its subset sums — or prove none exists and explain which ingredient of the frame lemma is missing.

Checkpoint 16.8. In the λ -lift, where exactly is $Y + Z \leq a - 1$ used, and what goes wrong without it? Construct a numerical instance: a box with $Y + Z = a$ whose budget margin degrades as it climbs.

17. REDUCTION TO THE HARD CORE

This section shrinks the problem from all alphabets to a two-parameter family of triples. The tools are one growth lemma of Graham and four reduction lemmas, each an explicit construction.

17.1. Graham's growth lemma.

Lemma 17.1 (Graham growth). *Suppose $\text{sums}(S)$ contains a run of ℓ consecutive integers, and γ is a positive integer with $\gamma \leq \ell$. Then $\text{sums}(S \cup \{\gamma\})$ (one new copy of γ) contains a run of $\ell + \gamma$ consecutive integers.*

Proof. Let $[u, u + \ell - 1] \subseteq \text{sums}(S)$. The enlarged multiset's sums include both the old run and the old run shifted by γ :

$$[u, u + \ell - 1] \cup [u + \gamma, u + \gamma + \ell - 1].$$

Since $\gamma \leq \ell$, the shifted run starts no later than $u + \ell$, so the two runs overlap or abut, forming $[u, u + \ell + \gamma - 1]$. $\square$

A run, once at least as long as a generator, can be extended by that generator forever at a price of one copy per γ gained. This ratchet is what lets us prove SHARP for a *smaller* maximum and then climb.

17.2. Discarding redundant elements. Call an alphabet G *irreducible* if $|G| = 3$, or $|G| \geq 4$ and G is *minimal*: no proper subset of size ≥ 3 has gcd equal to 1.

Lemma 17.2 (Graham reduction). *If SHARP holds for all irreducible alphabets, it holds for all alphabets.*

Proof. Induct on $|G|$. Let $|G| \geq 4$ be non-minimal: some $e^\dagger \in G$ is *redundant*, meaning $G \setminus \{e^\dagger\}$ still has gcd 1 (and size ≥ 3). Two cases.

If $e^\dagger \neq M$: the smaller alphabet has the same maximum M , so its SHARP witness — a multiset of size $\leq M - 1$ covering M consecutive integers — is literally also a witness for G .

If $e^\dagger = M$: let $M' := \max(G \setminus \{M\})$. By induction, a multiset of size $\leq M' - 1$ drawn from $G \setminus \{M\}$ covers a run of M' consecutive integers. Now ratchet with $a := \min G$: the run has length $M' > a$, so each added copy of a extends it by a (Lemma 17.1, the hypothesis holding throughout since runs only grow); after $\lceil (M - M')/a \rceil \leq M - M'$ copies the run has length $\geq M$. Total size $\leq (M' - 1) + (M - M') = M - 1$. $\square$

17.3. Cheap pairs. Three lemmas now dispose of every triple whose two smaller elements are arithmetically convenient. Throughout, G has $\min G = a \geq 2$ and $\max G = M$.

Lemma 17.3 (Small coprime pair). *If some coprime pair $\alpha, \gamma \in G$ has $\alpha + \gamma \leq M - 1$, then $m(G) \leq M - 1$.*

Proof. Take $x := M - \alpha$ copies of α and $\alpha - 1$ copies of γ — budget exactly $(M - \alpha) + (\alpha - 1) = M - 1$. The representatives $0, \gamma, 2\gamma, \dots, (\alpha - 1)\gamma$ cover all residues modulo α (coprimality), with top height $S = (\alpha - 1)\gamma$, so by the frame mechanism (Lemma 16.3 with $\nu = \alpha$) the sums cover $[S, \alpha x]$. Its length:

$$\alpha x - S + 1 = \alpha(M - \alpha) - (\alpha - 1)\gamma + 1 \geq \alpha(M - \alpha) - (\alpha - 1)(M - \alpha - 1) + 1 = M,$$

using $\gamma \leq M - \alpha - 1$ and then expanding (the cross terms telescope; do it — it lands exactly on M). $\square$

Lemma 17.4 (Pair at the boundary). *If some coprime pair $\alpha < \beta$ in G has $\alpha + \beta \in \{M, M + 1\}$, then $m(G) \leq M - 1$.*

Proof. For a parameter $t \geq 0$, take $x := \beta - 1$ copies of α and $y := \alpha - 1 + t$ copies of β . Lemma 16.1 applies and covers $[C, \alpha x + \beta y - C]$ with $C = (\alpha - 1)(\beta - 1)$; the length works out (expand!) to

$$\alpha x + \beta y - 2C + 1 = \alpha + \beta - 1 + t\beta,$$

at budget $x + y = \alpha + \beta - 2 + t$. If $\alpha + \beta = M + 1$: take $t = 0$; length M , budget $M - 1$. If $\alpha + \beta = M$: $t = 0$ gives length $M - 1$ — *one short* — so take $t = 1$: length $M - 1 + \beta \geq M$, budget $M - 1$ exactly. The budget absorbs precisely one extra copy; nothing is wasted, nothing is spare. $\square$

Example 17.5. $(\alpha, \beta, M) = (2, 3, 5)$, the $\alpha + \beta = M$ case. At $t = 0$ the multiset $\{2, 2, 3\}$ has sums $\{0, 2, 3, 4, 5, 7\}$: longest run $[2, 5]$, length $4 < 5$, insufficient exactly as the bookkeeping predicts. At $t = 1$, $\{2, 2, 3, 3\}$ has sums $\{0, 2, 3, 4, 5, 6, 7, 8, 10\} \supseteq [2, 8]$: seven consecutive, at budget $4 = M - 1$.

Lemma 17.6 (Non-coprime pair, $|G| = 3$). *Let $G = \{a, b, M\}$ with $d := \gcd(a, b) \geq 2$. (Then $\gcd(d, M) = 1$ automatically, since $\gcd(G) = 1$.) Then $m(G) \leq M - 1$.*

Proof. Write $a = d\alpha$, $b = d\beta$, $\gcd(\alpha, \beta) = 1$, $\alpha < \beta$. The multiset:

$$x := \beta - 1 \text{ copies of } a, \quad y := M - d - \beta + 1 \text{ copies of } b, \quad d - 1 \text{ copies of } M,$$

of size $(\beta - 1) + (M - d - \beta + 1) + (d - 1) = M - 1$ exactly.

The scaled grid. Sums using only a 's and b 's are d times sums from the (α, β) -grid. The hypotheses of Lemma 16.1: $x = \beta - 1$ exactly, and $y \geq \alpha - 1$ because $M > b = d\beta$ forces $M \geq d\beta + 1$, so $y \geq d\beta + 1 - d - \beta + 1 = (d - 1)(\beta - 1) + 1 \geq \beta > \alpha - 1$. With $C := (\alpha - 1)(\beta - 1)$, the covered set is the arithmetic progression $d \cdot [C, \alpha x + \beta y - C]$, of step d and

$$L := \alpha x + \beta y - 2C + 1 \text{ terms.}$$

The grid is long enough: $L \geq M + 1$. L grows with M (through y), so the worst case is the smallest legal $M = d\beta + 1$, where the claim rearranges (a few lines of algebra with $u := \beta(d - 1)$) to

$$(\beta - 1)(u - \alpha + 2) \geq u + 1,$$

which at $d = 2$ (i.e. $u = \beta$) reduces to $\beta(\beta - \alpha) + \alpha - 3 \geq 0$ — true, since $\beta(\beta - \alpha) \geq \beta \geq 2$ and $\alpha \geq 1$ — and only strengthens as d (hence u) grows.

Bridging the d residues. The $d - 1$ copies of M contribute $0, M, 2M, \dots, (d - 1)M$, which — since $\gcd(d, M) = 1$ — run through *all* d residues modulo d . Adding them to the step- d progression fills every gap: for $n \in [dC + (d - 1)M, dC + (L - 1)d]$, pick $\kappa \in [0, d - 1]$ with $\kappa M \equiv n - dC \pmod{d}$; then $n - \kappa M$ is a step- d progression member (it is $\equiv dC$, and lies in the progression's range). The solid interval has length $(L - 1)d - (d - 1)M + 1 \geq M$, which unwinds precisely to $L \geq M + 1$, just proved. $\square$

17.4. Minimal alphabets of four or more elements.

Lemma 17.7 (Structure of minimal alphabets). *Let $G = \{g_1, \dots, g_m\}$ be minimal with $m \geq 4$, and set $d_i := \gcd(G \setminus \{g_i\})$. Then: each $d_i \geq 2$; the d_i are pairwise coprime; $d_j \mid g_i$ whenever $j \neq i$; consequently every $g_i \geq 30$ when $m = 4$ and ≥ 210 when $m \geq 5$; and every pair of elements of G has $\gcd \geq 2$.*

Proof. $d_i \geq 2$ is the definition of minimality (the subset $G \setminus \{g_i\}$, of size ≥ 3 , must not have gcd 1). For $i \neq j$, any common divisor of d_i and d_j divides every element of $(G \setminus \{g_i\}) \cup (G \setminus \{g_j\}) = G$, hence divides $\gcd(G) = 1$. That $d_j \mid g_i$ for $j \neq i$ is immediate ($g_i \in G \setminus \{g_j\}$). So g_i is divisible by the product of the $m - 1$ pairwise coprime numbers $\{d_j\}_{j \neq i}$, each ≥ 2 ; the least possible such product is $2 \cdot 3 \cdot 5 = 30$ ($m = 4$) or $2 \cdot 3 \cdot 5 \cdot 7 = 210$ ($m \geq 5$). Finally, a coprime pair inside G would sit inside a proper size-3 subset of gcd 1 (add any third element), contradicting minimality. $\square$

Lemma 17.8 (Minimal alphabets). *If G is minimal with $|G| \geq 4$, and SHARP holds for all alphabets of maximum $< M$, then $m(G) \leq M - 1$.*

Proof. Descend. Remove $a := \min G$. By minimality $d := \gcd(G \setminus \{a\}) \geq 2$; note $\gcd(a, d) = 1$ (a common factor would divide all of G). The rescaled alphabet $H := (G \setminus \{a\})/d$ has $\gcd(H) = 1$, $|H| \geq 3$, and maximum $M/d < M$ — so SHARP applies to it by hypothesis: a run of M/d consecutive integers within H 's subset sums, at cost $\leq M/d - 1$.

Extend. Let $a' := \min H$ and ratchet the run (Lemma 17.1, inside H) up to length

$$\ell := \left\lceil \frac{M - 1 + (d - 1)a}{d} \right\rceil + 1,$$

at an extra cost of at most $\lceil (\ell - M/d)/a' \rceil$ copies of a' .

Scale and bridge. Multiplying the run's multiset by d (i.e. using the original elements of $G \setminus \{a\}$) turns the run into a step- d progression of ℓ terms; the d residues are bridged by $0, a, 2a, \dots, (d - 1)a$ — all residues modulo d , as $\gcd(a, d) = 1$ — yielding a solid interval of length $(\ell - 1)d - (d - 1)a + 1 \geq M$ by the choice of ℓ . Total budget:

$$T \leq \left(\frac{M}{d} - 1 \right) + \left\lceil \frac{\ell - M/d}{a'} \right\rceil + (d - 1).$$

Count. From $a < \min(G \setminus \{a\}) = da'$, i.e. $a \leq da' - 1$:

$$\ell - \frac{M}{d} \leq \frac{(d - 1)a}{d} + 2 < (d - 1)a' + 2,$$

so the middle term is $\leq d + 1$ and $T \leq M/d + 2d - 1$. The requirement $T \leq M - 1$ rearranges to $M(1 - \frac{1}{d}) \geq 2d$, i.e. $M \geq 2d^2/(d - 1)$. For $d \geq 3$ this is at most $3d$, and $M \geq 3d$ because $G \setminus \{a\}$ contains three distinct multiples of d . For $d = 2$ it demands $M \geq 8$, — and every element of G is ≥ 30 (Lemma 17.7). $\square$

17.5. The hard core. Assemble the reductions. By Lemma 17.2 we may take G irreducible; Lemma 17.8 absorbs the minimal $|G| \geq 4$ case (its appeal to smaller maxima will be organized as a strong induction in §23); so $G = \{a, b, M\}$ is a triple, with $a \geq 2$ after the $1 \in G$ observation of §15. If $\gcd(a, b) \geq 2$: Lemma 17.6. So $\gcd(a, b) = 1$; and now Lemmas 17.3 and 17.4 applied to the pair (a, b) dispose of every triple with $a + b \leq M + 1$. What survives is the *hard core*:

$$G = \{a, b, M\}, \quad \gcd(a, b) = 1, \quad h := M - b \in [1, a - 2],$$

(the condition $a + b \geq M + 2$ is $h \leq a - 2$), which forces $a \geq 3$. In words: the two large generators crowd within $a - 2$ of the top, and the small generator is at least 3. The standing notation of §16.3 is in force; add to it the mod- a data

$$\bar{e} := e \bmod a \in [1, a - 1], \quad \bar{\mu} := M \bmod a, \quad \eta := \bar{\mu} \cdot \bar{e}^{-1} \bmod a \quad (\text{when } \bar{\mu} \neq 0),$$

where $\bar{e}^{-1}$ is the inverse modulo a (existing since $\gcd(e, a) = \gcd(b - a, a) = \gcd(b, a) = 1$; in particular $\bar{e} \neq 0$). Three observations, each a one-liner worth doing:

$$\eta = 0 \iff a \mid M; \quad \eta = a - 1 \iff a \mid b + M; \quad \eta = 1 \text{ is impossible.}$$

(For the last: $\eta = 1$ would mean $M \equiv e \pmod{a}$, i.e. $a \mid M - b + a - \dots$ — unwind it: $M - e = M - b + a$, so $a \mid h$; but $1 \leq h \leq a - 2$.) The first two conditions are disjoint on the hard core (both would give $a \mid b$).

Every hard-core triple is now routed through the ordered six-branch tree of §15.2 — first matching condition wins — and §§18–22 close the branches one by one.

Checkpoint 17.9. First check each of these *is* a hard-core triple, then route it through the ordered tree (compute e, h, μ, g, η): $(5, 12, 15)$, $(3, 4, 5)$, $(5, 7, 9)$, $(13, 22, 27)$, $(5, 8, 9)$, $(5, 16, 17)$. You should hit all six branches, in the order D, P, L, E, T, B — several of these are the coming sections' running examples. Then find a triple that is *not* hard-core and say which reduction lemma catches it (try $(12, 25, 36)$).

Checkpoint 17.10. Show that the hard core is nonempty for every $a \geq 3$ (find its smallest member for each a), and that $(3, 4, 5)$ is the smallest hard-core triple of all.

18. CASES D AND L: TWO CLEAN FORMULAS

The first two branches are warm-ups with real content: each is closed by one explicit multiset whose budget check is a two-line inequality.

18.1. Case D: $a \mid M$.

Lemma 18.1 (Case D). *Let (a, b, M) be a hard-core triple with $a \mid M$; write $M = qa$ ($q \geq 2$). Put $x_{\text{eff}} := \lceil (M - 1 + (a - 1)b)/a \rceil$ and $z := \lceil (x_{\text{eff}} - (q - 1))/q \rceil$. Then the multiset of*

$$a - 1 \text{ copies of } b, \quad q - 1 \text{ copies of } a, \quad z \text{ copies of } M$$

has size $\leq M - 1$ and its subset sums contain M consecutive integers.

Proof. Coverage. The copies of b give the representatives $0, b, 2b, \dots, (a - 1)b$ — all residues modulo a , coprimality again — with top height $S = (a - 1)b$. For the sliding phase we need enough “effective copies of a ”: the pads $ia + \kappa M = a(i + q\kappa)$ with $i \leq q - 1$, $\kappa \leq z$ realize au for every $u \in [0, (q - 1) + qz]$ — the blocks $[q\kappa, q\kappa + q - 1]$ abut as κ steps — and $(q - 1) + qz \geq x_{\text{eff}}$ by the choice of z . Now run the frame argument (Lemma 16.3's proof verbatim, with the effective copies in place of literal ones): every $n \in [S, S + M - 1]$ is (residue representative) $+ au$ for some $u \leq \lceil (M - 1 + S)/a \rceil = x_{\text{eff}}$.

Budget. Since $b \leq M - 1$,

$$\frac{M - 1 + (a - 1)b}{a} \leq \frac{M - 1 + (a - 1)(M - 1)}{a} = M - 1,$$

so $x_{\text{eff}} \leq M - 1 = qa - 1$, whence

$$z \leq \left\lceil \frac{qa - 1 - (q - 1)}{q} \right\rceil = \left\lceil \frac{q(a - 1)}{q} \right\rceil = a - 1.$$

The total is then

$$(a - 1) + (q - 1) + z \leq 2a + q - 3 \leq qa - 1 = M - 1,$$

the last step being exactly $(a - 1)(q - 2) \geq 0$: true for all $q \geq 2$, $a \geq 3$, with equality only at $q = 2$ — the tightest members of the branch. $\square$

Example 18.2. $(a, b, M) = (5, 12, 15)$: $q = 3$, $S = 4 \cdot 12 = 48$, $x_{\text{eff}} = \lceil 62/5 \rceil = 13$, $z = \lceil 11/3 \rceil = 4$. Multiset: four 12s, two 5s, four 15s — size $10 \leq 14 = M - 1$. The four 12s hit residues $0, 2, 4, 1, 3$ modulo 5; the pads realize $5u$ for $u \in [0, 14] \ni 13$; and the sums cover $[48, 62] \supseteq$ fifteen consecutive integers.

18.2. Case L: equal spacings, $e = h$. When $e = h$ the triple is an arithmetic progression $\{a, a + g, a + 2g\}$ with $g = e = h$ (so $g \leq a - 2$ and $\gcd(a, g) = 1$), and the staircase machinery of §16 closes it directly.

Lemma 18.3 (Case L). *For $G = \{a, a + g, a + 2g\}$ with $\gcd(a, g) = 1$ and $1 \leq g \leq a - 2$, the multiset*

$$(x, y, z) = \left(\left\lfloor \frac{a-2}{2} \right\rfloor + 2g, \quad 1, \quad \left\lceil \frac{a-2}{2} \right\rceil \right)$$

(copies of a, b, M respectively) has budget exactly $a + 2g - 1 = M - 1$ and covers M consecutive integers.

Proof. Here $e = g$, $\mu = 2g$, so $(e', \mu') = (1, 2)$, $C' = 0$, and with $y = 1$: $V' = 1 + 2z$, $c = z + 1$. The budget is $x + 1 + z = (a - 2) + 2g + 1 = M - 1$ in all parity cases. Three sub-cases, one per applicable staircase clause.

a even. Then $z = \frac{a-2}{2}$ and $x - c = 2g - 1 \geq g$, so $x \geq c + g$; the frame-merge condition $V' - C' = 2z + 1 = a - 1 \geq a - 1$ holds *with equality*. The extended form (Lemma 16.4(d)) covers a solid interval of length $(x - c - 2g + 2)a + g(2z + 1) + 1 = a + g(a - 1) + 1 \geq a + 2g = M$ (i.e. $g(a - 3) \geq -1$: always).

a odd, $g \geq 2$. Then $z = \frac{a-1}{2}$, $x - c = 2g - 2 \geq g$, and $V' - C' = a \geq a - 1$ with room to spare. The extended form covers length $ga + 1 \geq a + 2g = M$ (i.e. $(g - 1)a \geq 2g - 1$: true for $g \geq 2, a \geq 3$).

a odd, $g = 1$. Then $x = c$ exactly, and the short merge (Lemma 16.4(c)) applies: level c covers $ca + [0, a]$ (as $V' = 2z + 1 = a$), level $c + 1$ covers $(c + 1)a + [1, a]$, and the merge condition $V' \geq a + \max(C', 1) - 1 = a$ holds with equality; the run $[ca, (c + 1)a + a]$ has length $2a + 1 \geq a + 2 = M$. $\square$

Note the pattern, which will repeat: the constructions are not delicate to *find* — they are the obvious staircases — but their budgets close against $M - 1$ with equality at the extremes. The margin throughout Part IV is zero at the corners; that is what “sharp” means concretely.

Checkpoint 18.4. Run Case L on $(5, 7, 9)$ ($g = 2$; odd a): write out the multiset, the levels, and the covered interval, and reconcile with Remark 16.5 — why does the Case-L multiset succeed where $\{5, 5, 5, 7\}$ failed?

19. CASE P: PAIRS AS MOVERS

Now the line $a \mid b + M$: here $b + M = ra$ for an integer r , and $r \geq 3$ — because $a < b < M$ gives $b + M > 2a$. This branch has the prettiest mechanism in Part IV. Modulo a , the two big generators are *negatives of each other*: $M \equiv -b$. So one copy of b plus one copy of M is residue-neutral — a *mover*: it costs 2 elements and advances the total by exactly $b + M = ra$, i.e. r grid units of a , without touching the residue. Unpaired copies of b (or of M) set the residue; movers and copies of a then slide the frame.

Lemma 19.1 (Case P). *Let (a, b, M) be a hard-core triple with $b + M = ra$, $r \geq 3$. Then the multiset*

$$(x, y, z) = \left(r - 1, \quad \left\lceil \frac{M-r}{2} \right\rceil, \quad \left\lfloor \frac{M-r}{2} \right\rfloor \right)$$

(copies of a, b, M), of budget exactly $(r-1) + (M-r) = M-1$, has subset sums containing M consecutive integers.

Proof. Coordinates. A selection of i copies of a, j of b, κ of M is worth

$$i a + j b + \kappa M = i a + w b + \kappa r a, \quad w := j - \kappa$$

(substituting $M = ra - b$): κ full movers, plus w unpaired copies of b if $w > 0$ (or $|w|$ unpaired copies of M if $w < 0$, by the symmetric substitution). Set

$$p := \left\lfloor \frac{a}{2} \right\rfloor, \quad q := \left\lfloor \frac{a-1}{2} \right\rfloor \quad (\text{so } p+q = a-1),$$

and declare the run to start at $\text{low} := \max(pb, qM)$. We must realize every $n \in [\text{low}, \text{low} + M - 1]$.

Three preliminary facts.

- (P1) $M \geq a + r - 1$. From $b < M$: $ra = b + M < 2M$, so $2M \geq ra + 1$; and $a, r \geq 3$ give $a(r-2) \geq 3(r-2) \geq 2r-3$, whence $2M \geq ra + 1 = a(r-2) + 2a + 1 \geq 2a + 2r - 2$.
- (P2) $M \leq a(r-1) - 1$. From $b > a$: $M = ra - b < a(r-1)$.
- (P3) $y \geq p$ and $z \geq q$. By (P1), $M - r \geq a - 1$, so $z = \lfloor (M-r)/2 \rfloor \geq \lfloor (a-1)/2 \rfloor = q$ and $y = \lceil (M-r)/2 \rceil \geq \lceil (a-1)/2 \rceil = p$.

The two branches. Fix a target n . Since $\gcd(a, b) = 1$ there is a unique $w \in [0, a-1]$ with $w b \equiv n \pmod{a}$.

Branch 1: $w \leq p$ — spend w unpaired bs. Then $w b \leq p b \leq \text{low} \leq n$, so $A := (n - w b)/a$ is a nonnegative integer. Split $A = i + r\kappa$ with $i := A \bmod r \leq r-1 = x$ and $\kappa := \lfloor A/r \rfloor$: take i copies of a, κ movers, and the w unpaired bs — that is, i copies of $a, w + \kappa$ copies of b, κ copies of M , worth exactly n (check: $ia + (w + \kappa)b + \kappa M = ia + w b + \kappa r a = aA + w b = n$). Availability requires

$$\kappa \leq z \quad \text{and} \quad w + \kappa \leq y.$$

For the first, since $aA \leq n \leq \text{low} + M - 1$, it suffices that $\text{low} + M - 1 \leq a(rz + x)$ (then $A \leq rz + x$, so $\kappa \leq z$). For the second, note $a(A + r w) = (n - w b) + w(b + M) = n + w M \leq (\text{low} + M - 1) + p M$, so it suffices that

$$\text{low} + M - 1 + p M \leq a(r y + x). \quad (\text{C}_y)$$

Branch 2: $w > p$ — spend $v := a - w \in [1, q]$ unpaired Ms. Since $M \equiv -b, v M \equiv -v b \equiv w b \equiv n \pmod{a}$, and $v M \leq q M \leq \text{low} \leq n$: so $A' := (n - v M)/a$ is a nonnegative integer, split as $A' = i' + r\kappa'$, giving i' copies of a, κ' copies of $b, v + \kappa'$ copies of M . Availability: $\kappa' \leq y$, which follows from $\text{low} + M - 1 \leq a(r y + x)$ (weaker than (C_y)); and $v + \kappa' \leq z$, for which — from $a(A' + r v) = n + v b \leq (\text{low} + M - 1) + q b$ — it suffices that

$$\text{low} + M - 1 + q b \leq a(r z + x). \quad (\text{C}_z)$$

(C_z also implies the interim condition of Branch 1.)

Four extremes. Each of $(\text{C}_y), (\text{C}_z)$ splits in two according to which term achieves $\text{low} = \max(pb, qM)$; using $p+q = a-1$ and $b+M = ra$, the four resulting inequalities are

$$\begin{array}{ll} (\text{E1}) & (a-1)b + M - 1 \leq a(rz + x) & [\text{C}_z, \text{low} = pb] \\ (\text{E2}) & qra + M - 1 \leq a(rz + x) & [\text{C}_z, \text{low} = qM] \\ (\text{F1}) & pra + M - 1 \leq a(ry + x) & [\text{C}_y, \text{low} = pb] \\ (\text{F2}) & aM - 1 \leq a(ry + x) & [\text{C}_y, \text{low} = qM] \end{array}$$

(for (E1): $pb + qb = (a - 1)b$; for (F1): $pb + pM = pra$; for (F2): $qM + pM + M = aM$). Verifying these four suffices for every w and every n , because w entered only through $w \leq p$ and $v \leq q$ — the extremes used.

(F2). It suffices that $M \leq ry + r - 1$. From $y \geq (M - r)/2$ and $r \geq 3$: $ry + r - 1 \geq \frac{3}{2}(M - r) + r - 1 = \frac{1}{2}(3M - r - 2) \geq M \iff M \geq r + 2$, and (P1) with $a \geq 3$ gives $M \geq a + r - 1 \geq r + 2$.

(F1). $pra \leq ar y$ by (P3), so (F1) reduces to $M - 1 \leq ax = a(r - 1)$, which follows from (P2).

(E2). Identical: $qra \leq ar z$ by (P3), and the remainder is (P2) again.

(E1) — *the binding one*. Substitute $b = ra - M$ and use $z \geq (M - r - 1)/2$; it suffices that

$$2[(a - 1)(ra - M) + M - 1] \leq ar(M - r - 1) + 2a(r - 1).$$

Expanding both sides and collecting (worth doing on paper — every term is tame), the inequality rearranges to

$$0 \leq M(ar + 2a - 4) - ar^2 - 2ra^2 + 3ar - 2a + 2.$$

The coefficient of M is positive, so we may insert (P1)'s bound $2M \geq ra + 1$; after multiplying through by 2, the right side is at least $a(ar^2 - 2ar - 2r^2 + 3r - 2)$, so (E1) holds whenever

$$ar(r - 2) \geq 2r^2 - 3r + 2. \quad (*)$$

For $r \geq 4$: $a \geq 3$ gives $ar(r - 2) \geq 3r^2 - 6r$, and $3r^2 - 6r \geq 2r^2 - 3r + 2 \iff r^2 - 3r - 2 \geq 0$, true from $r = 4$ on. For $r = 3$: (*) reads $3a \geq 11$, i.e. $a \geq 4$.

So (*) can fail only at $(a, r) = (3, 3)$ — where $b + M = 9$ with $3 < b < M$ forces $(a, b, M) = (3, 4, 5)$, and we check that one triple directly: $(x, y, z) = (2, 1, 1)$, and (E1) reads $2 \cdot 4 + 5 - 1 = 12 \leq 3(3 + 2) = 15$. True. (Concretely the multiset is $\{3, 3, 4, 5\}$ of Example 14.2, whose sums cover the run $[3, 12]$ — our very first example was the tightest triple on the P-line.) $\square$

Case P needs no certificate table and no computer: one closed-form multiset, four inequalities, one directly-checked triple. (The parent paper notes that its Lean formalization proves this case by a slightly different route — instantiating small r separately — a difference of route, not of content.)

Checkpoint 19.2. Play out Case P on $(3, 4, 5)$ by hand: for each $n \in [\text{low}, \text{low} + 4]$, find the branch, the residue count w , and the selection realizing n . Where is the mover used?

Checkpoint 19.3. Derive the display in (E1) from (C_z) with $\text{low} = pb$: check $pb + qb = (a - 1)b$ and trace each substitution. (This is the kind of bookkeeping the formalization checks so humans don't have to — but doing it once by hand is how you learn what the machine is sparing you.)

20. CASE E: THE η -BOX

The remaining triples have $\eta \notin \{0, 1, a - 1\}$ — neither divisibility holds — and split by size. This section handles the roomy regime: both a and $\mu = M - a$ at least 12. Here a single box works, built directly from the invariant η , and the whole case is one budget computation. The case operates entirely modulo a : recall $b \equiv \bar{e}$ and $M \equiv \bar{\mu} = \eta \bar{e} \pmod{a}$.

Lemma 20.1 (Case E). *Let (a, b, M) be a hard-core triple with $a \nmid M$, $a \nmid b + M$, $a \geq 12$ and $\mu \geq 12$. Put*

$$t := \min(\eta, a - \eta) \in [2, a/2], \quad Z := \left\lceil \frac{a - t}{t} \right\rceil, \quad \sigma := t + Z.$$

Then the box $(Y, Z) = (t - 1, Z)$ covers all residues modulo a , and its budget is $\leq M - 1$.

Proof. Coverage. Work modulo a . If $t = \eta$: a box pair (j, k) contributes $jb + kM \equiv \bar{e}(j + \eta k) = \bar{e}(j + tk)$, and over $j \leq t - 1, k \leq Z$ the quantity $j + tk$ takes every value in $[0, t(Z + 1) - 1]$ — the t -blocks abut — which has length $t(Z + 1) \geq a$ by the choice of Z . If $t = a - \eta$: then $M \equiv -t\bar{e}$, the contribution is $\bar{e}(j - tk)$, and $j - tk$ covers the interval $[-tZ, t - 1]$, again of length $\geq a$. Either way $j + tk$ (or $j - tk$) sweeps a full residue system, and multiplying by the unit $\bar{e}$ permutes residues: all of $\mathbb{Z}_a$ is covered.

Budget. Every representative height is at most $S_0 := (t - 1)b + ZM$, and the box size is $K := t - 1 + Z = \sigma - 1$. It suffices to show

$$M - 1 + S_0 \leq a(M - 1 - K),$$

for then $\lceil (M - 1 + S)/a \rceil \leq M - 1 - K$ and the budget is $\leq M - 1$. Using $b \leq M - 1$: $M - 1 + S_0 \leq t(M - 1) + ZM \leq \sigma M - t$, while the right side is $a(M - \sigma)$; so it suffices that $\sigma M + a\sigma \leq aM + t$, and — discarding $t \geq 0$ — that

$$M(a - \sigma) \geq a\sigma.$$

Now bound σ : $Z \leq (a - 1)/t$ gives $\sigma \leq t + (a - 1)/t$, and the function $t \mapsto t + (a - 1)/t$ on $[2, a/2]$ peaks at an endpoint (it is convex), so $\sigma \leq \max(\frac{a+3}{2}, \frac{a}{2} + 2) = \frac{a}{2} + 2$. Hence

$$\frac{a\sigma}{a - \sigma} \leq \frac{a(a + 4)}{a - 4} = a + 8 + \frac{32}{a - 4} \leq a + 12 \quad (a \geq 12),$$

and $M = a + \mu \geq a + 12$ delivers exactly this. Both hypotheses $a \geq 12$ and $\mu \geq 12$ are spent in this last line. $\square$

Example 20.2. $(a, b, M) = (13, 22, 27)$: $e = 9, h = 5, \mu = 14, \bar{e} = 9, \bar{\mu} = 1$, and $\eta = 9^{-1} \bmod 13 = 3$ (as $9 \cdot 3 = 27 \equiv 1$). So $t = 3, Z = \lceil 10/3 \rceil = 4$: box $(2, 4)$. The offsets $\bar{e}(j + 3k)$ with $j \leq 2, k \leq 4$ have $j + 3k$ sweeping $[0, 14] \supseteq [0, 12]$. Corner height $S_0 = 2 \cdot 22 + 4 \cdot 27 = 152$; budget $\leq \lceil (26 + 152)/13 \rceil + 6 = 14 + 6 = 20 \leq 26 = M - 1$, with plenty to spare — Case E is the *comfortable* regime, which is exactly why it tolerates a one-size-fits-all box.

Checkpoint 20.3. Why is $t \geq 2$? ($t = \min(\eta, a - \eta)$, so you need both $\eta \geq 2$ and $\eta \leq a - 2$; recall which values of η the earlier branches removed, and why.) And why work with $t = \min(\eta, a - \eta)$ rather than η itself? Find a triple where η is close to a , and compare the budgets of the two candidate boxes.

21. CASE T: THIN μ , AND THE FIRST TABLE

Now $\mu = e + h \leq 11$ (with $e \neq h$, neither divisibility, and a unbounded). Since $e \leq 10$ and $h \leq 10$, there are finitely many pairs (e, h) ; each pair defines a *line* of triples $(a, a + e, a + e + h)$ parameterized by a alone. On each line we run one explicit staircase; its *coverage* works for every a , and only its *budget* can fail — and only finitely often, which is where the certificates of Table A come in. Note $\mu' \geq 3$ throughout this case: if $\mu' = 2$ then $e' = 1$, giving $e = g$ and $\mu = 2g$, i.e. $h = \mu - e = g = e$ — but $e \neq h$ here.

Construction T.. Take $y := \mu' - 1$ and:

- *Variant A* ($g = 1$; here $e' = e, \mu' = \mu$, and write $C = C', V = V'$):

$$z := \max\left(e - 1, \left\lceil \frac{\max(a, \mu) - 1 + 2C - ye}{\mu} \right\rceil\right), \quad x := c + 1 \quad (c = y + z).$$

- Variant $g \geq 2$:

$$z := \max \left(e' - 1, \left\lceil \frac{a + \lceil (\mu - 1)/g \rceil + 2C' - ye'}{\mu'} \right\rceil \right), \quad x := c + g.$$

In both, the max has two jobs: the constant branch maintains the staircase's standing hypothesis $z \geq e' - 1$, and the ceiling branch buys run length.

Lemma 21.1 (Case T, coverage). *On every Case-T line, Construction T's subset sums contain M consecutive integers. Its budget is $2c + 1$ (variant A) or $2c + g$ (variant $g \geq 2$).*

Proof. Variant A. The ceiling branch gives $z\mu \geq \max(a, \mu) - 1 + 2C - ye$, so

$$V - C = ye + z\mu - 2C \geq \max(a, \mu) - 1 \geq a - 1 :$$

the two-frame merge condition. With $x = c + 1$, Lemma 16.4(b) covers $[ca + C, (c + 1)a + V]$, of length $a + (V - C) + 1 \geq a + \max(a, \mu) \geq a + \mu = M$. Budget $x + y + z = 2c + 1$.

Variant $g \geq 2$. The ceiling branch gives $V' - C' \geq a + \lceil (\mu - 1)/g \rceil \geq a + (\mu - 1)/g$. With $x = c + g$, the *base form* of Lemma 16.4(d) — no merge hypothesis needed — covers a solid interval of length

$$g(V' - C') - (g - 1)a + 1 \geq ga + (\mu - 1) - (g - 1)a + 1 = a + \mu = M.$$

Budget $x + y + z = 2c + g$. □

So coverage is unconditional; everything now rides on the inequality

$$\text{budget} \leq M - 1 = a + \mu - 1.$$

The budget grows like $2z \approx 2a/\mu'$, i.e. with slope $2/\mu' \leq 2/3$ in a , while the target grows with slope 1 — so far out on every line the margin is positive and widening. Failures can only be *early* on a line.

Example 21.2 (The line $(e, h) = (2, 1)$). Here $\mu = 3$, $g = 1$, $(e', \mu') = (2, 3)$, $C = 2$: the triples $(a, a + 2, a + 3)$. Variant A takes $y = 2$ and $z = \max(1, \lceil (a - 1)/3 \rceil)$, budget $2c + 1$ against target $a + 2$:

a	z	c	budget $2c+1$	target $a+2$	margin
7	2	4	9	9	0 (tight)
13	4	6	13	15	2
19	6	8	17	21	4
25	8	10	21	27	6

The margin grows at rate $\approx 1/3$; on this lucky line it never goes negative. Other lines are less lucky: on $(e, h) = (5, 1)$, the triples $(4, 9, 10)$, $(7, 12, 13)$ and $(8, 13, 14)$ (among others) have construction budgets *exceeding* $M - 1$, and each needs a replacement certificate.

Proposition 21.3 (T-tail). *Across all Case-T lines, the applicable variant's budget exceeds $M - 1$ for precisely 172 triples, all with $a \leq 29$; each of these carries an explicit box certificate — a triple (x, Y, Z) with $Y + Z \leq a - 1$ and budget $\leq M - 1$, verified by the two-line computation of Lemma 16.3 — listed as Table A of the parent repository. For $a > 3000$ no failures occur; the range $a \leq 3000$ was checked exhaustively by computer.*

Proof. with the computation's role made explicit. Two separate claims.

The finite range $a \leq 3000$. For each of the finitely many lines and each $a \leq 3000$, the applicable variant's budget is a closed formula; a computer evaluated all of them exactly (integer arithmetic, no floating point) and found the budget to exceed $M - 1$ for exactly the

172 triples recorded in Table A, the largest at $a = 29$. This is a finite computation, redone independently three ways (§25); each failing triple’s replacement certificate is itself re-verified mechanically. Nothing here asks you to trust a search heuristic: the claims are “this explicit formula, at these explicit inputs, exceeds that explicit bound” and “this explicit multiset covers that explicit run”, both checkable by hand for any single row — 172 rows is merely more than a human wants to do.

The tail $a > 3000$. Here no computer is needed. On a fixed line, once $a \geq 3000$ the ceiling branch of z dominates its constant branch (the ceiling’s argument exceeds $(3000 - 1 - 100)/11 > 260$, while the constant branch is at most 9), so the budget equals its ceiling-branch value, which is bounded above by a linear function $\beta(a)$ of slope $2/\mu' \leq 2/3$; the target $\tau(a) = a + \mu - 1$ has slope 1. Therefore the margin $\tau - \beta$ is nondecreasing in a (rate $\geq 1/3$), and it suffices that it is positive at $a = 3000$ on every line — a final finite evaluation (in exact rational arithmetic; the minimum over all lines is 993, attained at $(e, h) = (2, 1)$). Positive there, nondecreasing after: positive forever. $\square$

The logical shape deserves a look: the *infinite* part of Case T is a slope comparison a student can check; the *finite* part is large but mindless. This is the pattern of every computer-assisted step in this work — the machine never supplies an idea, only stamina — and §25 describes the checking apparatus in full.

Checkpoint 21.4. Verify the $a = 7$ row of Example 21.2: build Construction T’s multiset for $(7, 9, 10)$, list its levels, confirm coverage of 10 consecutive integers and budget exactly $9 = M - 1$ (tight, but not failing).

Checkpoint 21.5. The triple $(5, 8, 9)$ (line $(e, h) = (3, 1)$, $g = 1$, $\mu = 4$) is one of the 172 failures. Compute Construction T’s budget and watch it exceed $M - 1 = 8$; then verify its Table-A replacement certificate $(x, Y, Z) = (5, 2, 1)$: the multiset $\{5, 5, 5, 5, 5, 8, 8, 9\}$ has budget 8, its parts $8j + 9k$ hit all residues mod 5, and its sums cover $[17, 25]$ — nine consecutive integers.

22. CASE B: FINITE BASES, LIFTED FOREVER

The last branch: $a \leq 11$ with $\mu \geq 12$. Now a is bounded but b (hence e , hence M) is not — each small a carries infinitely many triples. The finiteness that saves us is *modular*: group the triples into *classes*

$$[a; \bar{e}, h], \quad \bar{e} \in [1, a - 1], \quad h \in [1, a - 2],$$

by the residue of e modulo a and the exact value of h . Within a class, members differ only in $\lambda := \lfloor e/a \rfloor$, and stepping λ by one is exactly the map $(a, b, M) \mapsto (a, b + a, M + a)$ of the λ -lift (Lemma 16.6). So:

certify the smallest Case-B member of each class by an explicit box; the λ -lift then certifies every larger member, forever.

There are finitely many classes ($a \leq 11$, $\bar{e} < a$, $h \leq a - 2$), and the enumeration — machine-checked, §25 — yields exactly 178 classes whose members reach Case B. Their certified bases are Table B of the parent repository.

Example 22.1 (A class and its lift). The class $[5; 1, 1]$: triples with $a = 5$, $e \equiv 1 \pmod{5}$, $h = 1$. Its base in Table B is $(a, b, M) = (5, 16, 17)$ with box $(x, Y, Z) = (10, 1, 2)$: the parts $16j + 17k$ ($j \leq 1$, $k \leq 2$) have residues $\{0, 1, 2, 3, 4\}$ modulo 5 — check: $0, 16 \equiv 1, 17 \equiv 2, 33 \equiv 3, 34 \equiv 4, 50 \equiv 0$ — and the budget is $10 + 1 + 2 = 13 \leq 16 = M - 1$. The class members are $(5, 21, 22), (5, 26, 27), (5, 31, 32), \dots$, and Lemma 16.6 hands the *same* box up the whole

ladder: the residues never change, each lift grows the budget by at most $Y + Z + 1 = 4$ while the target $M - 1$ grows by exactly 5. The margin widens forever; one finite check settles an infinite family.

Two administrative points, each with a one-line proof idea (the parent paper’s Descent lemma).

Class membership of cases D and P is λ -invariant. Both conditions ($a \mid M$; $a \mid b + M$) depend only on residues modulo a , which the lift preserves. So a class is “a D-class”, “a P-class”, or neither, wholesale — no class straddles branches as λ climbs, except for one wrinkle:

The $e = h$ wrinkle. The condition $e = h$ (Case L) is *not* λ -invariant — e grows with λ while h stays fixed — but it can only hold at the class’s smallest member ($e \geq a > h$ once $\lambda \geq 1$). Exactly six classes (all on the diagonal $\bar{e} = h$, with $a \in \{9, 10, 11\}$) have a $\lambda = 0$ member that satisfies $\mu \geq 12$ but is diverted to Case L; those classes are simply *re-based* at their first genuine Case-B member, from which the lift proceeds as usual.

Why most bases sit at $\lambda \geq 1$. A class’s $\lambda = 0$ member often has $\mu = e + h < 12$ (small e), landing in Case T instead; the class enters Case B only once e has grown past $12 - h$. Example 22.1’s base has $\lambda = 2$ for exactly this reason. The enumeration accounts for every class and every member — completeness of the routing is part of what the harnesses check.

Checkpoint 22.2. Verify the base certificate of the class $[5; 1, 1]$ end to end: residues, heights, budget, and the covered run of 17 consecutive integers. Then perform one lift by hand — write the certificate’s guarantee for $(5, 21, 22)$ and check the budget bound $13 + 4 \leq 21 - 1$... and note that the lemma’s bound is generous: recompute the exact budget at $(5, 21, 22)$ directly.

23. THE PROOF OF SHARP, AND ITS SHARPNESS

Proof of Theorem 15.1. By strong induction on M : proving the statement for maximum M , we may assume it for every alphabet with a smaller maximum. (Exactly one place consumes this: Lemma 17.8 passes to the rescaled alphabet H of maximum $M/d < M$. No hard-core branch invokes the hypothesis, and no branch re-enters Lemma 17.8 at a larger maximum, so the recursion terminates.)

If $1 \in G$: done (§15). By Lemma 17.2, reduce to irreducible G ; by Lemma 17.8, dispose of minimal $|G| \geq 4$; by Lemma 17.6, assume the triple $\{a, b, M\}$ has $\gcd(a, b) = 1$; by Lemmas 17.3–17.4, assume $a + b \geq M + 2$: the hard core. Route:

if $a \mid M$: D $\rightarrow$ else if $a \mid b + M$: P $\rightarrow$ else if $e = h$: L $\rightarrow$ else if $a, \mu \geq 12$: E $\rightarrow$ else if $\mu \leq 11$: T $\rightarrow$ else B.

Each branch delivers a multiset of size $\leq M - 1$ covering M consecutive integers: Lemmas 18.1, 19.1, 18.3, 20.1, 21.1 with Proposition 21.3 (Table A), and Table B with Lemma 16.6.

Exhaustiveness and consistency of the routing. D and P are disjoint on the hard core (both would force $a \mid b$). A triple with $e = h$ and $a \mid b + M$ would have $b + M = 2a + 3e$, so $a \mid 3e$; with $\gcd(a, e) = 1$ this forces $a \mid 3$, i.e. $a = 3$, $e = h = 1$: the single triple $(3, 4, 5)$, which the ordered routing sends to P (and which L would also handle — no conflict, the router just picks first). Every triple surviving to the last three branches has $\eta \notin \{0, 1, a - 1\}$, and the conditions “ $a \geq 12 \wedge \mu \geq 12$ ”, “ $\mu \leq 11$ ”, “ $a \leq 11 \wedge \mu \geq 12$ ” partition the remaining (a, μ) -space. Every hard-core triple lands in exactly one branch. $\square$

The debt of Part III is paid: SHARP holds, so Proposition 14.5 holds unconditionally, and with it the non-existence half. It remains only to record sharpness — the checks were exhibited in §15:

Proposition 23.1 (Non-improvability). $m(\{3, 4, 5\}) = 4 = M - 1$: the constant $M - 1$ cannot be improved to $M - 2$. And for $G = \{2, 3\}$ ($\gcd 1$, $M = 3$), no multiset of size ≤ 2 covers 3 consecutive integers: the hypothesis $|G| \geq 3$ cannot be dropped. $\square$

Checkpoint 23.2. The routing sends $(3, 4, 5)$ to P, yet L’s construction also covers it. Compute both multisets and compare. Then find a triple where the D-condition and the E-size-conditions both hold, and explain why the ordered routing makes the overlap harmless.

Part 5. Assembly, and what the computer checked

24. ASSEMBLY: PROOF OF THE DICHOTOMY

All the pieces exist; this section only has to bolt them together.

Proof of Theorem A. Part (1), existence. This is Theorem 6.1 (proved there for all $k \geq 2$, hence for the problem’s $k \geq 3$): if $d_2 \geq k + 1$, the announcement $r = 192 d_2$ wins against every lacunary B of that ratio.

Part (2), strong non-existence. Let $d_2 \leq k$ (with $k \geq 3$) and let $r_1, r_2, \dots$ be any prescribed sequence of ratio lower bounds. Build the certificate B of Lemma 7.5 for the sequence $R(i) := r_i$; it satisfies $b_{i+1} \geq r_i b_i$ for every i . Let A be any (d_1, d_2) -sequence. By Proposition 14.5 — now unconditional, since Part IV proved SHARP (Theorem 15.1) — A is tail-covering. Retracing which case fired, for the record:

The walk’s tail alphabet (rescaled)	Case closed by
one letter	Lemma 8.3
two letters, gap word eventually periodic	Lemma 9.1
two letters, k even and $d_2 = k$	Proposition 13.5
two letters, some $W_2 \geq 2$ (k odd or $d_2 \leq k-1$)	Lemmas 11.3(a), 11.2
two letters, balanced	Theorem 12.8, then Lemma 9.1 or 13.1
three or more letters	Lemma 14.3 + SHARP

By Lemma 7.5(2), $(kA) \cap B \neq \emptyset$. As A was arbitrary, the single B defeats every admissible walk; as (r_i) was arbitrary, this is the strong form.

The dichotomy. For any $k \geq 3$ and $1 \leq d_1 < d_2$, exactly one of $d_2 \geq k + 1$ and $d_2 \leq k$ holds; part (1) gives a winning strategy in the first case, part (2) denies one in the second (a winning r would in particular defeat the certificate built for the constant sequence $r_i \equiv r$). So the answer to Erdős problem 1112 is: *yes precisely when $d_2 \geq k + 1$* . $\square$

Every promise of the honest ledger (§2.4) has now been redeemed: gap (1) in Part II, gaps (2) and (3) in Part III, gap (4) in Part IV. The remaining section is not part of the proof; it is about how the proof is *checked*.

Checkpoint 24.1 (Grand tour). Fix $k = 3$, $(d_1, d_2) = (1, 3)$, and the ratio sequence $r_i = 10^i$. Follow the entire non-existence machine for the specific walk with gap word $1, 2, 1, 1, 2, 1, 2, \dots$ (the Fibonacci word’s letters +1): which case of the table fires, what modulus results, and where in the certificate B ’s construction does the collision get guaranteed? (Nothing new to prove — the point is to watch every part of Parts III–IV touch one concrete example.)

25. WHAT THE COMPUTER CHECKED, AND HOW YOU CAN TOO

This proof is computer-assisted, and this document has flagged every place the assistance occurs: the 172 Table-A certificates and the $a \leq 3000$ scan (Case T), the 178 Table-B

base certificates and the class enumeration (Case B), and the minimal-alphabet enumeration ($M \leq 130$) behind Lemma 17.7’s use in §23. This closing section explains what, exactly, was computed, why you need not take anyone’s word for it — and what such checking can and cannot promise. Nothing here is needed to *understand* the proof; it is needed to *trust* it, which is a different and equally honorable activity.

25.1. Certificates, not searches. Every computed object in this work is a *certificate*: a small tuple whose correctness is a finite, mindless computation. A Table-A or Table-B row $(a, b, M; x, Y, Z)$ asserts two things — the multiset of x copies of a , Y of b , Z of M has size $\leq M - 1$, and its subset sums contain M consecutive integers — and *you* can check any row with pencil and paper in minutes (Checkpoints in §§21–22 did exactly this). How the rows were *found* is irrelevant to their truth: a certificate severs the proof from the search that produced it. The same holds for the scans: “this formula exceeds that bound at exactly these 172 inputs” is a statement any programmer can re-derive in an afternoon, in any language.

25.2. Three independent checks. The parent repository verifies the finite layer three separate ways, sharing no code:

- (1) *Two Python harnesses.* One reconstructs every designated witness across all 83,251 hard-core triples with $M \leq 120$ and verifies each in exact integer arithmetic; the other is a from-scratch re-implementation working only from the lemma statements. Both fail closed — any invalid row, missing class, or deviant count aborts with an error — and both report identical branch tallies (2392 D, 2605 P, 1420 L, 71,421 E, 3254 + 172 T, 1987 B). Agreement rules out coding slips; it cannot, and does not claim to, rule out an error in the shared mathematical design — that is what the proofs are for.
- (2) *The table generator.* The typeset tables in the parent paper’s supplement are *generated* from one canonical data file by a script that re-verifies every row (budget and coverage, by exact bitmask arithmetic) on every build, so no printed row can silently diverge from the data that is checked.
- (3) *The Lean kernel.* The entire proof — not just the finite layer — is formalized in Lean 4 and machine-checked; the certificate tables are transcribed into the formal development and re-decided by Lean’s kernel.

25.3. What a Lean formalization is. Lean is a *proof assistant*: a programming language in which definitions, theorem statements and proofs are all written formally, and a small program — the *kernel* — checks that every proof step follows from the rules of logic and the axioms. The human (or, in this work, largely the AI systems noted below) supplies the proof; the kernel is the referee that cannot be sweet-talked. Three facts summarize the state of this development:

- It is *complete*: all three headline theorems — the dichotomy, the explicit ratio $192 d_2$, and the strong non-existence form — are proved, over frozen definitions, from start to finish.
- It is *gap-free*: no `sorry` (Lean’s “I owe you a proof” placeholder) anywhere, and the axiom audit reports only Lean’s three standard foundational axioms.
- Its finite checks run *in the kernel*: no appeal to compiled external code (no `native_decide`), so even the $172 + 178$ certificate rows and the $a \leq 3000$ scan are validated by the same small trusted checker as every logical step.

Anyone can reproduce this: clone the repository, run `lake build`, and watch it compile with no complaints (the parent README gives the exact commands, and links a public log of a complete run).

25.4. What machine-checking does not promise. Honesty requires the other half. A formal verification establishes: *these formal statements follow from these axioms*. It does not establish:

- *That the formal statement means what the English problem says.* If the Lean definition of “lacunary” or “gap window” subtly mismatched the problem, the kernel would happily verify the wrong theorem. This correspondence must be audited by humans reading both — which is why the parent paper devotes a section to walking through each modeling choice (for instance: over the natural numbers, careless subtraction truncates at zero, and a naively phrased gap condition would silently admit decreasing walks; the formalization phrases the window additively to dodge exactly this). The definitions are few and short by design, to make that audit as small as possible.
- *Novelty or attribution.* Whether SHARP is genuinely new is a question about the literature, addressed in the parent repository by a documented search — no computer can settle it.
- *That nothing outside the trusted base is wrong.* The kernel, the pinned compiler and library versions — these are trusted. The trust base is small and standard, but it is not zero; no honest account says otherwise.

One story worth retelling from the parent paper: an early draft of the staircase lemma omitted the frame-merge hypothesis, and the *formalization process* caught it — the counterexample (5, 7, 9) of Remark 16.5 is now permanently enshrined in the development as a guard. The moral runs both ways: human-written mathematics slips, and machine-checking catches real slips — and what the machine certifies is exactly the statement it was given, neither more nor less.

25.5. Provenance. Like its parent, this document was produced with substantial help from automated systems — the mathematics was worked out and drafted by AI models directed and audited by the author, the formal Lean development was likewise AI-produced and kernel-checked, and the author takes responsibility for all of it. The parent paper states the division of labour in full; nothing about this friendly version changes it. If you are a student reading this: the theorem is true because of the proofs you have just read and can check — every one of them, with enough patience, by hand — and not because any person or machine says so. That is the whole point of mathematics, and it is why a document like this one can exist at all.

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